

SGA 7-I

Monodromy Groups in Algebraic Geometry

English translation

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SGA 7-I, Introduction and Exposé I

Introduction

Let S be a scheme and let $f : X \rightarrow S$ be a morphism of schemes. If f is proper and smooth, and the prime number ℓ is invertible on S , then the ℓ -adic cohomology groups

$$H^i(X_{\bar{s}}, \mathbb{Z}_\ell)$$

of the geometric fibres of f form a locally constant sheaf on S . The problem considered here is to understand what remains of this statement when the smoothness hypothesis fails at a special fibre, and in particular to describe the specialization morphism, vanishing cycles, and monodromy.

We shall only consider the case where S is a henselian trait, that is, the spectrum of a henselian discrete valuation ring. This is in practice the essential case. Let

$$S = \text{Spec}(V),$$

with generic point η and closed point s . The fundamental exact sequence is

$$\cdots \longrightarrow H^i(X_s, \mathbb{Z}_\ell) \xrightarrow{\text{sp}} H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell) \longrightarrow R^i\Phi \mathbb{Z}_\ell \longrightarrow H^{i+1}(X_s, \mathbb{Z}_\ell) \longrightarrow \cdots.$$

The $\text{Gal}(\bar{s}/s)$ -module $H^i(X_s, \mathbb{Z}_\ell)$ is regarded as a $\text{Gal}(\bar{\eta}/\eta)$ -module with trivial inertia action, and sp is the specialization map.

Criteria are also given, at present only in characteristic zero, for the vanishing of the small-degree sheaves $R^i\Phi$. For example, if the special fibre is locally a complete intersection, then

$$R^i\Phi = 0 \quad (i \neq n)$$

in the isolated-singularity situation of I, 4.5.

The monodromy theorem asserts that the action of the inertia subgroup I of $\text{Gal}(\bar{\eta}/\eta)$ is quasi-unipotent: for $T \in I$ there exist integers $N > 0$ and $M > 0$ such that

$$(T^N - 1)^M = 0 \quad \text{on} \quad H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell).$$

Applied to the H^1 of abelian varieties, i.e. to their Tate modules, these results allow one to study reduction modulo p . Two proofs are given of the stable reduction theorem for abelian varieties: after a finite extension of the base trait, the neutral component of the special fibre of the Néron model is an extension of an abelian variety by a torus.

Bures-sur-Yvette, May 1972
P. Deligne

Exposé I – Summary of the first exposés of A. Grothendieck

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1. Arithmetic proof of the monodromy theorem.
2. Vanishing cycles.
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4. Vanishing criteria for sheaves of vanishing cycles.
5. Action of monodromy on fundamental groups.
6. Appendix by P. Deligne: arithmetic proof of the stable reduction theorem.

0. Preliminaries

Terminology

We recall the following definitions.

A *trait* is the spectrum of a discrete valuation ring. Its generic and closed points are often denoted by η and s .

An *strictly local* ring is a henselian local ring with separably closed residue field. A strictly local scheme is the spectrum of such a ring.

A *geometric point* of S , in this exposé, is a morphism

$$\bar{x} : \mathrm{Spec}(k) \rightarrow S$$

with image $x \in S$, where k contains a separable closure of $k(x)$. For a henselian trait $S = \mathrm{Spec}(V)$, one often writes $\bar{\eta}$ for a geometric point over the generic point and $\bar{s}$ for one over the closed point.

The *strict localization* of X at a geometric point $\bar{x}$ is understood in the sense of SGA 4, VIII, 4.4.

A local, henselian local, or strictly local S -scheme is said to be *essentially of finite type over S* if it is the spectrum of a local ring, respectively the henselization of such a local ring, respectively the strict localization, of a scheme of finite type over S .

Reminders on ℓ -adic cohomology

Let ℓ be a prime invertible on the schemes considered. We use the usual notation

$$H^i(X_{\bar{x}}, \mathbb{Z}_{\ell}) = \varprojlim_n H^i(X_{\bar{x}}, \mathbb{Z}/\ell^n \mathbb{Z}).$$

The functors $R^i\Psi$ and $R^i\Phi$ denote, respectively, nearby cycles and vanishing cycles. For S a henselian trait and $X \rightarrow S$, the stalks of $R^i\Psi$ at a geometric point $\bar{x}$ of the special fibre are

computed on the strict localization of X at $\bar{x}$ after passage to the geometric generic fibre. The exact sequence displayed in the introduction is the long exact sequence associated with the specialization triangle.

Reminders on traits

Let S be a henselian trait with generic point η and closed point s . The Galois group $\text{Gal}(\bar{\eta}/\eta)$ has inertia subgroup I , and the quotient is $\text{Gal}(\bar{s}/s)$. Its wild inertia subgroup is denoted by P . The tame quotient is

$$I/P \simeq \prod_{\ell' \neq p} \mathbb{Z}_{\ell'}(1),$$

where p is the residual characteristic exponent.

The Néron desingularization

A standard ingredient is Néron's desingularization. In the form used here, it permits one, after replacing a morphism by an étale neighbourhood or by a suitable model, to reduce many local questions to the case of regular schemes, or to schemes with normal crossings in the special fibre.

Proposition 0.1. Let S be a trait and let X be of finite type over S . After replacing X locally by a suitable modification which is an isomorphism on the generic fibre, one may arrange that the singularities relevant for the vanishing-cycle calculation occur on a controlled normal-crossings model, whenever resolution of singularities is available.

Proposition 0.2. In equal characteristic zero, or under the corresponding resolution hypotheses, one can reduce the monodromy calculation to the case where the special fibre is a divisor with normal crossings.

1. Arithmetic proof of the monodromy theorem

Proposition 1.1. Let K be a complete discretely valued field and let V be a finite-dimensional $\mathbb{Q}_{\ell}$ -vector space with continuous action of $\text{Gal}(\bar{K}/K)$, arising from the cohomology of a scheme of finite type over K . After replacing K by a finite extension, the action of inertia on V is unipotent.

Theorem 1.2 (Monodromy theorem). Let X be proper and smooth over the generic point η of a henselian trait S , and let ℓ be invertible on S . For every i , the action of inertia on

$$H^i(X_{\bar{\eta}}, \mathbb{Q}_{\ell})$$

is quasi-unipotent. Equivalently, for T in an open subgroup of inertia,

$$(T - 1)^M = 0$$

for some M depending on i .

The arithmetic proof proceeds by reducing to representations arising from the cohomology of varieties over finite fields and using the constraints imposed by weights. The unipotent part of the inertia action is then detected by comparing the possible weights of the graded pieces.

2. Vanishing cycles

Let S be a strictly local trait and let $f : X \rightarrow S$ be of finite type. The sheaves $R^i\Psi$ of nearby cycles and $R^i\Phi$ of vanishing cycles on the special fibre measure the failure of specialization

$$H^i(X_s) \rightarrow H^i(X_{\bar{\eta}})$$

to be an isomorphism.

For a geometric point $\bar{x}$ of X_s , $R^i\Psi_{\bar{x}}$ is the cohomology of the geometric generic fibre of the strict localization of X at $\bar{x}$, and $R^i\Phi_{\bar{x}}$ is obtained from the cone of the natural map from the special fibre to the nearby fibre.

Proposition 2.1. If f is smooth in a neighbourhood of $\bar{x}$, then

$$R^0\Psi_{\bar{x}} \simeq \mathbb{Z}_\ell, \quad R^i\Psi_{\bar{x}} = 0 \quad (i > 0),$$

and $R^i\Phi_{\bar{x}} = 0$ for all i .

Proposition 2.2. The sheaves $R^i\Psi$ and $R^i\Phi$ commute with étale localization on X and with strict localization at geometric points of the special fibre.

Proposition 2.3. If X is replaced by the schematic closure of its generic fibre, then $R^i\Psi$ and $R^i\Phi$ are unchanged away from the nilpotent and vertical excess components. This often reduces the study to the flat case.

3. Geometric proof of the monodromy theorem

Purity conjecture

Let X be regular, D a divisor with normal crossings, and $U = X - D$. The purity statement needed here says that the tame fundamental group of U near a point of D is controlled by the local branches of D . Equivalently, the cohomology of the corresponding tame coverings has the expected exterior-algebra form.

Suppose that, étale locally at a point of the special fibre, the divisor is

$$X_s = \sum_{i \in B} m_i D_i$$

with normal crossings. Let $\bar{x}$ be a geometric point of X localized at x , let

$$C = \{ i \mid x \in D_i \},$$

and let K be the complex concentrated in degrees 0 and -1

$$K : \bigoplus_{i \in C} \mathbb{Z}(1) \xrightarrow{d} \mathbb{Z}(1),$$

where $d = \sum a_i$.

Theorem 3.3. Assuming purity, the tame vanishing-cycle groups $R^i\Psi_{\bar{x}}$ are described as follows:

- i) There is a canonical isomorphism of graded A -algebras

$$R^*\Psi_{\bar{x}} \simeq H^*(K \otimes A) \otimes A[t],$$

where t has degree 2.

- ii) The inertia action on $R^1\Psi_{\bar{x}}$ is described by a transitive action on a finite set whose cardinality is the prime-to- p part of

$$m = \gcd(m_i).$$

The proof proceeds as follows. Let $X_{\bar{x}}$ be the strict localization of X at $\bar{x}$, and let

$$U = X_{\bar{x}} - (X_s \cap X_{\bar{x}})$$

be the complement of the normal-crossings divisor. If t_i is a local equation of D_i , define for $(n, p) = 1$

$$U_n = \text{Spec}(\mathcal{O}_{X_{\bar{x}}}[T]/(T^n - \prod t_i^{a_i})),$$

and let $\tilde{U}_n$ be the inverse image of U in U_n . Purity gives the cohomology of these covers. Passing to the projective limit over the tame coverings gives a pro-covering of U with group

$$\mathbb{Z}(1)(\overline{k(s)})^C.$$

By definition,

$$R^i\Psi_{\bar{x}} = H^i(\tilde{U}, A).$$

The connected components are permuted transitively by tame inertia, and the cohomology of each component is computed from the complex K .

Corollary 3.4. Assume purity and suppose f proper. Let N be the least common multiple of the multiplicities m_j . Let $q \geq 0$, and let q' be the minimum of q and the maximum number of

components D_j passing through a single point. For T in tame inertia I/P , one has

$$(T^N - 1)^{q'+1} = 0 \quad \text{on} \quad R^q \Psi_{\bar{x}}.$$

Theorem 3.5. Let C be a projective smooth curve over the fraction field of a discrete valuation ring V . The inertia group acts on

$$H^1(C_{\bar{\eta}}, \mathbb{Q}_\ell)$$

by quasi-unipotent automorphisms of level 2: there exists N such that, for $T \in I$,

$$(T^N - 1)^2 = 0.$$

Remark 3.6. If A is an abelian variety over $k(\eta)$, then, since A is a quotient of a Jacobian, the same conclusion holds for the inertia action on $H^1(A_{\bar{\eta}}, \mathbb{Q}_\ell)$, or on the Tate module $T_\ell(A)$.

Remark 3.7. For every scheme X of finite type over η , the action of inertia on

$$H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)$$

is in fact quasi-unipotent of level 2.

When purity and resolution of singularities are available, for example in characteristic zero, the preceding method proves the monodromy theorem and gives bounds for the nilpotence exponent. In equal characteristic zero, for X proper and smooth over η and T in an open subgroup of I , one finds

$$(T - 1)^{i+1} = 0 \quad \text{on} \quad H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell).$$

4. Vanishing criteria for sheaves of vanishing cycles

Let S be a strictly local trait and let $f : X \rightarrow S$ be of finite type. Set

$$n = \dim(X_\eta).$$

If X' is the schematic closure of X_η in X , then:

- a) $R^0 \Phi \mathcal{O}_{X'} = \mathcal{O}_{X_s} \otimes \mathbb{Z}_\ell$ and the other $R^i \Phi$ vanish;
- b) on X' , the $R^i \Phi$ and $R^i \Psi$ associated with X/S and X'/S agree.

Since X' is flat over S , this often reduces the problem to the flat case.

Theorem 4.2. One has

$$R^i \Phi = 0 \quad (i > n).$$

More precisely, if x is a point of X_s whose closure has dimension k , and if $\bar{x}$ is a geometric point localized at x , then

$$R^i\Phi_{\bar{x}} = 0 \quad (i > n - k).$$

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension $< d$, then the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$$

is an isomorphism for $i > n + d + 1$ and an epimorphism for $i = n + d + 1$.

To prove, in some cases, that the small-degree $R^i\Phi$ vanish, one uses local duality, either directly or through the local Lefschetz theorems of SGA 2, XIV. At present this tool is available in characteristic zero, or in equal characteristic p when resolution of singularities in the relevant dimensions is available.

Theorem 4.5. Assume that S has characteristic zero. Let $f : X \rightarrow S$ be of finite type, and let x be a closed point of X_s . Suppose:

- (a) x is an isolated non-smooth point in its fibre;
- (b) X has relative geometric depth at least n at x : in a neighbourhood of x , the scheme X is defined by k equations in a smooth scheme of relative dimension N at x , with

$$n < N - k.$$

Then

$$R^i\Phi_{\bar{x}} = 0 \quad (i < n).$$

Corollary 4.6. Let S be a henselian trait of characteristic zero, let X/S be a relative complete intersection of relative dimension n , and let $x \in X_s$ be an isolated non-smooth point in its fibre. Then

$$R^i\Phi_{\bar{x}} = 0 \quad (i \neq n).$$

Corollary 4.7. Under the hypotheses of 4.6, $R^n\Phi$ is a flat $\mathbb{Z}_\ell$ -module.

Variant 4.8. In Theorem 4.5, omit condition (a), and let Z be the non-smooth locus of f . Then one still obtains

$$R^i\Phi = 0 \quad (i < n - \dim Z).$$

Proposition 4.10. Under the general hypotheses of 4.5, suppose:

- (a) x is an isolated non-smooth point in its fibre;

(b) for every point y of $X_{\bar{\eta}}$, with closure of dimension $d(y)$ and specializing to x , one has

$$\mathrm{prof}_{\mathrm{et},y}(X_{\bar{\eta}}) \geq n - d(y);$$

(c) $\mathrm{prof}_{\mathrm{et},x}(X_s) \geq n$.

Then

$$R^i \Phi_{\bar{x}} = 0 \quad (i < n - 1).$$

5. Action of monodromy on π_1

In this section, corresponding to Grothendieck's exposé of 19 November 1968, we use the semistable reduction theorem for curves. Artin and Winters gave a direct proof; it can also be deduced from the analogous theorem for abelian varieties. We also use Schlessinger's theory as presented in Rim's Exposé VI.

Let k be algebraically closed, let C be a projective curve over k , and let $x_0, \dots, x_n$ be finitely many closed points of C . The pointed curve

$$X = (C; x_0, \dots, x_n)$$

is called *stable* if:

- (a) C is reduced and its only singularities are ordinary double points with distinct tangents;
- (b) the points x_i are distinct smooth points of C ;
- (c) if C_j is an irreducible component of C , and E is the set of points of the normalization $\tilde{C}_j$ lying over a singular point of C or over one of the x_i , then if C_j has genus 0, respectively 1, the set E has at least 3, respectively at least 1, elements.

Condition (c) is equivalent to the absence of non-trivial infinitesimal automorphisms. It is also equivalent to ampleness of the dualizing sheaf twisted by the marked points:

$$\omega\left(\sum x_i\right).$$

A curve C over a scheme S , equipped with sections $x_0, \dots, x_n : S \rightarrow C$, is a stable curve over S if its geometric fibres are stable. The extension from the unmarked case to the marked case is formal:

- (1) for X/k stable, ω^n is very ample for $n \geq 3$;
- (2) the formal moduli scheme of X is smooth, and the singularity locus of C lies over a normal-crossings divisor in the moduli space;

- (3) for stable curves X and Y over S , the scheme $\text{Isom}(X, Y)$ is finite and unramified over S , with isomorphisms preserving the marked points;
- (4) for X/k stable, $\text{Aut}_S(X) \rightarrow \text{Aut}_S(\text{Pic}^0(X))$ is injective.

Proposition 5.1.1. Let

$$X = (C; x_0, \dots, x_n)$$

be stable over the generic point η of a trait S , and assume C_η smooth. There exists a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X'_{\eta'} = X_\eta \otimes_\eta \eta'.$$

Let X be a geometrically connected scheme of finite type over the fraction field of a strictly local trait S . For a geometric point ξ of $X_{\bar{\eta}}$ there is an exact sequence

$$0 \longrightarrow \pi_1(X_{\bar{\eta}}, \xi) \longrightarrow \pi_1(X, \xi) \longrightarrow \text{Gal}(\bar{\eta}/\eta) \longrightarrow 0, \quad (5.2.1)$$

and hence a homomorphism from $\text{Gal}(\bar{\eta}/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar{\eta}})$. Passing to the maximal prime-to- p quotient gives

$$\text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Aut ext}(\pi_1^{(p')}(\bar{\eta})). \quad (5.2.2)$$

If X_η has a rational point x , with geometric point $\bar{x}$ over it, then the exact sequence splits canonically and gives

$$[x] : \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Aut}(\pi_1^{(p')}(X_{\bar{\eta}}, \bar{x})). \quad (5.2.3)$$

Theorem 5.3. The image of wild inertia P under (5.2.2), or equivalently under (5.2.3), is finite.

For the proof, one first reduces to the case where X is geometrically normal and integral. Let $\tilde{X}_\eta$ be the normalization of X_η ; after a finite extension of $k(\eta)$, the connected components of $\tilde{X}_\eta \times_\eta \bar{\eta}$ may be assumed geometrically normal and integral with rational base points, and the connected components of $X_{\bar{\eta}}$ geometrically connected with rational base points. If X is normal and $U \subset X$ is open, then $\pi_1(U)$ maps onto $\pi_1(X_{\bar{\eta}})$. Thus one may suppose X_η smooth and affine. Cutting by generic hyperplane sections and using Bertini, one reduces to X smooth, geometrically connected, and of dimension one.

Theorem 5.4. Let S be strictly local. Let $f : X \rightarrow S$ be proper and flat, with special fibre a curve whose only non-smooth points are ordinary double points, and let Y be a finite union of disjoint sections contained in the smooth locus. Let x be a rational point of

$$X_\eta = X - Y.$$

Let $U \subset S$ be the open over which X is smooth, and let ξ be a geometric point of U . Then the image of

$$\pi_1(U, \xi)$$

in

$$\text{Aut}(\pi_1^{(p')}(X_{\bar{\eta}}, \bar{x}))$$

is abelian and prime to p .

One must note that $\pi_1(U, \xi)$ can act on the prime-to- p fundamental group only because the prime-to- p fundamental groups satisfy the specialization theorem.

6. Appendix: arithmetic proof of the stable reduction theorem

Let S be a trait and let A_η be an abelian variety over $k(\eta)$ of dimension $d \neq 0$. For a local morphism of traits $S' \rightarrow S$, denote by $A_{\eta'}$ the abelian variety obtained by extension of scalars, by $A_{S'}$ its Néron model over S' , by $A_{s'}$ its special fibre, and by $A_{s'}^0$ the neutral component of $A_{s'}$.

Theorem 6.1. For a suitable S' , the variety $A_{\eta'}$ has stable reduction: the group $A_{s'}^0$ is an extension of an abelian variety by a torus.

It follows fairly easily that one may take S' so that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field. Let ℓ be a prime different from the residual characteristic exponent p , let

$$T_\ell(A)$$

be the Tate module, let

$$V_\ell(A) = T_\ell(A) \otimes \mathbb{Q}_\ell,$$

and let

$$\rho : \text{Gal}(\bar{\eta}/\eta) \rightarrow \text{GL}(V_\ell(A))$$

be the natural representation. After a preliminary scalar extension, assume that inertia acts unipotently, hence through its largest pro- ℓ quotient $\mathbb{Z}_\ell(1)$. Let T be the image in $\text{GL}(V_\ell(A))$ of a generator of $\mathbb{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that

$$(T - 1)^r \neq 0.$$

Lemma 6.2. The map

$$(c, v) \in I \times V_\ell(A) \longmapsto (\rho(c) - 1)^r v$$

induces morphisms and an isomorphism

$$V_\ell(A) \otimes \mathbb{Q}_\ell(r) \longrightarrow V_\ell(A) / \text{Ker}(T - 1)^r \otimes \mathbb{Q}_\ell(r) \xrightarrow{\sim} V_\ell(A). \quad (6.2.1)$$

An ℓ -adic representation $\rho : \text{Gal}(\bar{s}/s) \rightarrow \text{GL}(V)$ is said to have *weight* n if, for a suitable model X of $k(s)$ and every closed point x of X , the eigenvalues of geometric Frobenius F_x are algebraic numbers all of whose complex conjugates have absolute value

$$|k(x)|^{n/2}.$$

If V has weight n and W has weight $m \neq n$, then

$$\text{Hom}_{\text{Gal}(\bar{s}/s)}(V, W) = 0.$$

By the universal property of the Néron model, $V_\ell(A_s^0)$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2 , where a is the dimension of the largest abelian quotient of A_s^0 and m the dimension of its multiplicative part. For the dual abelian variety A^* , the duality

$$V_\ell(A^*) \times V_\ell(A) \rightarrow \mathbb{Q}_\ell(1)$$

shows that $V_\ell(A)$ is an extension of a representation of dimension m and weight 0 by one of dimension $2a^*$ and weight -1 .

Since $\mathbb{Q}_\ell(r)$ has weight $-2r$ and $\text{Im}(T-1)^r \neq 0$, the preceding weight argument and (6.2.1) imply $r \leq 1$. If $r = 0$, then $m = 0$, $a = d$, and A has good reduction. If $r = 1$, then $\text{Im}(T-1)$ has weight -2 and the corresponding quotient has weight 0 ; the two spaces have the same dimension $m \leq d$. One obtains

$$\dim V_\ell(A^*) = 2a^* + m^* = 2a + m,$$

and

$$2 \dim A_{s'}^0 \geq 2(a + m) > 2a + m + m = 2d = 2 \dim A_{\eta'}.$$

Thus $A_{s'}^0$ has stable reduction, and at the same time one gets $m^* = m$.

For the general case, one reduces to a situation in which the appropriate hypotheses of the preceding equal- or unequal-characteristic propositions hold, the group $\text{Gal}(\bar{\eta}/\eta)$ acts trivially on A_ℓ , and inertia acts unipotently on $T_\ell(A)$. The same argument then applies.

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SGA 7-I, Exposé II through Exposé VII.2.1

SGA 7-I, Expose II – Finiteness Properties of the Fundamental Group

Michele Raynaud

Let k be a separably closed field of characteristic exponent $p \geq 0$, let X be a connected scheme of finite type over k , and let

$$\pi_1^{(p')}(X)$$

be the largest quotient of $\pi_1(X)$ which is prime to p . We prove in this expose that, if X is a surface, $\pi_1^{(p')}(X)$ is topologically finitely presented, and that the same conclusion holds for arbitrary X if one assumes resolution of singularities in the relevant dimensions.

In the case of a surface, the proof, due to J.-P. Murre, reduces to the case where one has a fibration over $\mathbb{P}^1$ satisfying the three conditions in the following proposition.

Proposition 0.1. *Let k be algebraically closed, and let S be a projective, irreducible, smooth surface over k . Then one can find a blow-up*

$$g : S' \rightarrow S,$$

where S' is a regular surface endowed with a fibration

$$f : S' \rightarrow \mathbb{P}^1,$$

such that f has a section σ and the following conditions hold:

- (i) *f is smooth at the points of $\sigma(\mathbb{P}^1)$;*
- (ii) *the fibres of f are geometrically integral curves having only ordinary singularities;*
- (iii) *there exists a non-empty open subset U of $\mathbb{P}^1$ such that the fibres of f at the points of U are smooth.*

The proposition follows from the existence of a projective embedding $S \hookrightarrow \mathbb{P}^r$ for which one can find a Lefschetz pencil. Let

$$D = \{H_t\}_{t \in \mathbb{P}^1}$$

be such a pencil of hyperplanes. Recall that it has the following properties: its axis A meets S transversally, so that the closed subscheme $A \cap S$ is reduced and consists of finitely many closed points $x_1, \dots, x_n$ of S ; there is a non-empty open set $U \subset \mathbb{P}^1$ such that, for $t \in U$, H_t meets S transversally; and for $t \in \mathbb{P}^1 - U$, H_t meets S transversally except at one point, which is an ordinary quadratic singular point.

Through each point x of S not lying on A there passes a unique hyperplane H_t . The map which sends x to this $t \in \mathbb{P}^1$ is a rational map from S to $\mathbb{P}^1$ with domain of definition $S - (A \cap S)$.

Let $g : S' \rightarrow S$ be the blow-up of the closed subscheme $S \cap A$. The scheme S' is regular and the fibre S'_i over a point x_i identifies with the projective bundle defined by the tangent space to S at x_i , hence is isomorphic to $\mathbb{P}^1$. Moreover the rational map $f = a \circ g$ is everywhere defined. Finally, to each point x_i one associates the section σ_i of f which sends a point $t \in \mathbb{P}^1$ to the tangent vector at x_i to $H_t \cap S$.

We verify conditions (i), (ii), and (iii). For each $t \in \mathbb{P}^1$, the curves S_t and $S'_t = g^{-1}(S_t)$ are isomorphic: the morphism $S'_t \rightarrow S_t$ is an isomorphism except possibly at the points x_i , and these are regular points of S_t ; since g is proper, it follows that S'_t is geometrically integral. This proves (ii). Assertion (i) follows from the fact that f is smooth along $\sigma(\mathbb{P}^1)$, since g is an isomorphism in a neighbourhood of $\sigma(\mathbb{P}^1)$ and a is smooth at the points where the corresponding hyperplane meets S transversally.

Corollary 0.2. *With the notation of the proposition, let Y be a normal-crossings divisor in S , and let $Y' = Y \times_S S'$. Then f , g , and U may be chosen so that Y' is a normal-crossings divisor relative to U .*

Let G be a profinite group. Recall that G is said to be *topologically finitely generated* if there is an integer n such that G is isomorphic to a quotient of the free profinite group $F(n)$ on n generators. Let $K = \text{Ker}(F(n) \rightarrow G)$. We say that G is *topologically finitely presented* if, in addition, there are finitely many elements $k_1, \dots, k_m$ of K such that the smallest closed normal subgroup of $F(n)$ containing $k_1, \dots, k_m$ is equal to K .

Theorem 0.3. *Let k be a separably closed field of characteristic exponent $p \geq 0$, and let X be a connected scheme of finite type over k . Suppose that schemes of finite type and of dimension at most $\dim(X)$ over an algebraic closure of k are strongly desingularizable. This hypothesis holds for $\dim X = 2$ by Abhyankar, and for $k = \mathbb{C}$ by Hironaka. Then the fundamental group*

$$\pi_1^{(p')}(X)$$

is topologically finitely presented.

By SGA 1, IX, 4.10, one may suppose that k is algebraically closed.

Reduction to the case where X is a regular surface. If $f : X' \rightarrow X$ is a morphism of finite type, put $X'' = X' \times_X X'$, and denote by $\pi_0(X')$ and $\pi_0(X'')$ the sets of connected components of X' and X'' , respectively. When f is a morphism of effective descent for the category of étale coverings, the fundamental group $\pi_1(X)$ can be computed explicitly in terms of the fundamental groups of the connected components of X' and X'' . It follows from SGA 1, IX, 5.2 and 5.3 that, if the $\pi_1(X'_\alpha)$ are topologically finitely generated, then so is $\pi_1(X)$; and if the $\pi_1(X'_\alpha)$ are topologically finitely presented while the $\pi_1(X''_\beta)$ are topologically finitely generated, then $\pi_1(X)$ is topologically finitely presented.

Since X is desingularizable, one can find a regular scheme X' and a proper birational morphism $f : X' \rightarrow X$. This morphism is of effective descent for etale coverings, so it is enough to prove the theorem under the additional hypothesis that X is regular.

Let $(U_i)_{i \in I}$ be a finite cover of X by affine open subsets, and let $f : \coprod U_i \rightarrow X$ be the canonical morphism. Since this is again a morphism of effective descent for etale coverings, it is enough to prove the theorem for the U_i . Thus one is reduced to the case where X is affine and smooth.

Assume from now on that X is affine and smooth over k , with $\dim X > 2$. One can apply the Lefschetz theorem for quasi-projective schemes: if Y is a sufficiently general hyperplane section of X , then

$$\pi_1(Y) \xrightarrow{\sim} \pi_1(X)$$

is an isomorphism. Hence it is enough to prove the theorem for Y , and this reduces the argument to the case $\dim X = 2$.

The case where X is a smooth surface over k . By the strong desingularization theorem for surfaces, one can find a smooth integral projective surface S and an open immersion $i : X \hookrightarrow S$ such that the divisor $Y = S - X$ has normal crossings. Apply the preceding corollary and keep its notation.

It is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically finitely presented. Since X and X' are regular and

$$\mathrm{codim}_{X'} \left(X' - \bigcup_{1 \leq i \leq n} x_i \right) \geq 2,$$

every etale covering of X' induces on $X' - \bigcup_{1 \leq i \leq n} x_i$ an etale covering which extends uniquely to an etale covering of X by SGA 2, 1.11. This gives an isomorphism

$$\pi_1(X') \xrightarrow{\sim} \pi_1(X).$$

Consider the cartesian diagram

$$\begin{array}{ccc} X' & \longleftarrow & X \\ \downarrow h & & \downarrow \\ \mathbb{P}^1 & \xleftarrow{j} & U. \end{array}$$

Here $h = f|_{X'}$. Let L be the set of primes distinct from p , and let $\bar{\eta}$ be a geometric point above the generic point η of $\mathbb{P}^1$. We verify the hypotheses of SGA 1, XIII, 4.8. The morphism h has a section because f does. Condition a) of SGA 1, XIII, 4.7 follows as follows. Since the fibres of h are geometrically integral, h is 0-acyclic and locally 0-acyclic. It remains to show that, for every locally constant sheaf of sets F on X' , the pair (F, h) is cohomologically proper in dimension ≤ 0 . If ξ is a closed point of $\mathbb{P}^1$ and if Z denotes the integral closure of $\mathrm{Spec} \mathcal{O}_{\mathbb{P}^1, \xi}$ in $\bar{\eta}$, then $\mathcal{X} = X' \times_{\mathbb{P}^1} Z$ satisfies condition (S_2) at the closed points of X' and is regular at every other point, hence is normal. It follows that an etale covering of $\mathcal{X}_\xi$ whose restriction to X'_ξ is connected

is itself necessarily connected.

Since X' is the complement in S' of a normal-crossings divisor relative to U , the morphism h_η is locally 1-aspherical for L , and, for every finite constant sheaf of L -groups F on X' , the sheaf $R^1 h_* (F_\eta)$ is cohomologically proper in dimension ≤ 1 . This gives condition b). Because $\mathbb{P}^1$ is normal and $T = \mathbb{P}^1 - U$ has codimension 1, condition c) follows from the etale depth inequality at $\bar{s}$; the topological depth condition at the corresponding points of X' follows from purity; and since h is smooth at $\sigma(t)$ for every $t \in \mathbb{P}^1$, the local 1-asphericity condition along the section gives condition d).

By SGA 1, XIII, 4.7, if a is a geometric point of X' , one has an exact sequence

$$1 \longrightarrow \pi_1(X'_\eta, a) \longrightarrow \pi_1(X', a) \longrightarrow \pi_1(U, \bar{\eta}) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K be the image of $\pi_1(U, a)$ in $\text{Aut}(\pi_1^{(p')}(X'_\eta, a))$. The coinvariant group $\pi_{1,\text{ad}}^{(p')}(X'_\eta, a)$ is the quotient of $\pi_1^{(p')}(X'_\eta, a)$ by the closed normal subgroup generated by elements of the form $\alpha(x)x^{-1}$, with $x \in \pi_1^{(p')}(X'_\eta, a)$ and $\alpha \in K$. By loc. cit. 4.7.4, one has an isomorphism

$$\pi_1^{(p')}(X', a) \xrightarrow{\sim} \pi_{1,\text{ad}}^{(p')}(X'_\eta, a).$$

Let $\mathbb{P}^1 - U = \{t_1, \dots, t_n\}$. For each i , let T_i be the strict localization of $\mathbb{P}^1$ at a geometric point $\bar{t}_i$ above t_i , and let η_i be the generic point of T_i . An etale covering of U whose restriction to every η_i is trivial extends to an etale covering of $\mathbb{P}^1$, hence is trivial. Equivalently, $\pi_1(U, a)$ is the closed normal subgroup generated by the images of the groups $\text{Gal}(\bar{k}(\bar{t}_i)/k(t_i))$ in $\pi_1(U, a)$. By Expose I, 5.3, the image K_i of $\text{Gal}(\bar{k}(\bar{t}_i)/k(t_i))$ in $\text{Aut}(\pi_1^{(p')}(X'_\eta, a))$ is finite. Since K is the smallest closed normal subgroup containing all K_i , and since $\pi_1^{(p')}(X'_\eta, a)$ is topologically finitely presented by SGA 1, XIII, 3.2, this proves that $\pi_1^{(p')}(X', a)$, and hence $\pi_1^{(p')}(X)$, is topologically finitely presented.

Remark 0.4. Resolution of singularities is not needed for the theorem in the case where X is the open complement of a normal-crossings divisor in a smooth projective scheme over k .

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SGA 7-I, Expose VI – Formal Deformation Theory

by D. S. Rim

1. Formal existence theorem

We recall the notion of cofibered category (S.G.A. 1 IV). Let $\mathcal{C}$ be a fixed category. A *cofibered category* A over $\mathcal{C}$ is, by definition, a category A together with a functor $P : A \rightarrow \mathcal{C}$ such that :

Ax. 1. For any map f in $\mathcal{C}$ and any object a in A with $P(a) =$ the source of f , there exists an f -morphism $\alpha : a \rightarrow b$ which is cocartesian (i.e. for any f -morphism $\alpha' : a \rightarrow b'$ there exists a unique 1-morphism $\tau : b \rightarrow b'$ in A such that $\alpha' = \tau\alpha$ where $1 =$ the target of f).

Ax. 2. A composite of cocartesian morphisms in A is again cocartesian.

The following properties of a cofibered category $P : A \rightarrow \mathcal{C}$ follow immediately from the definitions :

(cancellation) Let $\alpha, \beta : a \rightrightarrows a'$ be any two cocartesian maps such that $P(\alpha) = P(\beta)$. If there exists a cocartesian map $\gamma : b \rightarrow a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(factorization) Given cocartesian maps $\alpha : a \rightarrow a', \beta : a \rightarrow a''$, there exists a cocartesian map $\delta : a' \rightarrow a'''$ such that $\delta\alpha = \gamma\beta$.

0.0.1

Ex. 1.1. (a) Let $F : \mathcal{C} \rightarrow (\text{categories})$ be a functor. Define the category $\int F$ as follows : an object in $\int F$ is a pair (S, a) where $S \in \text{ob}(\mathcal{C})$ and $a \in \text{ob } F(S)$, and a map $(S, a) \rightarrow (S', a')$ is a pair (f, u') where $f : S \rightarrow S'$ is a map in $\mathcal{C}$ and $u' : F(f)(a) \rightarrow a'$ is a map in $F(S')$. Composition of two maps $(S, a) \xrightarrow{(f, u')} (S', a') \xrightarrow{(g, u'')} (S'', a'')$ is given by $(gf, u'' \cdot F(g)(u')) : (S, a) \rightarrow (S'', a'')$. With respect to the functor $P : \int F \rightarrow \mathcal{C}$ given by $(S, a) \mapsto S$, $\int F$ becomes a cofibered category over $\mathcal{C}$. If $F(S)$ is a groupoid for every $S \in \text{ob}(\mathcal{C})$, then $\int F$ is a cofibered groupoid over $\mathcal{C}$.

In particular, a functor $F : \mathcal{C} \rightarrow (\text{groups})$ yields a cofibered group, and a functor $F : \mathcal{C} \rightarrow (\text{Ens})$ yields a cofibered discrete groupoid. We also note that a natural transformation $\theta : F \rightarrow G$, where $F, G : \mathcal{C} \rightarrow (\text{categories})$ are functors, determines a cocartesian functor $\int F \rightarrow \int G$.

Ex. 1.1. (b) Let X_k, Y_k be schemes over k and let there be given an isomorphism $X \otimes k \xrightarrow{\sim} Y \otimes k$. Then the functor $\text{Isom}_{X,Y} : \mathcal{C}_\Lambda \rightarrow (\text{groupoids})$ given by

$$\text{Isom}_{X,Y}(S) = \text{the set of isomorphisms } X \otimes S \xrightarrow{\sim} Y \otimes S \text{ such that } \dots$$

yields a cofibered groupoid $P : \int \text{Isom}_{X,Y} \rightarrow \mathcal{C}_\Lambda$.

Ex. 1.1. (c) Let X_0 be a fixed scheme over k , and let $\mathcal{M}_X$ be the category consisting of deformations of X_0 over $\mathcal{C}_\Lambda$ (see §4). Thus an object in $\mathcal{M}_X$ is a pair (R, X) where R is in $\mathcal{C}_\Lambda$, and X is a deformation of X_0 to R . Then the functor $P : \mathcal{M}_X \rightarrow \mathcal{C}_\Lambda$ given by $(R, X) \mapsto R$ defines a cofibered groupoid over $\mathcal{C}_\Lambda$.

Ex. 1.1. (d) For any groupoid $\mathcal{X}$ we denote by $[\mathcal{X}]$ the discrete groupoid consisting of the isomorphism classes of objects in $\mathcal{X}$. If $P : F \rightarrow \mathcal{C}$ is a cofibered groupoid, we obtain a functor $[F] : \mathcal{C} \rightarrow (\text{Ens})$ given by $[F](S) = [F(S)]$, and in turn a cofibered discrete groupoid, which is, by abuse of notation, denoted by $[F]$.

Definition 0.5. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is called *semi-homogeneous* if the following properties hold :

- (1) every diagram of solid arrows in A where $a' \rightarrow a$ is surjective can be completed into a commutative diagram including dotted arrows.
- (2) Let $R \times_k k[\epsilon]$ be the projection maps, where $k[\epsilon]$ is the ring of dual numbers over k , and let $a' \rightrightarrows a$ be π_1 - π_2 -map in A . Then there exists a $\mathcal{C}_\Lambda$ -map $\gamma : a' \rightarrow a''$ with $\alpha = \beta\gamma$ if and only if $(\pi_1)_*(a') = (\pi_2)_*(a'')$.

Remark 0.6. A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). We note that if A is semi-homogeneous, then so is $[A]$ and in particular 1.2(2) entails the canonical map

$$[A(R \times_k k[\epsilon])] \longrightarrow [A(R)] \times [A(k[\epsilon])]$$

to be bijective.

Remark 0.7. Consider a commutative diagram in a cofibered category A over $\mathcal{C}_\Lambda$. Set $R = P(a)$, $R' = P(a')$ and $R_i = P(a_i)$ for $i = 1, 2$. If we set a''' to be the “direct image” under the map $(P(\nu_1), P(\nu_2)) : R' \rightarrow R_1 \times R_2$, we obtain a commutative diagram. Thus 1.2(1) can be replaced by :

- (1)' every diagram in A above $R_1 \leftarrow R' \rightarrow R_2$ in $\mathcal{C}_\Lambda$, where $R' \rightarrow R_1 \times R_2$ is surjective can be completed to a commutative diagram, where μ_i ($i = 1, 2$) are projections.

Remark 0.8. If A is a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, then $[A(k[\epsilon])]$ is canonically provided with a vector space structure over k , where $k[\epsilon]$ is the ring of dual numbers over k . Indeed, let $\mathcal{V}$ be the category of finite-dimensional vector spaces over k . Then any functor $H : \mathcal{V} \rightarrow (\text{Ens})$ preserving the product is canonically factored through $\mathcal{V} \rightarrow (\text{Ens})$, where $\mathcal{V} \rightarrow (\text{Ens})$ is the forgetful functor. On the other hand, the functor $G : \mathcal{V} \rightarrow \mathcal{C}_\Lambda$ given by $W \mapsto k[W] = k \oplus W$, with $w^2 = 0$, transforms products in $\mathcal{V}$ into fibre products over k in $\mathcal{C}_\Lambda$. If A is a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, then $[A(k[V] \times_k k[W])] \rightarrow [A(k[V])] \times [A(k[W])]$ is bijective by 1.3(a), and the functor $V \mapsto [A(k[V])]$ commutes with products, hence the result.

Definition 0.9. (1) Let $R \in \text{ob}(\mathcal{C}_\Lambda)$. A *small extension* of R in $\mathcal{C}_\Lambda$ is a surjective map $R' \rightarrow R$ in $\mathcal{C}_\Lambda$ such that $(\text{Ker } f)^2 = 0$ and $\text{Ker } f \cong k$.

In other words, $R' \rightarrow R$ in $\mathcal{C}_\Lambda$ is a small extension of R if

$$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$.

- (2) Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$, and let a be an object in A . A *small prolongation* of a is an arrow $a' \rightarrow a$ in A such that $P(a' \rightarrow a)$ is a small extension in $\mathcal{C}_\Lambda$. An arrow $a' \rightarrow a$ in A will be called *versal for small prolongations of a* if, for every small prolongation $a_1 \rightarrow a$ in A , there exists a commutative diagram

$$\begin{array}{ccc} a' & \longrightarrow & a_1 \\ \searrow & & \swarrow \\ & a & \end{array}$$

Proposition 0.10. *Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$.*

- (1) *Let $R \times_k k[\epsilon] \xrightarrow{\pi_1} R$ be projection maps in $\mathcal{C}_\Lambda$, and let $a \xrightarrow{\phi} a'$ be a 1-map. Then there exists a map $a \xrightarrow{\psi} a'$ such that $\phi \otimes \beta = 1_a$ if and only if there exists a map $a' \xrightarrow{\theta} (\pi_2)_*(a'')$ such that $\dots$*
- (2) *Let $R' \rightarrow R$ be a small extension in $\mathcal{C}_\Lambda$, and consider a commutative diagram*

$$\begin{array}{ccc} R' \times_R R' & & \\ \downarrow & \searrow & \\ a'' & \longrightarrow & a \\ \downarrow & & \downarrow \\ R' & \longrightarrow & R \end{array}$$

Define the canonical map $\mu : R' \times_R R' \rightarrow k[\text{Ker } f]$ by $(r_1, r_2) \mapsto r_2 - r_1$. Then, if there exists a map $a' \rightarrow (f)_(a'')$, then there exists a map $\gamma : a' \rightarrow a$ such that $\alpha = a\gamma$.*

Proof. (1) The part “only if” is clear. Now assume that there exists a map $\phi : a \rightarrow (\pi_2)_*a'$. If we set $h : R \rightarrow R \times_k k[\epsilon]$ by $h = 1 \times P(\phi)$, then the composite map $R \xrightarrow{h} R \times_k k[\epsilon] \xrightarrow{\pi_1} R$ is the identity, and therefore we get maps $a \xrightarrow{\gamma} a'$ such that $\pi_1\gamma = 1_a$ where $P(\gamma) = h$ and $P(a') = \pi_1$.

On the other hand, $P(\alpha) = P(\pi_2\phi)$ and hence there exists a 1-map $\psi : a' \rightarrow (\pi_2)_*a''$. This means that $(\pi_2)_*\psi\alpha = (\pi_2)_*\phi$, and therefore by 1.2(2) we have a $\mathcal{C}_\Lambda$ -map $u : h_*a \rightarrow a'$ such that $\alpha u = a_1$. Define $\beta : a \rightarrow a'$ by $\beta = u\gamma$. Then $\phi\beta = \alpha u\gamma = \pi_1\gamma = 1_a$.

- (2) The map $1 \times f : R' \times_R R' \rightarrow R' \times_k k[\text{Ker } f]$ is an isomorphism compatible with the projections on the first factor, and $\pi_2 \circ (1 \times f) = \mu$ where $\pi_2 : R' \times_k k[\text{Ker } f] \rightarrow k[\text{Ker } f]$ is the projection map. We note that $k[\text{Ker } f] \cong k[\epsilon]$ since $R' \rightarrow R$ is a small extension. Therefore, if there exists a map $a'' \rightarrow (\mu)_*(a')$, then by (1) above there exists a map $\gamma' : a' \rightarrow a''$ such that $a' \cdot \gamma' = 1_{a'}$. Define $\gamma : a' \rightarrow a$ by $\gamma = \beta\gamma'$. We then have $a\gamma = a$. $\square$ $\square$

Corollary 0.11. *Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$ and consider a diagram in A where a_1 is a small prolongation. Assume that a is versal for small prolongations of e . Then there exists a map $a' \rightarrow a_1$ such that $a = a_1\theta$ in A if and only if there exists a map $f : P(a') \rightarrow P(a_1)$ such that $P(a) = P(a_1) \circ f$ in $\mathcal{C}_\Lambda$.*

Proof. Assume that $P(a) = P(a_1) \circ f$ where $f : P(a') \rightarrow P(a_1)$. Replacing a' by its direct image under f , we may and shall assume that $P(a') = P(a_1)$. Set $R' = P(a_1)$, $R = P(a)$. Since A is

semi-homogeneous, we get a commutative diagram. Since a is versal for small prolongations of e , so is a' and therefore our statement follows from 1.5(2). $\square$ $\square$

Definition 0.12. For each object R in $\mathcal{C}_\Lambda$ we put $\Omega_R = \mathfrak{m}_\Lambda / (\mathfrak{m}_\Lambda R + \mathfrak{M})$ where $\mathfrak{m}_\Lambda, \mathfrak{M}$ are the maximal ideals of Λ, R respectively. It is a finite-dimensional vector space over k , and $R \mapsto \Omega_R$ defines a functor $\Omega : \mathcal{C}_\Lambda \rightarrow \mathcal{V}$ where $\mathcal{V}$ is the category of finite-dimensional vector spaces over k . If A is a cofibered category over $\mathcal{C}_\Lambda$, then the composite functor $A \xrightarrow{P} \mathcal{C}_\Lambda \xrightarrow{\Omega} \mathcal{V}$ is, by abuse of notation, denoted by Ω_A again.

Proposition 0.13. *Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$. Then, given an object a in A versal for small prolongations of e , there exists $a' \rightarrow a$ such that :*

- (i) a' is versal for small prolongations of a ;
- (ii) $P(a') \rightarrow P(a)$ is surjective ; and the kernel is annihilated by the maximal ideal of $P(a)$.
- (iii) $\Omega_A : e \rightarrow \Omega_a$ is an isomorphism.

Proof. Set $R = P(a)$ and consider an exact sequence $0 \rightarrow J \rightarrow S \rightarrow R \rightarrow 0$ where S is a formal power-series ring over Λ in a finite number of variables, and $J \subset \mathfrak{M}$ where $\mathfrak{M}$ is the maximal ideal of S . Let $\mathcal{F}$ be the set of ideals J_α in S such that (i) $J \supset J_\alpha \supset \mathfrak{M}J$ and (ii) there exists π_α -morphism $a_\alpha \rightarrow a$ in A where $\pi_\alpha : S/J_\alpha \rightarrow R$ is the canonical map. The property 1.2(1) of A together with the fact that $J/\mathfrak{M}J$ is a finite dimensional vector space over k entails that there exists a smallest ideal in the family $\mathcal{F}$. Let it be J' and choose a π' -morphism $a' \rightarrow a$. $\square$ $\square$

0.0.2

We now consider the category $\text{pro}(\mathcal{C}_\Lambda)$ of pro-objects in $\mathcal{C}_\Lambda$. If A is a cofibered category over $\mathcal{C}_\Lambda$, we obtain a cofibered category $\text{pro}(A)$ over $\text{pro}(\mathcal{C}_\Lambda)$ (in groupoids). For a cofibered category A over $\mathcal{C}_\Lambda$, we define :

$$\hat{A} = \text{the category of formal objects of } A \text{ over } \text{pro}(\mathcal{C}_\Lambda).$$

Definition 0.14. An object ξ in $\hat{A}$ is called a *versal formal object* of A if every surjective map $q : a' \rightarrow a$ in $\mathcal{C}_\Lambda$ and every map $\xi \rightarrow a$ in $\hat{A}$ can be completed to a commutative diagram.

Definition 0.15. A versal formal object ξ in $\hat{A}$ is called *minimally versal* if the morphism $\bar{\xi} : \text{Spf}(\hat{R}) \rightarrow [A]$ induced by ξ is a minimal embedding, where $\hat{R} = P(\xi)$.

Theorem 0.16 (Formal Existence Theorem). *A semi-homogeneous cofibered groupoid A over $\mathcal{C}_\Lambda$ which satisfies the conditions of Proposition ?? admits a minimally versal formal object, unique up to (non-unique) isomorphism.*

0.0.3

Comparison with Schlessinger's conditions. Let $F : \mathcal{C}_\Lambda \rightarrow (\text{Ens})$ be a functor preserving the final object. The conditions (S1') and (S2) of Schlessinger are equivalent to saying that $\int F$ is semi-homogeneous. The condition (H3) of Schlessinger is equivalent to saying that $[A(k[\epsilon])]$ is a finite-dimensional vector space over k .

2. Deformation of algebraic schemes

Let X_0 be a scheme over k , locally of finite type. We define the category $\mathcal{M}_{X_0}$ of deformations of X_0 over $\mathcal{C}_\Lambda$ as follows :

Definition 0.17. An object of $\mathcal{M}_{X_0}$ over $R \in \mathcal{C}_\Lambda$ is a pair (X, ϕ) where X is a scheme flat over R and $\phi : X \otimes_R k \xrightarrow{\sim} X_0$ is an isomorphism of k -schemes. A morphism $(X, \phi) \rightarrow (X', \phi')$ over $R \rightarrow R'$ is a morphism $f : X \rightarrow X'$ over $R \rightarrow R'$ such that the diagram

$$\begin{array}{ccc} X \otimes_R k & \xrightarrow{\sim} & X_0 \\ \downarrow & & \parallel \\ X' \otimes_{R'} k & \xrightarrow{\sim} & X_0 \end{array}$$

commutes.

With the projection functor $P(X, \phi) = R$, $\mathcal{M}_{X_0}$ is a cofibered groupoid over $\mathcal{C}_\Lambda$.

Proposition 0.18. *If X_0 is proper over k , then $\mathcal{M}_{X_0}$ is semi-homogeneous. Moreover, $[\mathcal{M}_{X_0}(k[\epsilon])]$ is canonically isomorphic to $\text{Ext}_{X_0}^1(\Omega_{X_0/k}^1, \mathcal{O}_{X_0})$.*

0.0.4

For a morphism $f : X_0 \rightarrow Y_0$ of schemes over k , one can define the category of deformations of f and prove analogous results. The obstructions to deforming X_0 lie in $\text{Ext}_{X_0}^2(\Omega_{X_0/k}^1, \mathcal{O}_{X_0})$.

3. The cotangent complex

Let $f : X \rightarrow S$ be a morphism of schemes. The cotangent complex $\mathbb{L}_{X/S}$ is an object of the derived category $D^-(X)$ defined up to unique isomorphism.

Theorem 0.19. *The cotangent complex satisfies the following properties :*

(i) *For any commutative diagram*

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

there is a canonical base-change morphism $\mathbb{L}_{X/S}^ \mathbb{L}_{X'/S'} \rightarrow \mathbb{L}_{X'/S'}$ which is an isomorphism when the diagram is cartesian and $X \rightarrow S$ is flat.*

(ii) For any composition $X \xrightarrow{f} Y \xrightarrow{g} Z$, there is a distinguished triangle

$$g^* \mathbb{L}_{Y/Z} \longrightarrow \mathbb{L}_{X/Z} \longrightarrow \mathbb{L}_{X/Y} \longrightarrow g^* \mathbb{L}_{Y/Z}[1].$$

(iii) $\mathcal{H}^0(\mathbb{L}_{X/S}) \cong \Omega_{X/S}^1$.

(iv) If f is smooth, then $\mathbb{L}_{X/S} \cong \Omega_{X/S}^1[0]$.

4. Deformations of ordinary quadratic singularities

Let X_0 be a scheme over k with only ordinary quadratic singularities. An ordinary quadratic singularity is, locally for the étale topology, isomorphic to the singularity at the origin of the cone $\operatorname{Spec} k[x_1, \dots, x_n]/(x_1^2 + \dots + x_n^2)$.

Proposition 0.20. *Let X_0 be as above. Then :*

(i) *The deformation functor of the singularity is smooth (i.e., unobstructed).*

(ii) *The miniversal deformation space is isomorphic to $\operatorname{Spf}(k[[t]])$, where the parameter t corresponds to the deformation of the singularity.*

(iii) *In a miniversal deformation, the singular locus is a divisor with normal crossings.*

Proof. Locally, the singularity is defined by $f = x_1^2 + \dots + x_n^2 = 0$. The tangent space to the deformation functor is $\operatorname{Ext}^1(\Omega_{X_0}^1, \mathcal{O}_{X_0})$ which is one-dimensional, generated by the class of f itself. The obstruction space $\operatorname{Ext}^2(\Omega_{X_0}^1, \mathcal{O}_{X_0})$ vanishes, proving (i) and (ii). For (iii), one notes that the total space of the miniversal deformation is smooth and the singular fiber is a divisor. $\square \quad \square$

Conclusion

5. Semi-homogeneous groupoids (continued)

We assume that $\underline{A}(K) = \{e\}$.

Definition 0.21. A cofibered groupoid $\underline{A}$ over $\underline{C}_{\Lambda, K}$ is called *semi-homogeneous* if the following properties hold:

(I) Every diagram

$$\begin{array}{ccc} a'' & \dashrightarrow & a' \\ \downarrow & & \downarrow \\ a_1 & \longrightarrow & a \end{array}$$

of solid arrows in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed into a commutative diagram including dotted arrows.

(II) For any R , the natural map

$$[\underline{A}(R \times_K D_{K'}(K))] \longrightarrow [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))] \simeq [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

is bijective.

If $\underline{A}$ is semi-homogeneous, so is $[\underline{A}]$.

1.17. Let us first consider the restriction of $\underline{A}$ to the full subcategory of $\underline{C}_{\Lambda, K}$ whose objects are the $D_V(K)$. Let Ω be the vector space $\Omega_{K/\Lambda}^1 \otimes_K K'$, and $d : K \rightarrow \Omega$ the derivative. A map $f : D_V(K) \rightarrow D_W(K)$ can be written:

$$f(k + v\varepsilon) = k + (u(v) + D \cdot dk)$$

with $u \in \text{Hom}_{K'}(V, W)$ and $D \in \text{Hom}_{K'}(\Omega, W)$. If we identify f with (u, D) , the composition law is

$$(u, D) \circ (u', D') = (uu', D + uD') .$$

The condition (II) entails

(II') The functor from K' -vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

Lemma 0.22. *Assume condition (II') is satisfied by $\underline{A}$. Then*

(1) $[\underline{A}(D_{K'}(K))]$ carries a unique vector space structure, such that

$$[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))]$$

as a functor in V .

(2) There is a linear map $\gamma : \text{Hom}(\Omega, K') \rightarrow [\underline{A}(D_{K'}(K))]$ such that, for any $D \in \text{Hom}(\Omega, K')$,

$$[\underline{A}((k, D))] : [\underline{A}(D_{K'}(K))] \longrightarrow [\underline{A}(D_{K'}(K))] \text{ is } x \longmapsto x + \gamma(D) .$$

(3) For any $(u, D) : D_V(K) \longrightarrow D_W(K)$, the map

$$\begin{aligned} (u, D) : [\underline{A}(D_V(K))] &\longrightarrow [\underline{A}(D_W(K))] \text{ i.e.} \\ (u, D) : V \otimes [\underline{A}(D_{K'}(K))] &\longrightarrow W \otimes [\underline{A}(D_{K'}(K))] \text{ is} \\ x &\longmapsto u(x) + \gamma(D) . \end{aligned}$$

Let us introduce the notations

$$\begin{aligned} F(V) &= [\underline{A}(D_V(K))] \\ A &= F(K') \\ F(u, D) : F(V) &\rightarrow F(W) = \text{the map induced by } (u, D) : D_V(K) \rightarrow D_W(K) \\ F(D) : F(V) &\rightarrow F(V) : F(D) = F(1, D) . \end{aligned}$$

By (II') and the fact that K' is a K' -vector space object in the category of K' -vector spaces, $F(K')$ is a K' -vector space in **(Set)**, hence a K' -vector space, and (1) is clear. Let us now identify $F(V)$ and $V \otimes A$.

We deduce from the identity

$$(u, D) = (1, D) \circ (u, 0)$$

that

$$F(u, D) = F(D) \circ (u \otimes 1_A) .$$

The functoriality of F amounts to

$$\begin{aligned} F(0) &= \text{Id} \\ F(D) \circ F(D') &= F(D + D') \\ (u \otimes 1) \circ F(D) &= F(uD) \circ (u \otimes 1_A) . \end{aligned}$$

For any $D : \Omega \rightarrow V$ and any $n \in \mathbb{N}$, if i and p are the maps

$$i : V \hookrightarrow V \oplus K'^n \quad p : V \oplus K'^n \longrightarrow K'^n ,$$

then $F(i)^{-1} \circ F(D \circ p) \circ F(i)$ does not depend on n and is equal to $F(D)$. In fact, let $\chi : K'^n \rightarrow V$ be a map, $\chi_i = \chi(e_i)$ and let a_i be a family of elements of A . We have $(1, \chi \circ p) = (1, \sum \chi_i \circ p_i)$. Hence, if

$$F(D) \left(\sum x_i \otimes a_i \right) = \sum x_i \otimes a_i + \alpha(D) ,$$

then

$$F(D)(y) = y + \alpha(D) \quad \text{with}$$

$$\begin{aligned} \alpha(D + D') &= \alpha(D) + \alpha(D') \\ \alpha(uD) &= F(uD) \quad (\text{for any } u : V \rightarrow W) . \end{aligned}$$

One easily checks that any such α is defined, for some $\gamma \in \text{Hom}(\Omega, A)$, by the formula

$$\alpha(D) = \sum D(1) \cdot \gamma(v_i)$$

as promised in (2) (3).

Definition 0.23. (1) An object ξ in $\underline{A}$ is called *versal formal object* if every surjective map $a : a' \rightarrow a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\underline{A}}(\xi, a') \longrightarrow \mathrm{Hom}_{\underline{A}}(\xi, a) .$$

(2) A formal versal object ξ in $\underline{A}$, with image (R, ξ) in $\mathrm{Ob}(\underline{A})$, is *minimal* if for any two maps $\varphi, \psi : R \rightarrow D_V(K)$, if $\varphi(\xi)$ is isomorphic to $\psi(\xi)$, then there exists a derivation D of K into K' such that $(\psi, D) \circ \xi = \xi$, i.e., $\psi = \varphi - D \circ d$.

Theorem 0.24. *If $\underline{A}$ is semi-homogeneous and if*

$$\dim_K[\underline{A}(D_K)] < \infty ,$$

then $\underline{A}$ admits a minimally versal formal object, and any two of them are isomorphic.

Proof. With the notations of 1.18, let H^* be a subspace of $[\underline{A}(D_{K'}(K))]$, supplementary to the image of γ , and let H be the dual of H^* . We define

$$R_1 = D_H(K) ,$$

let ξ_1 be an object in $\underline{A}(R_1)$, whose image in

$$[\underline{A}(R_1)] = H^* \otimes [\underline{A}(D_{K'}(K))] = \mathrm{Hom}(H^*, [\underline{A}(D_{K'}(K))])$$

is the inclusion of H^* in $[\underline{A}(D_{K'}(K))]$. Then

- (1) The maps $\mathrm{Hom}_{\underline{C}_\Lambda}(D_{K'}(K), D_V(K)) \longrightarrow [\underline{A}(D_V(K))]$ are surjective.
- (2) The minimality condition 1.19(2) is valid for (R_1, ξ_1) .

Choose now some $S \in \underline{C}_\Lambda$, formally smooth over Λ , and an epimorphism

$$\pi : S \rightarrow R_1$$

such that $\mathrm{Ker} \pi \cap \mathfrak{m}_S^2 = 0$. If $\mathfrak{m}$ is the maximal ideal of S , we inductively define, for $n \geq 1$,

$$R_n = S/\mathfrak{m}^n \quad \text{and} \quad \xi_n \in \mathrm{Ob}(\underline{A}(R_n))$$

by the properties that

- (a) $\mathfrak{m}^n$ is the intersection of the ideals $I \supset \mathfrak{m}^n$ of S , such that ξ_n extends to an object ξ in $\underline{A}(S/I)$.
- (b) ξ_n is an extension of ξ_{n-1} .

If $R = \varprojlim R_n$ and $\xi = \varprojlim \xi_n$, then one checks that (R, ξ) is a minimally versal formal object, and the only one up to isomorphism. $\square$

6. Pro-representable cofibered groupoids

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, admitting a minimally versal formal object ξ . In general we cannot recover $\underline{A}$ (up to $\underline{C}_\Lambda$ -equivalence) out of ξ alone.

A number of cofibered groupoids which usually occur in practice enjoy a slightly stronger property than those of semi-homogeneous ones, as it will be defined explicitly in 2.5. To investigate them it is convenient to rephrase what we have done so far in terms of “smooth functors”. All the notations and terminologies in the previous section remain in force throughout in this section.

Definition 0.25. Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We say that φ is *smooth* if every surjective map $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

$$\begin{array}{ccc} \varphi(a') & \longrightarrow & b' \\ \downarrow & & \downarrow \beta \\ \varphi(a) & \longrightarrow & \varphi(a) \end{array} \quad (*)$$

for some $\alpha : a' \rightarrow a$ in $\underline{A}$. We say that φ is *minimally smooth* if it is smooth and $\varphi(k[\varepsilon]) : [\underline{A}(k[\varepsilon])] \rightarrow [\underline{B}(k[\varepsilon])]$ is bijective.

Remark 0.26. (a) Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We note that “ φ is smooth” $\Leftrightarrow$ “every surjective map $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram $(*)$ such that $\varphi(a') \rightarrow b'$ is a $\underline{C}_\Lambda$ -isomorphism” $\Leftrightarrow$ “every small prolongation $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram $(*)$ ”.

(b) For any cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, the natural functor $\underline{A} \rightarrow [\underline{A}]$ is minimally smooth.

(c) For each object R in $\widehat{\underline{C}}_\Lambda$, we set $h_R : \underline{C}_\Lambda \rightarrow (\mathbf{Set})$ to be the functor defined by $h_R(S) = \text{Hom}_{\Lambda\text{-alg}}(R, S)$ for all S in $\underline{C}_\Lambda$. It associates a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$, and a map $R \rightarrow S$ in $\underline{C}_\Lambda$ induces a functor $\underline{h}_S \rightarrow \underline{h}_R$ of cofibered groupoids. The functor $\underline{h}_S \rightarrow \underline{h}_R$ induced by a map $R \rightarrow S$ in $\underline{C}_\Lambda$ is smooth if and only if S is a formal power-series ring over R .

(d) Given an object R in $\underline{C}_\Lambda$, let us denote by $(R, \underline{C}_\Lambda)$ the category of objects in $\underline{C}_\Lambda$ over R . We have a canonical functor $(R, \underline{C}_\Lambda) \rightarrow \underline{C}_\Lambda$. Given $\xi \in \text{Ob}(\underline{A})$ with $P(\xi) = R$, let $\underline{A} \rightarrow (R, \underline{C}_\Lambda)$ be the pull-back of $\underline{A} \rightarrow \underline{C}_\Lambda$ under the canonical functor $(R, \underline{C}_\Lambda) \rightarrow \underline{C}_\Lambda$ and choose a section ξ^* such that $\xi^*(1_R) = \xi$. (This simply means that we choose, for each map $R \rightarrow S$, an f -map $\xi \rightarrow \xi(f)$). We then obtain a functor $\xi^* : \underline{h}_R \rightarrow \underline{A}$ over $\underline{C}_\Lambda$ by sending $(R \rightarrow S) \mapsto \xi(f)$. Of course this functor does not depend on the choice of ξ^* up to natural equivalence of functors. Now “ $\xi^* : \underline{h}_R \rightarrow \underline{A}$ is (minimally) smooth” means that every surjective map $a' \rightarrow \xi(f)$ in $\underline{A}$ can be completed into a commutative diagram

$$\begin{array}{ccc} \xi(f'') & \longrightarrow & a' \\ \downarrow & & \downarrow \\ \xi(f) & \longrightarrow & \xi(f) \end{array}$$

(and $\underline{h}_R(k[\varepsilon]) \rightarrow \underline{A}(k[\varepsilon])$ is bijective), which means precisely that ξ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\Lambda$. Thus we conclude that the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor $\underline{h}_R \rightarrow \underline{A}$ over $\underline{C}_\Lambda$, for some R .

The following general properties on smoothness are trivial to verify.

Proposition 0.27. *Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\Lambda$.*

(a) *If φ and ψ are smooth, so is $\psi \circ \varphi$.*

(b) *If ψ and $\psi \circ \varphi$ are smooth, so is φ .*

(c) *If $\psi \circ \varphi$ is smooth and if φ is surjective on isomorphism classes of objects, then ψ is smooth.*

Remark 0.28. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, admitting a versal formal object ξ . This means, by 2.2(d), that $\xi^* : \underline{h}_R \rightarrow \underline{A}$ is smooth, where $R = P(\xi)$. Therefore we have, “ $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ (c.f. 1.13) i.e., R is a formal power-series algebra over Λ ” $\Leftrightarrow$ “the composite functor $\underline{h}_R \rightarrow \underline{A} \rightarrow \underline{C}_\Lambda$ is smooth” $\Leftrightarrow$ “ $\underline{A} \rightarrow \underline{C}_\Lambda$ is a smooth functor” by 2.3(b).

We now consider a stronger property than being semi-homogeneous.

Definition 0.29. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. It is called *homogeneous* if every diagram

$$\begin{array}{ccc} a' & \longrightarrow & a \\ \downarrow & & \downarrow \\ b' & \longrightarrow & b \end{array}$$

where $a' \rightarrow a$ is surjective admits a fibre-product $a' \times_a b'$ in $\underline{A}$ such that

$$P(a' \times_a b') = P(a') \times_{P(a)} P(b') .$$

Remark 0.30. (a) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$\begin{array}{ccc} a' & \longrightarrow & a \\ \downarrow & & \downarrow \\ a_1 & \longrightarrow & a_2 \end{array}$$

in $\underline{A}$ above the diagram

$$\begin{array}{ccc} R' & \longrightarrow & R \\ \downarrow & & \downarrow \\ R_1 & \longrightarrow & R_2 \end{array}$$

in $\underline{C}_\Lambda$. If $R' \rightarrow R$ is surjective, we obtain a unique map $\theta : a' \rightarrow a_1 \times_{a_2} a_2$ compatible with the projections. If $P(a_1 \times_{a_2} a_2) = R_1 \times_{R_2} R_2$, then $P(\theta)$ is the identity map on $R_1 \times_{R_2} R_2$ and therefore θ is $\underline{C}_\Lambda$ -isomorphism, i.e., $a' = a_1 \times_{a_2} a_2$. In other words, if $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, every diagram (*) defines a fibre-product $a_1 \times_{a_2} a_2$ provided $R_1 \rightarrow R_2$ is surjective in $\underline{C}_\Lambda$. This fact entails immediately that a homogeneous cofibered groupoid is, a fortiori, semi-homogeneous.

(b) Consider a diagram

$$\begin{array}{ccc} & & X \\ & & \downarrow \\ Y & \longrightarrow & Z \end{array}$$

of categories and functors. The ordinary fibre-product $X \times_Z Y$ is a category in which an object is a pair (x, y) such that $\Phi(x) = \Psi(y)$, and a map $(x, y) \rightarrow (x', y')$ consists of a pair $\alpha : x \rightarrow x'$, $\beta : y \rightarrow y'$ such that $\Phi(\alpha) = \Psi(\beta)$. The fibre-product $X \times_Z Y$ in the sense of 2-categories is a category in which an object is a triple $(x, y; u)$, where $u : \Phi(x) \rightarrow \Psi(y)$ is an isomorphism, and a map $(x, y; u) \rightarrow (x', y'; u')$ consists of a pair $x \rightarrow x'$, $y \rightarrow y'$ with a commutative diagram

$$\begin{array}{ccc} \Phi(x) & \xrightarrow{u} & \Psi(y) \\ \downarrow & & \downarrow \\ \Phi(x') & \xrightarrow{u'} & \Psi(y') \end{array}$$

One can rephrase the notions of (semi-)homogeneous cofibered groupoid or smooth functor in terms of fibre-product of categories (in the sense of 2-category). Now let $P : \underline{A} \rightarrow \underline{C}_\Lambda$ be cofibered category, and consider an object $R \in \underline{C}_\Lambda$. If we choose a direct image functor $(\text{pr}_i)_! : \underline{A}(R_1 \times_R R_2) \rightarrow \underline{A}(R_i)$, where $\text{pr}_i : R_1 \times_R R_2 \rightarrow R_i$ is the projection for $i = 1, 2$, we obtain a functor

$$\underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_1) \times_{\underline{A}(R)} \underline{A}(R_2) .$$

It is clear from the definition of cofibered category that this functor, up to canonical natural equivalence, does not depend on the choice of direct image functors. It is now clear from the definition that a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is homogeneous over $\underline{C}_\Lambda$ if and only if

$$\underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_1) \times_{\underline{A}(R)} \underline{A}(R_2)$$

is an equivalence of categories whenever $R_i \rightarrow R$ is surjective in $\underline{C}_\Lambda$, and is semi-homogeneous if and only if

$$[\underline{A}(R_1 \times_R R_2)] \longrightarrow [\underline{A}(R_1)] \times_{[\underline{A}(R)]} [\underline{A}(R_2)]$$

is bijective whenever $R_i \rightarrow R$ is surjective, and is bijective when $R_1 = R_2 = k[\varepsilon]$.

Now let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. For each map $R' \rightarrow R$ in $\underline{C}_\Lambda$, φ induces a functor $\underline{A}(R') \rightarrow \underline{A}(R) \times_{\underline{A}(R)} \underline{B}(R')$ uniquely determined up to a (canonical) natural equivalence. It is again trivial to see that φ is smooth if and only if the functor $\underline{A}(R') \rightarrow \underline{A}(R) \times_{\underline{A}(R)} \underline{B}(R')$ is surjective on isomorphism classes of objects, whenever $R' \rightarrow R$ is surjective.

(c) If $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, the functor $\underline{A}(k[V \oplus W]) \rightarrow \underline{A}(k[V]) \times_{\underline{A}(k[W])} \underline{A}(k[W])$ is an equivalence of categories for any finite-dimensional vector spaces V, W over k (We recall our convention $\underline{A}(k) = \{e\}$). In particular, we have

$$\text{Aut}_{\underline{A}(k[V])}(1_V) \oplus \text{Aut}_{\underline{A}(k[W])}(1_W) \simeq \text{Aut}_{\underline{A}(k[V \oplus W])}(1_V \oplus 1_W)$$

where 1_V is the direct image of e under the unique map $k \rightarrow k[V]$ in $\underline{C}_\Lambda$. It follows that if $\underline{A}$ is homogeneous over $\underline{C}_\Lambda$, $\text{Aut}(1_V)$ as well as $[\underline{A}(k[\varepsilon])]$ are canonically provided with vector space structures over k .

(d) For any object R in $\underline{C}_\Lambda$, the cofibered (discrete) groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. However, the homogeneity of $\underline{A}$ does not imply in general

the homogeneity of $[\underline{A}]$. Indeed we have

Proposition 0.31. *Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$. The following statements are equivalent:*

- (i) $[\underline{A}]$ is homogeneous, i.e., $[\underline{A}(R' \times_R S)] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$ is bijective for every small extension $R' \rightarrow R$.
- (ii) For every small prolongation $a' \rightarrow a$ in $\underline{A}$, the map $\text{Aut}_{\underline{A}(R')}(a') \rightarrow \text{Aut}_{\underline{A}(R)}(a)$ is surjective, where $R' = P(a')$, $R = P(a)$.

Proof. (i) $\Rightarrow$ (ii). Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$ and let $a' \rightarrow a$ be an f -map in $\underline{A}$. Given $\tau \in \text{Aut}_{\underline{A}(R)}(a)$, we consider $a' \times_a a'$ and $a' \times_a a'_\tau$ where $a'_\tau \rightarrow a$ denotes the composite map $a' \rightarrow a \xrightarrow{\tau} a$. Since

$$[\underline{A}(R' \times_R R')] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(R')]$$

is bijective, we conclude the existence of an isomorphism $a' \times_a a' \xrightarrow{u} a' \times_a a'_\tau$, and u induces, via the projection on the second factor, an isomorphism $a' \xrightarrow{\tau'} a'$ which is a lifting of τ .

(ii) $\Rightarrow$ (i). Let a'_i ($i = 1, 2$) be objects in $\underline{A}(R' \times_R S)$ which have the same image under $[\underline{A}(R' \times_R S)] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$. We then obtain a diagram of solid arrows

$$\begin{array}{ccccc}
 & & a''_2 & & \\
 & \nearrow & & \searrow & \\
 & a''_1 & & b_1 & \xrightarrow{\tau} & b_2 \\
 \swarrow & & \searrow & \swarrow & & \searrow \\
 a'_2 & & & a_1 & & R' \times_R S \\
 & \searrow & \swarrow & & & \swarrow \\
 & & a_2 & & & R \\
 & & & & & \swarrow \\
 & & & & & S
 \end{array}$$

where $\bar{\tau} : a_1 \rightarrow a_2$ is the induced isomorphism from τ , so that the square box on the lower right corner is commutative. If the condition (ii) is verified, one can always replace the given isomorphism $a''_1 \xrightarrow{\sim} a''_2$ so that the square box on the left corner is also commutative. Since $a''_i = a'_i \times_{a_i} b_i$ for $i = 1, 2$ by 2.6(a), we obtain the isomorphism $a''_1 \xrightarrow{\sim} a''_2$. This proves that $[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$ is injective, and hence is bijective since $\underline{A}$ is homogeneous by hypothesis. $\square$

Now let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$. We have a canonical map

$$\text{Ker } f \longrightarrow R' \times_R R' \quad \text{given by} \quad (r_1, r_2) \longmapsto r_1 \cdot (0) + (r_2 - r_1),$$

and $D_f : \text{Ker } f \rightarrow R' \times_R R'[\text{Ker } f]$ is an isomorphism. Given any two f -maps $a_i \rightarrow a$ ($i = 1, 2$) in $\underline{A}$ above $R_1 \times_R R_2$, we may consider $\tau(a_1, a_2) = F(a_1 \times_a a_2)$ which is an object in $\underline{A}(k[\text{Ker } f])$.

Since $1 \times f : R_1 \times_R R_1 \rightarrow R' \times_R R'[\text{Ker } f]$ is an isomorphism, we have the isomorphism

$$a_1 \times_a a_2 = a_1 \times \tau(a_1, a_2)$$

compatible with the projections on the first factor.

The statement 1.5(2) takes a more definite form for a homogeneous groupoid.

Lemma 0.32. *Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and $a_i \rightarrow a$ ($i = 1, 2$) f -maps in $\underline{A}$. Then*

- (1) *There exists $p : a_1 \rightarrow a_2$ in $\underline{A}$ with $a_2 \circ p = a_1$ if and only if there exists an arrow $a_1 \rightarrow \tau(a_1, a_2)$ in $\underline{A}$.*
- (2) *There exists $p : a_1 \rightarrow a_2$ with $a_2 \circ p = a_1$, such that $P(p) = \text{Id}$ if and only if $\tau(a_1, a_2) = 0$ in $[\underline{A}(k[\text{Ker } f])]$.*

Proof. We prove both at the same time. There exists $p : a_1 \rightarrow a_2$ in $\underline{A}$ with $a_2 \circ p = a_1$ (such that $P(p) = \text{Id}$, $P(a_i) = R'$) $\Leftrightarrow$ there exists $h : a_1 \rightarrow a_1 \times \tau(a_1, a_2)$ in $\underline{A}$ with $\text{pr}_1 \circ h = \text{Id}$ (such that $P(h) = \text{diagonal map}$) $\Leftrightarrow$ there exists $h : a_1 \rightarrow a_1 \times \tau(a_1, a_2)$ in $\underline{A}$ with $\text{pr}_1 \circ h = \text{Id}$ (such that $P(h) : R' \rightarrow R' \times k[\text{Ker } f]$ is given by $r' \mapsto (r', r'(0))$) $\Leftrightarrow$ there exists an arrow $\gamma : a_1 \rightarrow \tau(a_1, a_2)$ in $\underline{A}$ (such that $P(\gamma) : R' \rightarrow k[\text{Ker } f]$ is given by $r' \mapsto r'(0)$). However, there exists an arrow $\gamma : a_1 \rightarrow \tau(a_1, a_2)$ in $\underline{A}$ such that $P(\gamma) : R' \rightarrow k[\text{Ker } f]$ is factored through $R' \rightarrow k \rightarrow k[\text{Ker } f]$ if and only if $\tau(a_1, a_2) = 0$. $\square$

Proposition 0.33. (a) *Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\Lambda$. Assume that*

- (i) *$\underline{B}$ is semi-homogeneous and ψ is homogeneous;*
- (ii) *$\psi \circ \varphi(k[\varepsilon]) : [\underline{C}(k[\varepsilon])] \rightarrow [\underline{C}(k[\varepsilon])]$ is injective.*

Then the composite functor $\psi \circ \varphi$ is smooth if and only if φ and ψ are both smooth.

(b) *Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is semi-homogeneous, so is $\underline{B}$. If $\underline{A} \rightarrow \underline{B}$ is smooth (i.e., for any surjective map $b' \rightarrow \varphi(a)$ in $\underline{B}$, there exists $a' : a' \rightarrow a$ in $\underline{A}$ and a $\underline{C}_\Lambda$ -isomorphism $p : \varphi(a') \rightarrow b'$ such that $\beta \circ p = \varphi(a)$).*

Proof. (a) Assume that the composite functor $\psi \circ \varphi$ is smooth. If φ is smooth, then so is ψ by 2.3(b), and therefore it suffices to show that φ is smooth. Let a small prolongation $b' \xrightarrow{g} \psi(a)$ in $\underline{B}$ be given. Since the composite functor $\psi \circ \varphi$ is smooth, there exists an arrow $a' \rightarrow a$ in $\underline{A}$ and a $\underline{C}_\Lambda$ -isomorphism $p : \psi\varphi(a') \rightarrow b'$ with a commutative diagram

$$\begin{array}{ccc} \psi\varphi(a') & \longrightarrow & b' \\ \downarrow & & \downarrow \\ \psi\varphi(a) & \longrightarrow & \psi\varphi(a) \end{array}$$

Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram

$$\begin{array}{ccc} \varphi(a') & \longrightarrow & b' \\ \downarrow & & \downarrow \\ \varphi(a) & \longrightarrow & \psi(a) \end{array}$$

above $R' \times_R R' \rightarrow R'$, which proves that φ is smooth. $\square$

Definition 0.34. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called (*minimally*) *pro-representable* if it is $\underline{C}_\Lambda$ -equivalent to $\underline{h}_{(R_0, R_1, \dots)}$ for some (normalized) cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$.

Proposition 0.35. *If $(R_0, R_1, \dots)$ is a normalized cogroupoid in $\widehat{\underline{C}}_\Lambda$, then every homomorphism $(\sigma_0, \sigma_1) : (R_0, R_1, \dots) \rightarrow (R_0, R_1, \dots)$ which induces $\underline{C}_\Lambda$ -equivalence $\underline{h}_{(\sigma_0, \sigma_1)} : \underline{h}_{(R_0, R_1, \dots)} \rightarrow \underline{h}_{(R_0, R_1, \dots)}$ is an isomorphism.*

Proposition 0.36. (a) *Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Assume that*

- (i) $\underline{A}$ *is homogeneous;*
- (ii) $[\underline{A}]$ *is homogeneous;*
- (iii) $\dim_k[\underline{A}(k[\varepsilon])] < \infty$.

Then $\underline{A}$ is minimally pro-representable.

(b) *Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a $\underline{C}_\Lambda$ -equivalence of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is minimally pro-representable, so is $\underline{B}$.*

(c) *If $\underline{A}$ is minimally pro-representable, so is $[\underline{A}]$.*

Proof. (c) Since $\underline{A}$ is minimally pro-representable, there exists a normalized cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$ and a $\underline{C}_\Lambda$ -equivalence $\underline{A} \simeq \underline{h}_{(R_0, R_1, \dots)}$. Then $[\underline{A}] \simeq [\underline{h}_{(R_0, R_1, \dots)}] = h_{R_0}$, which is minimally pro-representable.

(a) Since $\underline{A}$ is semi-homogeneous and $\dim[\underline{A}(k[\varepsilon])] < \infty$, $\underline{A}$ admits a minimally versal formal object. On the other hand, $(R_0, 1_{R_0})$ is a quasi-initial minimal object of $\widehat{\underline{h}}_{(R_0, R_1, \dots)}$ since $\underline{h}_{R_0}(k[\varepsilon]) \rightarrow [\widehat{\underline{h}}_{(R_0, R_1, \dots)}(k[\varepsilon])]$ is bijective, and therefore $(R_0, 1_{R_0})$ is a minimally versal formal object of $\widehat{\underline{h}}_{(R_0, R_1, \dots)}$ by 1.10(ii), which means that $e : \underline{h}_{R_0} \rightarrow \widehat{\underline{h}}_{(R_0, R_1, \dots)}$ is minimally smooth.

(c) Let (S_0, ξ_0) be a minimally versal formal object for $\widehat{\underline{h}}_{(R_0, R_1, \dots)}$ (1.11). For any object $\xi : R_0 \rightarrow T$ in $\widehat{\underline{h}}_{(R_0, R_1, \dots)}(T)$, the functor $\text{Isom}(\xi, \xi)$ from $\underline{C}_\Lambda/T \times_T$ to (sets) which to any R associates the sets of isomorphisms between the two images of ξ on $R : R_0 \xrightarrow{\xi} T \rightrightarrows T \otimes_\Lambda T \rightarrow R$, is represented by $R_1 \otimes_{R_0 \otimes_\Lambda R_0} (T \otimes_\Lambda T)$. By a passage to limit, one gets that $\text{Isom}(\xi_0, \xi_0)$ is pro-represented by some S_1 , and $\widehat{\underline{h}}_{(R_0, R_1, \dots)}$ is equivalent to $\widehat{\underline{h}}_{(S_0, S_1, \dots)}$. By (b), S_* is minimally smooth. $\square$

7. Cohomology

In this section, we fix a complete noetherian local ring Λ with residue field k . Let $\underline{C}_\Lambda$ be the category of artinian local Λ -algebras with residue field k . We also fix a functor G from $\underline{C}_\Lambda$ to the

category of abelian groups, which is a sheaf for the *fc* (faithfully coalgebraically flat) topology. For each $R \in \underline{C}_\Lambda$, we denote by $\underline{C}_R$ the category of R -algebras in $\underline{C}_\Lambda$, provided with the induced topology. We denote by $\underline{C}_R^\sim$ the category of sheaves of abelian groups on $\underline{C}_R$.

Definition 0.37. Let B be an R -algebra in $\underline{C}_\Lambda$. An *augmented simplicial object* over B is a simplicial object P_* in $\underline{C}_R$ together with an augmentation $P_* \rightarrow B$ such that

- (i) Each P_n is a polynomial algebra over R (in a finite number of variables).
- (ii) The underlying sequence of sets $P_n \rightarrow B$ is exact.

Proposition 0.38. *For any $B \in \underline{C}_R$, there exists an augmented simplicial object over B .*

For any augmented simplicial object $P_* \rightarrow B$, we define

$$\check{H}^n(P_*/R, F) = H^n(F(P_*))$$

for any sheaf F on $\underline{C}_R$. If $P_* \rightarrow B$ and $Q_* \rightarrow B$ are two augmented simplicial objects, then there exists a map $P_* \rightarrow Q_*$ over B , unique up to simplicial homotopy. Hence $\check{H}^n(P_*/R, F)$ does not depend on the choice of P_* , up to unique isomorphism. We shall write $\check{H}^n(B/R, F)$ instead of $\check{H}^n(P_*/R, F)$.

Proposition 0.39. *For any short exact sequence $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ in $\underline{C}_R^\sim$, we have a long exact sequence*

$$0 \rightarrow F'(B) \rightarrow F(B) \rightarrow F''(B) \rightarrow \check{H}^1(B/R, F') \rightarrow \check{H}^1(B/R, F) \rightarrow \cdots$$

Proposition 0.40. (a) $\check{H}^n(P/R, F) = 0$ for all $n > 0$ and for all presheaves F on $\underline{C}_R$, if P is a polynomial algebra over R .

(b) If B is an R -algebra of finite type, then we may take P to be of finite type over R so that $P \otimes_R \cdots \otimes_R P$ are all R -algebras of finite type. If G is of finite type, then $G(P \otimes_R \cdots \otimes_R P)$ are all A -modules of finite type, so that $\check{H}^n(B/R, G)$ is an A -module of finite type since A is noetherian.

(c) If $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ is an exact sequence in $\underline{C}_{A|R}^\sim$, then

$$0 \rightarrow F'(B) \rightarrow F(B) \rightarrow F''(B) \rightarrow \check{H}^1(B|R, F') \rightarrow \check{H}^1(B|R, F) \rightarrow \cdots$$

is exact. If $P \in \text{Ob}(\underline{C}_{A|R})$ is a polynomial algebra over R , then $\check{H}^1(P|R, F') = 0$ by 3.3(b) and hence $0 \rightarrow F'(P) \rightarrow F(P) \rightarrow F''(P) \rightarrow 0$ is exact. As for the other direction, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ in which

$$\text{Ob}(\underline{P}_{A|R}) = \{P \in \text{Ob}(\underline{C}_{A|R}) \mid P \text{ is a polynomial algebra over } R\},$$

provided with the induced topology, and let $p : \underline{P}_{A|R} \rightarrow \underline{C}_{A|R}$ be the inclusion functor. Since every object in $\underline{C}_{A|R}$ can be covered by an object in $\underline{P}_{A|R}$, it follows from a comparison lemma that $\tilde{p} : \underline{\tilde{C}}_{A|R} \rightarrow \underline{\tilde{P}}_{A|R}$ is an equivalence of categories. Consequently, if $0 \rightarrow F'(P) \rightarrow F(P) \rightarrow F''(P) \rightarrow 0$

is exact for every $P \in \text{Ob}(\underline{P}_{A|R})$, then $0 \rightarrow \tilde{p}F' \rightarrow \tilde{p}F \rightarrow \tilde{p}F'' \rightarrow 0$ is exact in $\tilde{\underline{P}}_{A|R}$ and hence $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ is exact in $\tilde{\underline{C}}_{A|R}$.

Proposition 0.41. (a) If F is an additive sheaf on $\underline{C}_R$, then

$$\check{H}^n(B/R, F) \simeq \check{H}^n(B/R, \tilde{F})$$

where $\tilde{F}$ is the associated sheaf.

(b) Let $R \rightarrow S$ be a map in $\underline{C}_\Lambda$. For any sheaf F on $\underline{C}_S$, we have a spectral sequence

$$E_2^{p,q} = \check{H}^p(B/R, \check{H}^q(F)) \implies \check{H}^{p+q}(B/S, F) .$$

(c) (communicated from Rinehart) Consider the forgetful functor $\beta : \underline{C}_{A|R} \rightarrow \underline{C}_{A|S}$. We have a spectral sequence

$$E_2^{p,q} = \check{H}^p(A|R, \beta^\# \check{H}_{A|S}^q(F)) \implies \check{H}^{p+q}(A|S, F)$$

where $\beta^\# = \underline{a}_{A|R} \circ \check{\beta}$, $\underline{a}_{A|R} : \check{\underline{C}}_{A|R} \rightarrow \tilde{\underline{C}}_{A|R}$ is “associated-sheaf” functor. Now let F be an additive sheaf. We claim that $\beta^\# \check{H}_{A|S}^q(F)$ is a constant sheaf for all $q > 0$. Indeed, if P is a polynomial algebra over R , then $P = R \otimes_S P'$ for a polynomial algebra P' over S and hence

$$[\beta^\# \check{H}_{A|S}^q(F)](P) = \check{H}^q(P|S, F) = \check{H}^q(R \otimes_S P'|S, F) \xrightarrow{\sim} \check{H}^q(R|S, F)$$

is an isomorphism for all $q > 0$ by 3.7(b), and surjective for $q = 0$. It follows from 3.3(c) that the canonical map $\beta^\# \check{H}_{A|S}^q(F) \rightarrow \check{H}^q(R|S, F) = \text{constant sheaf}$ is an isomorphism for all $q > 0$, and is surjective for $q = 0$. It follows that $E_2^{p,q} = \check{H}^p(A|R, \beta^\# \check{H}_{A|S}^q(F)) = 0$ for all $p, q > 0$, and hence the above spectral sequence yields the exact sequence

$$(*) \quad 0 \rightarrow E^{1,0} \rightarrow H^1 \rightarrow E^{0,1} \rightarrow E^{2,0} \rightarrow H^2 \rightarrow E^{0,2} \rightarrow E^{3,0} \rightarrow H^3 \rightarrow \dots$$

where we note that $E^{0,q} = [\beta^\# \check{H}_{A|S}^q(F)](A) \xrightarrow{\sim} \check{H}^q(R|S, F)$ for all $q > 0$ and $E^{p,0} = \check{H}^p(A|R, \tilde{\beta}F)$. On the other hand, the exact sequence $0 \rightarrow F_R \rightarrow \tilde{\beta}F \rightarrow F(R) \rightarrow 0$ gives us the exact sequence

$$(**) \quad 0 \rightarrow \check{H}^0(A|R, F_R) \rightarrow \check{H}^0(A|R, \tilde{\beta}F) \rightarrow \check{H}^0(A|R, F(R)) \rightarrow \check{H}^1(A|R, F_R) \rightarrow \check{H}^1(A|R, \tilde{\beta}F) \rightarrow 0$$

with $\check{H}^0(A|R, F_R) = \check{H}^0(A|S, F)$ and $\check{H}^0(A|R, F(R)) = \check{H}^0(R|S, F)$. The exact sequences $(*)$ and $(**)$ combined gives the desired exact sequence.

Definition 0.42. For an R -algebra A in $\underline{C}_\Lambda$ and an A -module M , we define

$$D^i(A/R, M) = \check{H}^i(A/R, \text{Der}_R(A, M)) .$$

Proposition 0.43. (a) $D^i(A/R, M)$ is an A -module.

(b) $D^i(A/R, M) = 0$ for all $i > 0$ if A is a polynomial algebra over R .

(c) If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an exact sequence of A -modules, then there is a long exact

sequence

$$\begin{aligned} 0 \rightarrow \operatorname{Der}_R(A, M') \rightarrow \operatorname{Der}_R(A, M) \rightarrow \operatorname{Der}_R(A, M'') \\ \rightarrow D^1(A/R, M') \rightarrow D^1(A/R, M) \rightarrow D^1(A/R, M'') \rightarrow \cdots \end{aligned}$$

(d) $D^i(A/R, M)$ does not depend on the base ring Λ .

Theorem 0.44. *Let A be an R -algebra of finite type and M an A -module of finite type. Then $D^i(A/R, M)$ is an A -module of finite type.*

Proposition 0.45. *Let $R \rightarrow S$ be a map in $\underline{C}_\Lambda$ and A an S -algebra. For any A -module M , we have a spectral sequence*

$$E_2^{p,q} = D^p(A/R, D^q(A/S, M)) \implies D^{p+q}(A/S, M) .$$

Theorem 0.46. (a) $D^1(A/R, M) \simeq \operatorname{Exalcomm}(A/R, M)$, the module of commutative algebra extensions of A by M over R .

(b) Let A be a local R -algebra of essentially finite type. Then $D^2(A/R, M) = 0$ for all A -modules M if and only if A is a relative complete intersection over R .

(c) Let R be noetherian and A a local R -algebra of essentially finite type. We denote by $\hat{A}$ the $\mathfrak{m}$ -adic completion of A where $\mathfrak{m}$ = the maximal ideal of A . Then for any A -module E of finite type we have a canonical isomorphism

$$D^1(\hat{A}/R, \hat{A} \otimes_A E) \simeq \hat{A} \otimes_A D^1(A/R, E) .$$

Proof. (a) follows immediately from $D^1(A/R, M) = \operatorname{Exalcomm}(A/R, M)$.

(b) Let P be a polynomial algebra over R in a finite number of variables and let $0 \rightarrow I \rightarrow P \rightarrow A \rightarrow 0$ be exact. Localizing if necessary, we may assume that I is generated by a P -regular sequence which would imply that $H_1(K_I) = 0$ and therefore $D^2(A/R, M) = \operatorname{Coker}(H^1(K_I, M) \rightarrow \operatorname{Hom}_A(H_1(K_I), M)) = 0$. Conversely assume that $D^2(A/R, M) = 0$ for every A -module M .

Let $0 \rightarrow I \rightarrow P \rightarrow A \rightarrow 0$ be exact where P is a polynomial algebra over R in a finite number of variables. We have

$$0 = D^2(A/R, M) = \operatorname{Coker}(H^1(K_I, M) \rightarrow \operatorname{Hom}_A(H_1(K_I), M))$$

i.e., $H^1(K_I, M) \rightarrow \operatorname{Hom}_A(H_1(K_I), M)$ is surjective for all A -module M . Choose an exact sequence $0 \rightarrow K \rightarrow F \rightarrow I \rightarrow 0$ where F is a free P -module of finite type. We note that $H_1(K_I) = K/K_0$ where $K_0 = \Im(\wedge^2 F \rightarrow \wedge^1 F)$. The fact that $D^2(A/R, H_1(K_I)) = 0$ i.e.,

$$H^1(K_I, H_1(K_I)) \rightarrow \operatorname{Hom}_A(H_1(K_I), H_1(K_I)) \text{ is surjective}$$

entails that $0 \rightarrow K/K_0 \rightarrow F/IF \rightarrow I/I^2 \rightarrow 0$ is exact and splits as A -modules and in particular I/I^2 is projective as A -module. Therefore, localizing A if necessary, we may assume that I/I^2

is A -free, and that $F/IF \rightarrow I/I^2$ is an isomorphism. Then the fact that $0 \rightarrow K/K_0 \rightarrow F/IF \rightarrow I/I^2 \rightarrow 0$ is exact and splits as A -modules entails that $H_1(K_I) = K/K_0 = 0$. This means that I is generated by a P -regular sequence i.e., A is a relative complete intersection over R .

(c) follows immediately from 3.12(4) and (6).

(d) Firstly it follows from 3.12(6) that

$$D^1(A/R, \hat{A} \otimes E) \simeq \hat{A} \otimes_A D^1(A/R, E)$$

is an isomorphism for all i , and therefore it suffices to show that $D^1(\hat{A}/R, \hat{A} \otimes_A E) \simeq \hat{A} \otimes_A D^1(A/R, E)$. $\square$

8. Deformations of schemes

Let k be a field and $\underline{C}_k$ the category of artinian local k -algebras with residue field k . Let X_0 be a scheme of finite type over k . By a *deformation* of X_0 to $R \in \underline{C}_k$, we mean a pair (R, X) where X is a scheme flat over R together with a morphism $X_0 \rightarrow X$ over R such that the induced morphism $X_0 \rightarrow X \times_{\text{Spec}(R)} \text{Spec}(k)$ is an isomorphism. We note that, since R is an artinian local ring, $i_X : X_0 \rightarrow X$ is necessarily a closed immersion which is a homeomorphism on underlying topological spaces. If (R, X) , (R', X') are deformations of X_0 to R , R' respectively, we define an arrow $(R', X') \rightarrow (R, X)$ to consist of a map $R' \xrightarrow{\varphi} R$ in $\underline{C}_k$ and a morphism $\Phi : X \rightarrow X'$ of schemes with a commutative diagram

$$\begin{array}{ccccc} X & & \xrightarrow{\Phi} & & X' \\ \downarrow & \searrow & & \swarrow & \downarrow \\ & X_0 & & & \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec}(R) & \xrightarrow{\text{Spec}(\varphi)} & & & \text{Spec}(R') \end{array}$$

We thus obtain the category $\underline{M}_{X_0}$ of (infinitesimal) deformations of X_0 in which the objects and the arrows are defined as above. The category $\underline{M}_{X_0}$ is provided with a functor $p : \underline{M}_{X_0} \rightarrow \underline{C}_k$ defined by $(R, X) \mapsto R$. Now let $(R, X) \in \text{Ob}(\underline{M}_{X_0})$ and a map $R \xrightarrow{\varphi} S$ in $\underline{C}_k$ be given. We then obtain a commutative diagram

$$\begin{array}{ccc} X \times_{\text{Spec}(R)} \text{Spec}(S) & \xrightarrow{p_1} & X \\ \downarrow & & \downarrow \\ \text{Spec}(S) & \xrightarrow{\text{Spec}(\varphi)} & \text{Spec}(R) \end{array}$$

and the arrow $(\varphi, p_1) : (R, X) \rightarrow (S, X \times_{\text{Spec}(R)} \text{Spec}(S))$ is obviously cocartesian in $\underline{M}_{X_0}$. It follows that the category $\underline{M}_{X_0}$ is a cofibered category over $\underline{C}_k$ via $p : \underline{M}_{X_0} \rightarrow \underline{C}_k$.

We observe the following facts:

Proposition 0.47. (a) $\underline{M}_{X_0}$ is a homogeneous cofibered groupoid over $\underline{C}_k$.

(b) If X_0 is proper over k (resp. affine), then for any deformation (R, X) of X_0 , X is proper (resp. affine) over R .

(c) If X_0 is proper over k , then $\underline{M}_{X_0}$ is semi-homogeneous.

Proposition 0.48. Assume that X_0 is proper over k . Then

$$[\underline{M}_{X_0}(k[\varepsilon])] \simeq H^1(X_0, \mathcal{T}_{X_0})$$

where $\mathcal{T}_{X_0}$ is the tangent sheaf of X_0 over k , and

$$\text{Aut}(k[\varepsilon]) \simeq H^0(X_0, \mathcal{T}_{X_0}) .$$

Theorem 0.49. Let X_0 be a proper scheme over k such that $H^1(X_0, \mathcal{T}_{X_0})$ and $H^2(X_0, \mathcal{T}_{X_0})$ are finite-dimensional k -vector spaces. Then $\underline{M}_{X_0}$ is minimally pro-representable.

Corollary 0.50. If X_0 is a non-singular proper scheme over k , then $\underline{M}_{X_0}$ is minimally pro-representable.

Proposition 0.51. Let X_0 be a proper scheme over k and Z_0 a closed subscheme of X_0 . Let $\underline{M}_{X_0, Z_0}$ be the category of deformations of the pair (X_0, Z_0) . Then $\underline{M}_{X_0, Z_0}$ is a homogeneous cofibered groupoid over $\underline{C}_k$. If X_0 and Z_0 are proper over k , then $\underline{M}_{X_0, Z_0}$ is minimally pro-representable, provided $H^1(X_0, \mathcal{T}_{X_0}(-Z_0))$ and $H^2(X_0, \mathcal{T}_{X_0}(-Z_0))$ are finite-dimensional.

9. Deformations of schemes (continued)

4.1(a): Let $(1_R, \Phi) : (R, X_1) \rightarrow (R, X_2)$ be an arrow in $\underline{M}_{X_0}$ i.e., $\Phi : X_2 \rightarrow X_1$ is a morphism over R such that $X_2 \otimes_R k \rightarrow X_1 \otimes_R k$ is an isomorphism. Since R is artinian and X_i is R -flat, it follows immediately that Φ is an isomorphism so that $(1_R, \Phi) : (R, X_1) \rightarrow (R, X_2)$ is an isomorphism in $\underline{M}_{X_0}$. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b): Suppose that X_0 is affine, say $X_0 = \text{Spec}(A_0)$. Then for any object $(R, X) \in \text{Ob}(\underline{M}_{X_0})$, X is also affine. Consequently, for any $R \in \text{Ob}(\underline{C}_\Lambda)$, the category $\underline{M}_{X_0}(R)$ is canonically equivalent, via the sections, to the category in which an object is a flat algebra A over R together with an R -algebra map $j_A : A \rightarrow A_0$ inducing an isomorphism $A \otimes_R k \xrightarrow{\sim} A_0$, and a map $\Phi : A_1 \rightarrow A_2$ is an R -algebra map with a commutative diagram

$$\begin{array}{ccc} A_1 & \xrightarrow{\Phi} & A_2 \\ j_{A_1} \downarrow & & \downarrow j_{A_2} \\ A_0 & \longrightarrow & A_0 \end{array}$$

4.1(c): Consider the cofibered category $\widehat{\underline{M}}_{X_0}$ in groupoids over $\widehat{\underline{C}}_\Lambda$. By definition, the category $\widehat{\underline{M}}_{X_0}(R)$ for any $R \in \text{Ob}(\widehat{\underline{C}}_\Lambda)$ is canonically equivalent to the category of formal schemes $\mathfrak{X}$, adic over the formal spectrum $\text{Spf}(R)$, such that $\mathfrak{X}_n = \mathfrak{X} \otimes_R R/\underline{m}^{n+1}$ is flat over $R/\underline{m}^{n+1}$ for every $n \geq 0$ (or what amounts to the same, if X_0 is noetherian, such that $\mathfrak{X}$ is flat over $\text{Spf}(R)$). Consequently,

if X_0 is noetherian, $\widehat{\underline{M}}_{X_0}$ is canonically $\widehat{\underline{C}}_\Lambda$ -equivalent to the category of deformations of X_0 over the objects in $\widehat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d): Let $(R, X) \in \text{Ob}(\underline{M}_{X_0})$ i.e., we have a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R) \end{array}$$

such that $X_0 \rightarrow X \times_{\text{Spec}(R)} \text{Spec}(k)$ is an isomorphism. Now let $U_0 \subset X_0$ be a fixed open subscheme of X_0 . Since $i_X : X_0 \rightarrow X$ is a closed immersion which is a homeomorphism on the underlying topological spaces, we obtain a commutative diagram

$$\begin{array}{ccc} U_0 & \xrightarrow{i_X|_{U_0}} & X|_{U_0} \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R) \end{array}$$

such that $U_0 \rightarrow (X|_{U_0}) \times_{\text{Spec}(R)} \text{Spec}(k) = X \otimes_R k|_{U_0}$ is an isomorphism, so that $(R, X|_{U_0})$ is a deformation of U_0 to R . Furthermore, for any arrow $(\varphi, \Phi) : (R, X) \rightarrow (R', X')$ in $\underline{M}_{X_0}$, $(\varphi, \Phi|_{U_0}) : (R, X|_{U_0}) \rightarrow (R', X'|_{U_0})$ is an arrow in $\underline{M}_{U_0}$. Consequently we have the restriction functor $\underline{M}_{X_0} \rightarrow \underline{M}_{U_0}$ over $\underline{C}_\Lambda$.

One may also consider a localization as well as a formalization at a point. Namely let x_0 be a fixed point in X_0 . If $(R, X) \in \text{Ob}(\underline{M}_{X_0})$, then $\text{Spec}(\underline{O}_{X, x_0})$ is a deformation of $\text{Spec}(\underline{O}_{X_0, x_0})$ to R and $\text{Spec}(\widehat{\underline{O}}_{X, x_0})$ is a deformation of $\text{Spec}(\widehat{\underline{O}}_{X_0, x_0})$ over R , where $\widehat{}$ stands for the completion with respect to the linear topology given by the powers of the maximal ideal. We thus obtain functors $\underline{M}_{X_0} \rightarrow \underline{M}_{\text{Spec}(\underline{O}_{X_0, x_0})} \rightarrow \underline{M}_{\text{Spec}(\widehat{\underline{O}}_{X_0, x_0})}$.

Firstly we establish that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. *Consider a diagram*

$$\begin{array}{ccc} & & R' \\ & & \downarrow \\ R & \xrightarrow{\text{surjective}} & \bar{R} \end{array}$$

in $\underline{C}_\Lambda$, and let E, E' be flat modules over R, R' respectively. Then for any R' -homomorphism $E' \rightarrow \bar{E} = E \otimes_R \bar{R}$, the fibre-product $E \times_{\bar{E}} E'$ is flat as $R \times_{\bar{R}} R'$ -module.

Proof: Let $0 \rightarrow I \rightarrow R \rightarrow \bar{R} \rightarrow 0$ be exact so that $0 \rightarrow I \rightarrow R \times_{\bar{R}} R' \rightarrow R' \rightarrow 0$ is exact. Since E is R -flat, $0 \rightarrow I \otimes_R E \rightarrow E \rightarrow \bar{E} \rightarrow 0$ is exact and hence we obtain the exact sequence

$$(*) \quad 0 \rightarrow I \otimes_R E \rightarrow E \times_{\bar{E}} E' \rightarrow E' \rightarrow 0.$$

Set $S = R \times_{\bar{R}} R$. Since I is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S -module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R' -module and $\text{Tor}_1^S(E \times_{\bar{E}} E', R') = 0$.

However, we note that $I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E')$ and $E' = R' \otimes_S (E \times_{\bar{E}} E')$ and therefore the

exact sequence $(*)$ can be rewritten as

$$0 \rightarrow I \otimes_S (E \times_{\bar{E}} E') \rightarrow E \times_{\bar{E}} E' \rightarrow R' \otimes_S (E \times_{\bar{E}} E') \rightarrow 0.$$

This means that $\mathrm{Tor}_1^S(R', E \times_{\bar{E}} E') = 0$ and $R' \otimes_S (E \times_{\bar{E}} E') = E'$ is flat as R' -module, and therefore $E \times_{\bar{E}} E'$ is S -flat.

Corollary 4.3. *Given a diagram*

$$\begin{array}{ccc} (R, X) & & (R', X') \\ & \searrow (\varphi, \Phi) & \swarrow (\varphi', \Phi') \\ & (\bar{R}, \bar{X}) & \end{array}$$

in $\underline{M}_{X_0}$, the amalgamated sum $X \amalg_{\bar{X}} X'$ (in the sense of schemes) is a flat scheme over $R \times_{\bar{R}} R'$, provided $\varphi : R \rightarrow \bar{R}$ is surjective. Consequently, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Proof: For each affine open $U_0 \subset X_0$, we have by definition that $X \amalg_{\bar{X}} X'|_{U_0} \simeq \mathrm{Spec}(\Gamma(U_0, X) \times_{\Gamma(U_0, \bar{X})} \Gamma(U_0, X'))$. Consequently, if $\varphi : R \rightarrow \bar{R}$ is surjective, $X \amalg_{\bar{X}} X'$ is a flat scheme over $R \times_{\bar{R}} R'$ by 4.2, and the morphism $i_X \amalg_{i_{X'}} : X_0 \rightarrow X \amalg_{\bar{X}} X'$ induces an isomorphism $X_0 \rightarrow (X \amalg_{\bar{X}} X') \otimes_S k$, where $S = R \times_{\bar{R}} R'$. Therefore $(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X') \in \mathrm{Ob}(\underline{M}_{X_0})$ and we obviously have that $(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X') = (R, X) \times_{(\bar{R}, \bar{X})} (R', X')$ in the category $\underline{M}_{X_0}$, which means that $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$.

Lemma 4.4. *For any scheme X_0 over k , we have a canonical exact sequence*

$$0 \rightarrow H^1(X_0, \underline{D}_{X_0}^0) \rightarrow [\underline{M}_{X_0}(k[\epsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1)$$

where $\underline{D}_{X_0}^i$ are as defined in paragraph 3.

Proof: (Special case) Assume that X_0 is affine and let $\Gamma(X_0, \underline{O}_{X_0}) = A_0$. Then as it was noted in 4.1(b), the category $\underline{M}_{X_0}(k[\epsilon])$ is canonically equivalent to the category $\underline{E}$, where

$\mathrm{Ob}(\underline{E})$: object in $\underline{E}$ is a flat algebra A over $k[\epsilon]$

together with a $k[\epsilon]$ -algebra map $j_A : A \rightarrow A_0$ inducing the isomorphism $k \otimes_{k[\epsilon]} j_A : k \otimes_{k[\epsilon]} A \xrightarrow{\sim} A_0$.

Alternatively, an object in $\underline{E}$ is a k -algebra A together with an exact sequence $0 \rightarrow A_0 \xrightarrow{\epsilon} A \xrightarrow{j_A} A_0 \rightarrow 0$ in which $\epsilon(A_0)^2 = 0$ as an ideal in A .

An arrow $A_1 \rightarrow A_2$ in $\underline{E}$ is a $k[\epsilon]$ -algebra map $\Phi : A_1 \rightarrow A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$. Therefore, an arrow in the alternative description is a k -algebra map $\Phi : A_1 \rightarrow A_2$ yielding a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_0 & \xrightarrow{\epsilon} & A_1 & \xrightarrow{j_{A_1}} & A_0 & \longrightarrow & 0 \\ & & \parallel & & \downarrow \Phi & & \parallel & & \\ 0 & \longrightarrow & A_0 & \xrightarrow{\epsilon} & A_2 & \xrightarrow{j_{A_2}} & A_0 & \longrightarrow & 0 \end{array}$$

It follows that $[\underline{M}_{X_0}(k[\epsilon])]$ is canonically isomorphic, via $\Gamma(X_0, -)$, to the set of isomorphism

classes of k -algebra extensions of A_0 by A_0 . In other words, we have a canonical bijection $[\underline{M}_{X_0}(k[\epsilon])] \xrightarrow{\sim} H^0(X_0, \underline{D}_{X_0}^1)$ via $\Gamma(X_0, -)$, which is trivially seen to respect the vector space structures on both sides.

(General case) For each affine open $U_0 \subset X_0$, we get a canonical map $[\underline{M}_{X_0}(k[\epsilon])] \rightarrow H^0(U_0, \underline{D}_{X_0}^1)$ defined as the composite map $[\underline{M}_{X_0}(k[\epsilon])] \rightarrow [\underline{M}_{U_0}(k[\epsilon])] \xrightarrow{\sim} H^0(U_0, \underline{D}_{X_0}^1)$ and hence we obtain a canonical map $[\underline{M}_{X_0}(k[\epsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1)$. Its kernel is the set of isomorphism classes of locally-trivial deformations of X_0 to $k[\epsilon]$, i.e., the set of isomorphism classes of deformations of X_0 to $k[\epsilon]$, locally (on X) isomorphic to $X_0 \otimes_k k[\epsilon]$. Since $\underline{D}_{X_0}^0 = \underline{\text{Der}}_k(\underline{O}_{X_0}, \underline{O}_{X_0}) = \underline{\text{Aut}}(X_0 \otimes_k k[\epsilon]|_{X_0}) =$ the sheaf of germs of automorphisms of $X_0 \otimes_k k[\epsilon]$ inducing the identity on the subscheme X_0 , it is canonically isomorphic to $H^1(X_0, \underline{D}_{X_0}^0)$, so that we obtain the desired exact sequence.

Theorem 4.5. *Let X_0 be a scheme of finite type over k for which either*

- (i) X_0 is proper over k or
- (ii) X_0 is affine but $X_0^{\text{sing}} = \{x \in X_0 \mid X_0 \text{ is non-smooth over } k \text{ at the point } x\}$ is a finite set.

Then

(a) $\underline{M}_{X_0}$ admits a formal versal object (called a formal versal deformation of X_0) over $\underline{C}_\Lambda$. If $(R, \mathfrak{X})$ is a formal versal deformation of X_0 , then for any $(S, \mathfrak{Y}) \in \text{Ob}(\widehat{\underline{M}}_{X_0})$, there is a map $(\varphi, \Phi) : (R, \mathfrak{X}) \rightarrow (S, \mathfrak{Y})$ in $\widehat{\underline{M}}_{X_0}$ such that the diagram below is cartesian

$$\begin{array}{ccc} \mathfrak{Y} & \xrightarrow{\Phi} & \mathfrak{X} \\ \downarrow & & \downarrow \\ \text{Spf}(S) & \xrightarrow{\text{Spf}(\varphi)} & \text{Spf}(R) \end{array}$$

Furthermore a formal versal deformation $(R, \mathfrak{X})$ of X_0 prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \rightarrow R$ in $\underline{C}_\Lambda$ and a deformation (R', X') of X_0 to R' , the homomorphism

$$\text{Aut}_{R'}(X'|_{X_0}) \rightarrow \text{Aut}_R(X' \otimes_R R|_{X_0})$$

is surjective.

(b) If X_0 is proper over k , then $\underline{M}_{X_0}$ is smoothly pseudo-representable over $\underline{C}_\Lambda$.

Proof: (a) Since $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$ by 4.3, it suffices to see that $[\underline{M}(k[\epsilon])]$ is a finite-dimensional vector space over k by 1.11 and 2.9(b).

Now consider the exact sequence

$$(*) \quad 0 \rightarrow H^1(X_0, \underline{D}_{X_0}^0) \rightarrow [\underline{M}_{X_0}(k[\epsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1).$$

If X_0 is proper over k , then $H^1(X_0, \underline{D}_{X_0}^0)$ and $H^0(X_0, \underline{D}_{X_0}^1)$ are both finite dimensional since $\underline{D}_{X_0}^i$ are coherent for all $i \geq 0$ by 2.13(a). Now assume that X_0 is affine such that X_0^{sing} is a finite set. Since $\underline{D}_{X_0}^1$ is a coherent sheaf on X_0 such that $\text{supp } \underline{D}_{X_0}^1 \subset X_0^{\text{sing}}$ by 2.13(a), we find that

$H^0(X_0, \underline{D}_{X_0}^1)$ is a finite-dimensional vector space over k , whereas $H^1(X_0, \underline{D}_{X_0}^0) = 0$ since X_0 is affine. Therefore, in either case, it follows from the exact sequence (*) that $[\underline{M}_{X_0}(k[\epsilon])]$ is a finite dimensional vector space over k . The second statement of (a) follows from 2.7.

(b) If X_0 is proper, $\text{Aut}(X_0 \otimes_k k[\epsilon]|_{X_0}) = H^0(X_0, \underline{D}_{X_0}^0)$ is a finite dimensional vector space and hence our statement follows from 2.13.

Remark 4.6(a): Let X_0 be affine. Then a formal versal deformation of X_0 exists if and only if $\underline{D}_{X_0}^1$ has its support on a finite number of points. However this does not imply that non-smooth points of X_0 are all isolated. There are affine schemes X_0 over k having a non-isolated non-smooth point but $\text{supp } \underline{D}_{X_0}^1$ is a finite set.

Remark 4.6(b): Henceforth a deformation of X_0 is meant by an object in $\underline{M}_{X_0}$. In case when there is a possible confusion, an object in $\underline{M}_{X_0}$ will be referred as an infinitesimal deformation of X_0 .

Remark 4.6(c): Let X_0 be a scheme over k . We have fixed a complete local noetherian ring Λ with the residue field k , and considered the deformations of X_0 over $\underline{C}_\Lambda$. However, if we drop the condition that a deformation is flat, everything which has been said in this paragraph remains valid. The condition of flatness was needed only for the existence of a formal versal deformation.

4.7. Let X_0 be a scheme of finite type over k . We define a functor $\underline{M}_{X_0}^{\text{loc}}$ from $\underline{C}_\Lambda$ to (Sets) as follows: For $R \in \text{Ob}(\underline{C}_\Lambda)$, $\underline{M}_{X_0}^{\text{loc}}(R)$ is the set of isomorphism classes of deformations of X_0 to R which are locally isomorphic to $X_0 \otimes_k R$. Similarly we define $\widehat{\underline{M}}_{X_0}^{\text{loc}}$.

Theorem 4.8. *Let X_0 be a scheme of finite type over k such that $H^0(X_0, \underline{D}_{X_0}^0) = 0$. Then*

- (a) $\underline{M}_{X_0}^{\text{loc}}$ is strictly prorepresentable.
- (b) If either X_0 is proper over k or X_0 is affine and X_0^{sing} is a finite set, then $\underline{M}_{X_0}$ is strictly prorepresentable.

Proof: (a) Since $H^0(X_0, \underline{D}_{X_0}^0) = 0$, there are no non-trivial automorphisms for any deformation of X_0 , and hence $\underline{M}_{X_0}^{\text{loc}}$ is equivalent to a functor from $\underline{C}_\Lambda$ to (Sets). By 4.4, $[\underline{M}_{X_0}^{\text{loc}}(k[\epsilon])] = H^1(X_0, \underline{D}_{X_0}^0)$ is a finite-dimensional vector space over k (under the assumption of properness or isolated singularities). The prorepresentability then follows from Schlessinger's criterion.

(b) follows from (a) and the fact that $H^0(X_0, \underline{D}_{X_0}^0) = 0$ implies the condition in 4.5(a) for prorepresentability.

4.9. We now consider the exact sequence from 4.4 more closely. Let

$$0 \rightarrow H^1(X_0, \underline{D}_{X_0}^0) \rightarrow [\underline{M}_{X_0}(k[\epsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1)$$

be the exact sequence. The map $[\underline{M}_{X_0}(k[\epsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1)$ is the obstruction map. Its image consists of those obstruction classes which vanish.

4.10. Let X_0 be an affine scheme of finite type over k .

- (a) If X_0 is locally a complete-intersection, then $\underline{M}_{X_0}$ is smooth over $\underline{C}_\Lambda$.
- (b) Assume that, for a fixed closed point $x \in X_0$, $X_0 - \{x\}$ is smooth over k . Then

$$\underline{M}_{X_0} \rightarrow \underline{M}_{\text{Spec}(\underline{Q}_{X_0,x})} \rightarrow \underline{M}_{\text{Spec}(\widehat{\underline{Q}}_{X_0,x})}$$

are both minimally smooth functors. In particular, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 , then $(R, \mathfrak{X}_{(x)})$ and $(R, \mathfrak{X}_{(\widehat{x})})$ are formal versal deformations of $\text{Spec}(\underline{Q}_{X_0,x})$ and $\text{Spec}(\widehat{\underline{Q}}_{X_0,x})$ respectively.

Proof: In this proof we go back to the notations of the cohomology of commutative algebras defined in the paragraph 3. We set $X_0 = \text{Spec}(A_0)$.

- (a) If X_0 is locally a complete-intersection, then the cotangent complex has certain vanishing properties which guarantee smoothness of the deformation functor. In particular, $D^2(A_0/k, A_0) = 0$, which implies that the deformation functor is unobstructed, hence smooth.
- (b) follows from 4.10 and 4.9(a).

4.14. We close this paragraph with an explicit construction of a formal versal deformation in a simple situation. In fact, if X_0 is affine and is a complete-intersection, it is trivial to write down explicitly. Thus let

$$A_0 = k[x_1, \dots, x_n] / (f_1, \dots, f_r) \quad S/J$$

where $S = k[x_1, \dots, x_n]$, $J = (f_1, \dots, f_r)$, and $f_1, \dots, f_r$ is an S -regular sequence. We then obtain the exact sequence

$$\cdots \rightarrow \text{Der}_k(S, A_0) \rightarrow \text{Hom}_{A_0}(J/J^2, A_0) \rightarrow D^1(A_0/k, A_0) \rightarrow 0$$

We note that J/J^2 is A_0 -free module of rank r generated by $f_1 \pmod{J^2}, \dots, f_r \pmod{J^2}$, since $f_1, \dots, f_r$ is an S -regular sequence. We now assume that $D^1(A_0/k, A_0)$ is a finite-dimensional vector space over k , say A has isolated non-smooth points only. Let $d = \dim_k D^1(A_0/k, A_0)$ and choose $p_j : J/J^2 \rightarrow A_0$ ($j = 1, 2, \dots, d$) which form a k -basis of $D^1(A_0/k, A_0)$. Now choose $P_{jk} \in \Lambda[x_1, \dots, x_n]$ such that $p_j(f_i) = P_{ji} \pmod{\underline{m}_\Lambda + J}$ where $\underline{m}_\Lambda$ is the maximal ideal of Λ . We then consider

$$\begin{aligned} R &= \Lambda[[t_1, \dots, t_d]] \\ B &= R[x_1, \dots, x_n] / (G_1, \dots, G_r) \end{aligned}$$

where $G_j = F_j - \sum_{i=1}^d t_i P_{ji}$ and F_j is a polynomial with coefficients in Λ gotten from f_j by lifting the coefficients from k to Λ . Since $R[x_1, \dots, x_n]$ is R -flat and $G_1, \dots, G_r$ modulo $\underline{m}_R$ (which becomes $f_1, \dots, f_r$) is a regular sequence in $k \otimes_R R[x_1, \dots, x_n]$, it follows that B is R -flat, and $k \otimes_R B = A_0$.

Furthermore,

$$\mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\epsilon]) \rightarrow D^1(A_0/k, A_0) = [\underline{M}_{\mathrm{Spec}(A_0)}(k[\epsilon])]$$

is bijective by our construction of B .

5. Fitting ideals

We now globalize the definition of Fitting ideals. Thus let X be any scheme, and $\underline{E}$ a quasi-coherent $\mathcal{O}_X$ -module of finite presentation. We then obtain a sequence of quasi-coherent ideal sheaves $\underline{I}_X^p(\underline{E})$ for $p \geq 0$ defined by

$$\Gamma(U, \underline{I}_X^p(\underline{E})) \simeq I_{\Gamma(U, \mathcal{O}_X)}^p(\Gamma(U, \underline{E}))$$

for every affine open subset $U \subset X$.

Proposition 5.1. *The Fitting ideals satisfy the following properties:*

- (a) $\underline{I}_X^0(\underline{E}) \subset \underline{I}_X^1(\underline{E}) \subset \dots$
- (b) *If $0 \rightarrow \underline{E}' \rightarrow \underline{E} \rightarrow \underline{E}'' \rightarrow 0$ is an exact sequence of quasi-coherent $\mathcal{O}_X$ -modules of finite presentation, then*

$$\underline{I}_X^p(\underline{E}) \supset \sum_{r+s=p} \underline{I}_X^r(\underline{E}') \cdot \underline{I}_X^s(\underline{E}'').$$

- (c) *If $\underline{E}$ is locally free of rank r , then $\underline{I}_X^p(\underline{E}) = \mathcal{O}_X$ for $p \geq r$ and the zero sheaf for $p < r$.*
- (d) *For any morphism $f : Y \rightarrow X$,*

$$\underline{I}_Y^p(f^* \underline{E}) \supset f^* \underline{I}_X^p(\underline{E}) \cdot \mathcal{O}_Y$$

with equality if f is flat.

Now let $X \rightarrow Y$ be a morphism of schemes. The relative Kahler-differentials $\Omega_{X/Y}^1 (= \underline{\Omega}_{X|Y}^1)$ is a quasi-coherent $\mathcal{O}_X$ -module. If the morphism $X \rightarrow Y$ is of essentially finite presentation, then $\underline{\Omega}_{X|Y}^1$ is a quasi-coherent $\mathcal{O}_X$ -module of finite presentation, and in turn we may consider the ideal sheaves $\underline{I}_X^p(\underline{\Omega}_{X|Y}^1)$.

Definition 5.2. *Let $X \rightarrow Y$ be a morphism of essentially finite presentation. For each integer $p \geq 0$, we define $J_X^p(X|Y)$ to be the closed subscheme of X defined by the ideal sheaf $\underline{I}_X^p(\underline{\Omega}_{X|Y}^1)$.*

The closed subscheme $J_X^p(X|Y)$ is called the p -th Jacobian subscheme of X over Y . Its underlying set consists of points $x \in X$ where $\Omega_{X/Y}^1$ requires more than p generators, i.e., points where the relative dimension (in the sense of minimal number of generators of Ω^1) is $> p$.

Proposition 5.3. *The Jacobian subschemes satisfy the following properties:*

- (i) $J_X^0(X|Y) \subset J_X^1(X|Y) \subset J_X^2(X|Y) \subset \dots$
- (ii) $J_X^p(X|Y) = X$ for $p \geq \dim(\Omega_{X/Y}^1 \otimes k(x))$ at all points x .

(iii) If $X \rightarrow Y$ is smooth, then $J_X^p(X|Y) = \emptyset$ for all $p \geq \dim(X/Y)$.

(iv) Formation of $J_X^p(X|Y)$ commutes with flat base change in Y .

5.4. Let S be a noetherian formal scheme, and $\mathfrak{X}$ an S -adic formal scheme of essentially finite type. We then define

$$I_{\mathfrak{X}}^p(\mathfrak{X}|S) = \varprojlim_n I_{X_n}^p(X_n|S_n)$$

$$J_{\mathfrak{X}}^p(\mathfrak{X}|S) = \varprojlim_n J_{X_n}^p(X_n|S_n)$$

where $X_n = \mathfrak{X} \times_{\mathrm{Spf}(R)} \mathrm{Spec}(R/\underline{m}^{n+1})$ and similarly for S_n .

$\mathfrak{X}_{(x)}$ is an S -adic formal scheme of essentially finite type and we have a commutative diagram

$$\begin{array}{ccc} \mathfrak{X}_{(\hat{x})} & \xrightarrow{i_{(\hat{x})}} & \mathfrak{X} \\ \downarrow & & \downarrow i_{(x)} \\ \mathfrak{X}_{(x)} & & \end{array}$$

We note that if x is an isolated point of $\mathfrak{X}$, then $i_{(x)}$, $i_{(\hat{x})}$ are both isomorphisms to the connected component of $\mathfrak{X}$ containing the point x .

In particular, if $|\mathfrak{X}| = \{x_1, \dots, x_m\}$ is a finite discrete space, then we have a canonical identification $\mathfrak{X} = \coprod_{1 \leq i \leq m} \mathfrak{X}_{(x_i)}$ (disjoint union).

Corollary 5.7. *Let S be a noetherian formal scheme, and $\mathfrak{X}$ an S -adic formal scheme of essentially finite type. We then have*

- (1) $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ commutes with base change i.e., $J_{\mathfrak{X}'}^p(\mathfrak{X}'|S') = J_{\mathfrak{X}}^p(\mathfrak{X}|S) \times_S S'$ where $\mathfrak{X}' = \mathfrak{X} \times_S S'$.
- (2) The connected components of $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ are in bijective correspondence with that of $J_{X_0}^p(X_0|S_0)$. For each point $x \in |J_{X_0}^p(X_0|S_0)|$ we have a canonical isomorphism $J_{\mathfrak{X}}^p(\mathfrak{X}|S)_{(x)} \simeq J_{\mathfrak{X}_{(x)}}^p(\mathfrak{X}_{(x)}|S)$. In particular if $|J_{X_0}^p(X_0|S_0)| = \{x_1, \dots, x_m\}$ is discrete, then

$$J_{\mathfrak{X}}^p(\mathfrak{X}|S) = \coprod_{1 \leq i \leq m} J_{\mathfrak{X}_{(x_i)}}^p(\mathfrak{X}_{(x_i)}|S)$$

(disjoint union).

6. Local complete intersections

We now specialize to the case of local complete intersections. The results of this section will be used in the study of deformations of curves with ordinary double points.

Proposition 6.1. *Let A_0 be a local k -algebra, essentially of finite type, such that*

$$\widehat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$$

where f is a non-degenerate quadratic form. Then $D^1(A_0/k, A_0)$ is a 1-dimensional vector space over k , and a formal versal deformation of $X_0 = \text{Spec}(A_0)$ is given by

$$R = \Lambda[[t]], \quad \hat{A} = R[[x_1, \dots, x_n]]/(F - t)$$

where F is a quadratic polynomial in $\Lambda[x_1, \dots, x_n]$ obtained from f by lifting the coefficients to Λ .

Proof: Since f is a non-degenerate quadratic form, the module of Kähler differentials $\Omega_{A_0/k}^1$ has a single relation given by the partial derivatives of f . The non-degeneracy implies that $D^1(A_0/k, A_0) \simeq k$. The explicit construction follows the method of 4.14 with $d = 1$.

Lemma 6.2. *Let A_0 be a local k -algebra, essentially of finite type such that*

$$\hat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \text{Spec}(A_0)$. Then

- (1) $J_{\mathfrak{X}}(\mathfrak{X}|R) \rightarrow J_R(\mathfrak{X}|R) \rightarrow \text{Spec}(\Lambda)$ are both isomorphisms.
- (2) $J_R(\mathfrak{X}|R)$ is a principal ideal in R generated by a free formal generator of R over Λ i.e. $I_R(\mathfrak{X}|R) = (t)$ and $R = \Lambda[[t]]$.
- (3) $\mathfrak{X} \rightarrow \text{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathfrak{X} = \text{Spf}(\hat{A})$. By 4.10(b), $(R, \text{Spf}(\hat{A}))$ is a formal versal deformation of $\hat{X}_0 = \text{Spec}(\hat{A}_0)$ where $\hat{A}_0 = \varprojlim_v A \otimes_R R_v$.

Now $\hat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$ and $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = (x_1, \dots, x_n)$ in $k[[x_1, \dots, x_n]]$. It follows readily that $[\underline{M}_{X_0}(k[\epsilon])] \simeq k$ and, by 4.14, we get

$$R \simeq \Lambda[[t]] \quad \text{and} \quad \hat{A} \simeq R[[x_1, \dots, x_n]]/(F - t)$$

where F is a quadratic polynomial in $\Lambda[x_1, \dots, x_n]$ ($\subset R[[x_1, \dots, x_n]]$) obtained from f by lifting the coefficients to Λ . Now

$$0 \rightarrow \hat{A} \xrightarrow{df} \hat{A}dx_1 \oplus \dots \oplus \hat{A}dx_n \longrightarrow \hat{\Omega}_{\hat{A}|R}^1 \rightarrow 0$$

so that $\hat{\Omega}_{\hat{A}|R}^1$ is the quotient of the free module by the single relation $df = 0$. Since f is non-degenerate, this relation is independent of the relations modulo t , and we find that the Fitting ideal $I_R(\mathfrak{X}|R) = (t)$.

Proposition 6.3. *With the same notation as in 6.2, we have the exact sequence*

$$0 \rightarrow A_0 \xrightarrow{(\partial f/\partial x_1, \dots, \partial f/\partial x_n)} A_0^n \rightarrow \Omega_{A_0/k}^1 \rightarrow 0$$

and $I_{A_0}^p(\Omega_{A_0/k}^1) = A_0$ for $p \geq n - 1$, while $I_{A_0}^{n-2}(\Omega_{A_0/k}^1) = \underline{m}_{A_0}$ is the maximal ideal.

Lemma 6.4. *Let X_0 be a k -scheme of finite type with isolated non-smooth points only. Denote by $\{z_1, \dots, z_r\}$ the set of non-smooth points in X_0 . Assume that each z_i is an ordinary quadratic singularity (i.e., $\widehat{\mathcal{O}}_{X_0, z_i} \simeq k[[x_1, \dots, x_n]]/(q_i)$ where q_i is a non-degenerate quadratic form). Then*

- (a) *R is formally smooth over Λ if and only if all q_i are non-degenerate.*
- (b) *$I_R(\mathfrak{X}|R) = \prod_{i=1}^r I_R(\mathfrak{X}_{z_i}|R)$ where each factor is a principal ideal.*

Proof: (a) follows from the local computation in 6.2 and the global-to-local exact sequence. (b) follows from 5.7 and the fact that the singularities are isolated.

Theorem 6.5. *Let X_0 be a k -scheme of finite type with isolated non-smooth points only. Let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . Assume that $\text{char } k \neq 2$, $\dim X_0$ is even, and all non-smooth points are ordinary quadratic singularities. Then*

- (1) *R is formally smooth over Λ .*
- (2) *$I_R(\mathfrak{X}|R)$ is a principal ideal and in turn so is $T_R(\mathfrak{X}|R) = \prod_{i=1}^r I_R(\mathfrak{X}_{z_i}|R)$. Furthermore, $I_R(\mathfrak{X}_{z_i}|R) = (t_i)$ and $t_1, \dots, t_r$ can be extended to a system of free formal generators of R over Λ .*
- (3) *$\mathfrak{X} \rightarrow \text{Spec}(\Lambda)$ is formally smooth.*
- (4) *If k is algebraically closed and X_0 is a complete irreducible curve, then $\dim R = 3g - 3 + a$ where $g = \dim_k H^1(X_0, \mathcal{O}_{X_0})$ and $a = \dim_k H^0(X_0, \underline{D}_{X_0}^1)$.*

Proof: The statement follows by combining 6.2, 6.4, and the computation of dimensions via the exact sequence of Lemma 4.4 and Riemann-Roch for curves.

Corollary 6.6. *Under the assumptions of 6.5, the universal deformation of a curve with r ordinary double points is obtained by smoothing each double point independently. The parameter space is smooth of dimension $3g - 3 + r$.*

6.7. We now consider the case of an ordinary quadratic singularity more explicitly. Let A_0 be a local k -algebra with $\widehat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$ where f is a non-degenerate quadratic form. Then the versal deformation is given by $R = \Lambda[[t]]$ and $\widehat{A} = R[[x_1, \dots, x_n]]/(F - t)$ where F lifts f . The singular locus of $\mathfrak{X}$ over R is defined by the vanishing of the $n \times n$ minors of the Jacobian matrix, which reduces to the single equation $t = 0$ since f is non-degenerate.

Corollary 6.8. *Let X_0 be a k -scheme of finite type with isolated non-smooth points only such that $H^0(X_0, \underline{D}_{X_0}^0) = 0$, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . Denote by $\{z_1, \dots, z_r\}$ the set of non-smooth points in X_0 , and assume that $\text{char } k \neq 2$, $\dim X_0$ is even, and all z_i 's are ordinary quadratic singularities. Then*

- (1) *R is formally smooth over Λ .*

- (2) $T_R(\mathfrak{X}|R) = \prod_{i=1}^r I_R(\mathfrak{X}_{z_i}|R)$ is a principal ideal. Furthermore, $I_R(\mathfrak{X}_{z_i}|R) = (t_i)$ and $t_1, \dots, t_r$ can be extended to a system of free formal generators of R over Λ .
- (3) $\mathfrak{X} \rightarrow \operatorname{Spec}(\Lambda)$ is formally smooth.
- (4) If k is algebraically closed and X_0 is a complete irreducible curve, then $\dim R = 3g - 3 + a$ where $g = \dim_k H^1(X_0, \mathcal{O}_{X_0})$ and $a = \dim_k H^0(X_0, \underline{D}_{X_0}^1)$.

SGA 7-I, Expose VII – Biextensions of Sheaves of Groups

A. Grothendieck

0. Introduction

The notion of a *biextension* was introduced by D. Mumford in the context of formal groups, in order to express conveniently the relations between the formal groups associated with an abelian scheme and with the dual abelian scheme. In the present expose we make a systematic study of this notion in the general context of sheaves of groups on a given site. In particular, we show that the notion fits in a remarkably simple way into the classical cohomological formalism of $\underline{\otimes}^L$ and $R\mathrm{Hom}$.

In the following expose we shall spell out the most interesting special cases, especially the case of biextensions of group schemes by the multiplicative group $\mathbb{G}_m$, working with sites such as the Zariski site, the fpqc site, or an intermediate site. One will see, in particular, that if A and B are abelian schemes over a scheme S , the biextensions of (A, B) by $\mathbb{G}_m$ have a remarkable interpretation: essentially, they are the classical divisorial correspondences on A and B . Thus the cohomological meaning of the notion of divisorial correspondence, already implicit in Tate duality for abelian varieties over p -adic fields, is made explicit in a convenient form.

This interpretation of divisorial correspondences is also useful for following what becomes of such a correspondence when A and B , supposed defined over the fraction field of a discrete valuation ring, are allowed to degenerate, using Neron theory. The results of Expose VIII will show, for example, that one obtains a biextension of the pair of identity components of the Neron models, and hence a biextension of the connected special fibres of the Neron models, playing the role of a specialization of the original divisorial correspondence. These results will be used in Expose IX, together with the monodromy theorem, to prove results on Neron models of abelian varieties, the most important being the semistable reduction theorem for the Neron model.

Throughout the sequel, unless explicitly stated otherwise, we work in a fixed U -topos, denoted $\mathcal{T}$. Thus $\mathcal{T}$ is a category equivalent to the category of sheaves on a suitable U -site, which may be taken to be $\mathcal{T}$ itself with its canonical topology. We shall often identify, terminologically, the objects of $\mathcal{T}$ with sheaves of sets on $\mathcal{T}$. The groups, extensions of groups, and so on that occur below are all relative to this topos $\mathcal{T}$.

1. Complements on extensions of groups

1.1. Let P and G be two groups of $\mathcal{T}$, and let

$$1 \longrightarrow G \xrightarrow{i} E \xrightarrow{j} P \longrightarrow 1 \tag{1.1.1}$$

be an extension of P by G . Thus E is a group, $j : E \rightarrow P$ is an epimorphism of groups, and i identifies G with $\mathrm{Ker} j$. We shall reinterpret the datum of such an extension in terms which will

be useful in the definition and study of biextensions.

1.1.1. The datum of an epimorphism $j : E \rightarrow P$ is the datum of a torsor over P , with structural morphism j . The fact that E is endowed with a group structure and that j is a homomorphism of groups is equivalent to saying that the torsor E is a group object in the topos of torsors over P , or equivalently in the topos $\mathcal{T}_P$ of objects of $\mathcal{T}$ endowed with an action of P . The datum of a homomorphism $i : G \rightarrow E$ making (1.1.1) exact is then the datum of a rigidification of this torsor by G , namely an isomorphism of G with the subgroup $\text{Ker } j$ of E . Thus an extension of P by G may be regarded as a torsor over P rigidified by G .

1.1.2. The homomorphism i lets G act on E in two ways, on the left and on the right, by

$$g \cdot x \cdot g' = i(g) x i(g')$$

for sections $g, g' \in G(S)$ and $x \in E(S)$. If E is viewed, via j , as an object over P , then this is the same as saying that the group induced by G on $\mathcal{T}_P$, namely $G_P = G \times P$, acts on E on the left and on the right. Each of the two actions makes E a torsor over P , with group G_P , and more precisely E is a bitorsor under (G_P, G_P) . This means that E is a right torsor for the right action of G_P , and the left action identifies G_P with the sheaf of automorphisms of this right torsor; equivalently, the left action makes E a left torsor and the right action identifies the opposite group G_P^0 with the automorphism group of the left torsor.

When $\mathcal{T}$ is the punctual topos, so that P and G are ordinary groups, a bitorsor under (G_P, G_P) is the same as, for each $p \in P$, a bitorsor E_p under (G, G) , namely the fibre of E over p . The same interpretation holds in the general topos if one says it after every base change $S \rightarrow e$: for every point $p \in P(S)$ one obtains a bitorsor E_p under (G, G) , these bitorsors being compatible with changes of base.

1.2. Let e_P be the unit section of P . The fibre E_{e_P} of E over e_P is canonically isomorphic to G , and this is an isomorphism of bitorsors under (G, G) when G is given its usual bitorsor structure by left and right translations. A bitorsor E under (G_P, G_P) is said to be *rigidified*, relative to e_P , when one is given an isomorphism of bitorsors

$$G \xrightarrow{\sim} E_{e_P},$$

or equivalently a section of this torsor.

The terminology is motivated by the case in which G is commutative and

$$\text{Hom}_{\text{pt}}(P, G) \simeq (e), \tag{1.3.1.1}$$

where the left hand side denotes morphisms carrying e_P to the unit section e_G of G . Equivalently, the natural homomorphism

$$\Gamma(G) \xrightarrow{\sim} \text{Hom}(P, G) \tag{1.3.1.2}$$

which assigns to a section g of G the constant morphism $P \rightarrow G$ with value g , is an isomorphism. These equivalent relations say that every automorphism of a torsor respecting the given rigidification is the identity. Under condition (1.3.1.1), the category of torsors under G_P rigidified relative to e_P is therefore rigid.

1.3.2. Let Q be an object of $\mathcal{T}$ endowed with a section e_Q , and let E be a torsor under $G_{P \times Q}$. Since e_Q defines a section of $P \times Q$ over P , one can consider rigidifications of E relative to this section in the induced topos $\mathcal{T}_P$; explicitly, these are sections of the torsor induced by E on $P \times e_Q$. Interchanging the roles of P and Q , one obtains partial rigidifications relative to $e_P \times Q$. Two such partial rigidifications are said to be compatible if their restrictions to $e_P \times e_Q$ coincide, that is, if they define the same section of the torsor which is the inverse image of E by the section (e_P, e_Q) of $P \times Q$.

A pair of compatible partial rigidifications is also called a *birigidification* of the torsor E relative to the pair of sections (e_P, e_Q) . If G is commutative, E admits a birigidification if and only if it admits the two partial rigidifications. Moreover one may prescribe either one of them in advance: one corrects the other by a morphism $Q \rightarrow G$ obtained by composing $Q \rightarrow e$ with the appropriate section of G .

Still in the commutative case, if E and E' are two birigidified torsors under $G_{P \times Q}$, then to say that they are isomorphic as plain $G_{P \times Q}$ -torsors is equivalent to saying that they are isomorphic as birigidified torsors. Indeed, if $u : E \rightarrow E'$ is a torsor isomorphism, the discrepancy between the two birigidifications is measured by morphisms $f : P \rightarrow G$ and $g : Q \rightarrow G$ whose values at the distinguished sections have the same product $a \in G(e)$. The morphism

$$h : P \times Q \longrightarrow G, \quad (p, q) \longmapsto f(p)ag(q)a^{-1}$$

then corrects u to an isomorphism respecting the birigidifications.

1.3.3. The terminology just used extends immediately to the case where one has a bitorsor E under (G_P, G_Q) . A rigidification of such a bitorsor is an isomorphism of the pull-back of E by the marked section with the trivial bitorsor. Equivalently, it is given by a central section, i.e. by a section commuting with the relevant action. The terminology of birigidifications is extended in the same way to bitorsors.

1.3.4. Return to the case of two groups $\mathcal{P}$ and G , and choose the unit section $e_{\mathcal{P}}$ in order to define rigidifications of bitorsors over $\mathcal{P}$, and birigidifications of bitorsors over $\mathcal{P} \times \mathcal{P}$. There are natural functors

$$\mathbf{EXT}(\mathcal{P}; G) \xrightarrow{i} \mathbf{BIEXTORS}_{\mathcal{P}}(\mathcal{P}; G) \xrightarrow{\delta} \mathbf{BIEXTORS}_{\mathcal{P} \times \mathcal{P}}(\mathcal{P}; G). \quad (1.3.4.1)$$

Here the three terms denote respectively the category of extensions of groups of $\mathcal{P}$ by G , the category of bitorsors under $(G_{\mathcal{P}}, G_{\mathcal{P}})$ rigidified at $e_{\mathcal{P}}$, and the category of bitorsors under $(G_{\mathcal{P} \times \mathcal{P}}, G_{\mathcal{P} \times \mathcal{P}})$

birigidified at $(e_{\mathcal{P}}, e_{\mathcal{P}})$.

The functor i sends an extension E of $\mathcal{P}$ by G to its corresponding bitorsor over $\mathcal{P}$, rigidified by the unit section of E . The functor δ is defined by

$$\delta(E) = \pi^*(E) \cdot \text{pr}_1^*(E)^{-1} \cdot \text{pr}_2^*(E)^{-1}, \quad (1.3.4.2)$$

where $\pi : \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$ is the group law and pr_1, pr_2 are the two projections. The bitorsor $\delta(E)$ has a canonical birigidification. For instance, its restriction to $\mathcal{P} \times e_{\mathcal{P}}$ is canonically the trivial bitorsor, using the given rigidification of E ; the other partial rigidification is obtained symmetrically, and the two are plainly compatible.

The composite δi is canonically isomorphic to the trivial functor, i.e. to the constant functor with value the trivial birigidified bitorsor over $\mathcal{P} \times \mathcal{P}$. Indeed, for every extension E of $\mathcal{P}$ by G , the usual multiplication isomorphism defining the extension gives a trivialization of $\delta i(E)$ compatible with its canonical birigidification.

Proposition 0.52. *Let $\mathcal{P}$ be a group of $\mathcal{T}$.*

(a) *Suppose that every morphism of sheaves of sets from $\mathcal{P} \times \mathcal{P}$ to G which is trivial on $\mathcal{P} \times e_{\mathcal{P}}$ and on $e_{\mathcal{P}} \times \mathcal{P}$ is trivial. Then the functor i in (1.3.4.1) is fully faithful. In particular, on a bitorsor E under $(G_{\mathcal{P}}, G_{\mathcal{P}})$, rigidified relative to $e_{\mathcal{P}}$, there is at most one structure of extension of $\mathcal{P}$ by G compatible with its bitorsor structure and having $e_{\mathcal{P}}$ as unit section.*

(b) *Suppose moreover that every morphism of sheaves of sets from $\mathcal{P} \times \mathcal{P} \times \mathcal{P}$ to G which is trivial on the three partial products*

$$\mathcal{P} \times \mathcal{P} \times e_{\mathcal{P}}, \quad \mathcal{P} \times e_{\mathcal{P}} \times \mathcal{P}, \quad e_{\mathcal{P}} \times \mathcal{P} \times \mathcal{P}$$

is trivial. Let E be a torsor under $G_{\mathcal{P}}$. There exists on E a structure of extension of $\mathcal{P}$ by G , compatible with its bitorsor structure, if and only if the underlying bitorsor of

$$\delta(E) = \pi^*(E) \cdot \text{pr}_1^*(E)^{-1} \cdot \text{pr}_2^*(E)^{-1}$$

over $\mathcal{P} \times \mathcal{P}$ is trivial. If this condition holds, then for every central section of E over $e_{\mathcal{P}}$, i.e. for every rigidification of E relative to $e_{\mathcal{P}}$, there is a unique extension structure on E compatible with the bitorsor structure and with the prescribed unit section. Equivalently, E , as a rigidified bitorsor, lies in the essential image of i . In addition, the birigidified bitorsor $\delta(E)$ has a unique trivialization.

Proof. For (a), let E and E' be two extensions of $\mathcal{P}$ by G , and let $f : E \rightarrow E'$ be a morphism of the underlying rigidified bitorsors. We must prove that f is a morphism of extensions, equivalently a group homomorphism. It is enough to prove $f(xy) = f(x)f(y)$. If $j' : E' \rightarrow \mathcal{P}$ is the structural morphism, then $j'f(xy) = j'(f(x)f(y))$, because both sides have the same image in $\mathcal{P}$. Hence

$$f(xy) = u(jx, jy) f(x)f(y)$$

for a well-defined morphism $u : \mathcal{P} \times \mathcal{P} \rightarrow G$. Taking successively $x = 1$ and $y = 1$, and using $f(1) = 1$, gives

$$u(p, 1) = u(1, p) = 1.$$

Thus u is trivial on $\mathcal{P} \times e_{\mathcal{P}}$ and on $e_{\mathcal{P}} \times \mathcal{P}$, and by hypothesis u is trivial. Therefore $f(xy) = f(x)f(y)$. For (b), the necessity is immediate from the preceding construction of δ . Conversely, choose a trivialization Φ of $\delta(E)$. Pulling it back by the unit section of $\mathcal{P} \times \mathcal{P}$ gives a rigidification of E relative to $e_{\mathcal{P}}$; fix this rigidification. The trivialization Φ need not be compatible with the induced birigidification of $\delta(E)$. Its discrepancies along $\mathcal{P} \times e_{\mathcal{P}}$ and $e_{\mathcal{P}} \times \mathcal{P}$ are given by central morphisms

$$f : \mathcal{P} \rightarrow G, \quad g : \mathcal{P} \rightarrow G$$

with the same value $a \in G(e)$ at the unit. The central morphism

$$h : \mathcal{P} \times \mathcal{P} \rightarrow G, \quad (p, q) \mapsto f(p)g(q)a^{-1}$$

has these two restrictions. Replacing Φ by $\Phi(p, q)h(p, q)^{-1}$, we get a trivialization of $\delta(E)$ compatible with its birigidification. Such a trivialization is unique, since any two differ by a central morphism $\mathcal{P} \times \mathcal{P} \rightarrow G$ trivial on the two partial products, hence trivial.

It remains to check associativity of the multiplication defined by Φ , knowing already that $e_{\mathcal{P}}$ is a two-sided unit. The two composites in the associativity condition differ by a central morphism

$$u : \mathcal{P} \times \mathcal{P} \times \mathcal{P} \rightarrow G$$

through the formula

$$(xy)z = u(jx, jy, jz)x(yz).$$

Putting successively $y = z = 1$, then $z = x = 1$, then $x = y = 1$, and using the fact that 1 is a two-sided unit, one sees that u is trivial on the three partial products. By hypothesis it is trivial, and hence $(xy)z = x(yz)$. $\square$

Corollary 0.53. *Suppose that every morphism of sheaves of sets*

$$\mathcal{P} \times G \longrightarrow G$$

which is trivial on $\mathcal{P} \times e_{\mathcal{P}}$ and on $e_{\mathcal{P}} \times G$ is trivial. Then, if G is commutative, G is central in every extension of groups of $\mathcal{P}$ by G . If $\mathcal{P}$ is also commutative and the hypothesis of part (a) of the proposition holds, then every extension E of $\mathcal{P}$ by G is commutative.

Indeed, to prove the first assertion one must show that the action of $\mathcal{P}$ on G defined by the extension is trivial. This action is represented by a morphism $u : \mathcal{P} \times G \rightarrow G$ whose restrictions to $\mathcal{P} \times e_{\mathcal{P}}$ and $e_{\mathcal{P}} \times G$ are trivial in the required sense; the hypothesis gives the assertion. For the second assertion, note that the opposite group E^0 of an extension E of $\mathcal{P}$ by G is an extension of $\mathcal{P}^0$ by G^0 , and the underlying bitorsor of $i(E^0)$ is obtained from $i(E)$ by exchanging left and right operations. If E is central, then $i(E) = i(E^0)$; if in addition $\mathcal{P}$ is commutative, uniqueness in the

proposition gives $E = E^0$, i.e. E is commutative.

1.3.7. A particularly useful case in which all the preceding hypotheses are satisfied is the case where

$$\underline{\mathrm{Hom}}_{\bullet}(\mathcal{P}, G) = e \quad (1.3.7.1)$$

where the left hand side denotes the sheaf of morphisms of pointed sheaves of sets from $\mathcal{P}$ to G . Independently of the marked sections, this condition may also be written in the form

$$G = \underline{\mathrm{Hom}}(\mathcal{P}, G), \quad (1.3.7.2)$$

where the right hand side is the sheaf of morphisms of underlying sheaves of sets. It also says that, for every object $\mathcal{Q}$ of $\mathcal{T}$, the map

$$u \longmapsto u \circ \mathrm{pr}_1 : \underline{\mathrm{Hom}}_{\bullet}(\mathcal{P}, G) \longrightarrow \underline{\mathrm{Hom}}_{\bullet}(\mathcal{P} \times \mathcal{Q}, G) \quad (1.3.7.3)$$

is bijective.

This property of $\mathcal{P}$, for fixed G , is stable under cartesian products. It therefore implies the same property for $\mathcal{P} \times \mathcal{P}$, for $\mathcal{P} \times \mathcal{P} \times \mathcal{P}$, and so on. Consequently it implies the hypotheses of the proposition and of the corollary. Moreover the categories occurring in (1.3.4.1) are all rigid, meaning that every automorphism of an object in one of these categories is the identity. Since condition (1.3.7.1) is stable under every base change, descent and uniqueness show that a rigidified bitorsor E over $\mathcal{P}$ corresponds to an extension of $\mathcal{P}$ by G as soon as this is true locally; equivalently, by the proposition, this is expressed by the vanishing of the section

$$\delta(E) \in \mathbf{H}^0(G_{\mathcal{P}})$$

defined by $\delta(E)$.

1.3.8. Let $p : \mathcal{P} \rightarrow S$ be a morphism of schemes, quasi-compact and quasi-separated and flat, such that

$$p_* \mathcal{O}_{\mathcal{P}} \xleftarrow{\sim} \mathcal{O}_S. \quad (1.3.8.1)$$

Using the fact that, by coherence of p , the formation of $p_* \mathcal{O}_{\mathcal{P}}$ commutes with flat base change, one sees immediately that these conditions on the relative scheme $\mathcal{P}/S$ are stable under cartesian products, in particular under the powers $\mathcal{P} \times \mathcal{P}$, $\mathcal{P} \times \mathcal{P} \times \mathcal{P}$, and so forth. They are also stable under every flat base change.

If G is an affine scheme over S , condition (1.3.7.2) follows at once from (1.3.8.1), since, writing $G = \mathrm{Spec}(A)$,

$$\underline{\mathrm{Hom}}_S(\mathcal{P}, G) = \mathrm{Hom}_S(A, p_* \mathcal{O}_{\mathcal{P}}) = \mathrm{Hom}_S(A, \mathcal{O}_S).$$

The same conclusion applies to the cartesian powers of $\mathcal{P}$, and, if G is flat over S , to the relative scheme $\mathcal{P}_G$ over G with respect to G_G .

When $\mathcal{P}$ and G are group schemes over S , and one works with a topology on $(\text{Sch})_S$ at least as fine as the canonical topology, so that $\mathcal{P}$ and G may be identified with sheaves on the site, every morphism from $\mathcal{P}$, $\mathcal{P} \times \mathcal{P}$, $\mathcal{P} \times \mathcal{P} \times \mathcal{P}$, and so on to G which is compatible with the marked sections is trivial. If G is flat, the same is true for every S -morphism $\mathcal{P} \times G \rightarrow G$ compatible with the marked sections, by interpreting it as a G -morphism $\mathcal{P}_G \rightarrow G_G$. Thus the hypotheses above apply, and the categories in (1.3.4.1) are rigid.

The most important case in which (1.3.8.1) holds, and remains true after every extension of the base, is that in which $\mathcal{P}$ is an abelian scheme over S . In this case the preceding conclusions hold for every affine group G over S . This is the case initially considered by J.-P. Serre, from which the preceding developments are visibly inspired. A case in which (1.3.8.1) holds without being universally true is when $\mathcal{P}$ is the Neron model of an abelian scheme over the fraction field of a discrete valuation ring with spectrum S .

1.4. Extensions of a free commutative group

Let I be an object of $\mathcal{T}$, and put

$$\mathcal{P} = \mathbb{Z}^{(I)}, \quad (1.4.1)$$

the free $\mathbb{Z}$ -module generated by I . To an extension E of $\mathcal{P}$ by a given $\mathbb{Z}$ -module G – in what follows only commutative extensions of groups will be considered – associate the corresponding torsor under $G_{\mathcal{P}}$, and then restrict it to I by the canonical morphism

$$I \longrightarrow \mathcal{P} = \mathbb{Z}^{(I)}.$$

This gives a canonical functor

$$\mathbf{EXT}(\mathcal{P}, G) \longrightarrow \mathbf{TORS}(I, G_I). \quad (1.4.2)$$

This functor is an equivalence of categories. Equivalently, for every G the maps

$$\text{Hom}_{\mathbb{Z}}(\mathcal{P}, G) \longrightarrow H^0(I, G_I) = G(I) \quad (1.4.3)$$

and

$$\text{Ext}_{\mathbb{Z}}^1(\mathcal{P}, G) \longrightarrow H^1(I, G_I) \quad (1.4.4)$$

are isomorphisms.

More generally, if A is a not necessarily commutative ring of $\mathcal{T}$ and $\mathcal{P} = A^{(I)}$ is the free A -module generated by I , then there are canonical isomorphisms, functorial in the variable A -module G ,

$$\text{Ext}_A^i(A^{(I)}, G) \xrightarrow{\sim} H^i(I, G), \quad i \geq 0, \quad (1.4.5)$$

which for $A = \mathbb{Z}$ and $i = 0, 1$ coincide with (1.4.3) and (1.4.4). To define (1.4.5), recall that the

left hand side is the derived functor of

$$G \longmapsto \operatorname{Hom}_A(A^{(I)}, G),$$

which by the definition of $A^{(I)}$ is the functor $G \mapsto G(I) = H^0(I, G)$. Since $G \mapsto G_I$ sends injective modules to injective modules in $\mathcal{T}_{/I}$, the desired conclusion follows. The announced compatibility with (1.4.3) and (1.4.4) is left to the reader.

The functor (1.4.2) is compatible with inverse images by a morphism of topoi

$$f : \mathcal{T}' \longrightarrow \mathcal{T}.$$

In rough terms, this is the commutativity of the diagram

$$\begin{array}{ccc} \mathbf{EXT}(\mathcal{P}, G) & \longrightarrow & \mathbf{TORS}(I, G_I) \\ \downarrow f^* & & \downarrow f^* \\ \mathbf{EXT}(f^*\mathcal{P}, f^*G) & \longrightarrow & \mathbf{TORS}(f^*I, (f^*G)_{f^*I}) \end{array}$$

up to the canonical transitivity isomorphisms

$$f^*(G_I) \xrightarrow{\sim} (f^*G)_{f^*I}, \quad f^*(\mathbb{Z}^{(I)}) \xrightarrow{\sim} \mathbb{Z}^{(f^*I)}.$$

We leave to the reader the precise formulation of these compatibilities.

2. The notion of a biextension of abelian sheaves

2.0. Throughout the present section, P , Q , and G denote commutative groups of the topoi $\mathcal{T}$. We shall often use the cartesian diagram

$$\begin{array}{ccc} P \times Q & \xrightarrow{\operatorname{pr}_1} & P \\ \downarrow \operatorname{pr}_2 & & \downarrow \\ Q & \longrightarrow & e. \end{array} \tag{2.0.1}$$

We denote by G_P , G_Q , and $G_{P \times Q}$ the inverse images of the group G on e on P , on Q , and on $P \times Q$ respectively. We identify $G_{P \times Q}$ indifferently with the inverse image $(G_P)_{P \times Q}$ of G_P by pr_1 and with the inverse image $(G_Q)_{P \times Q}$ of G_Q by pr_2 . In these notations the group structures of P , Q , and $P \times Q$ do not intervene. In fact, we shall practically never need to use the product group structure on $P \times Q$. On the other hand, it will be useful to consider on $P \times Q$ the structure of a P -group obtained by inverse image from the group Q on e ,

$$P \times Q = Q_P \longrightarrow P, \tag{2.0.2}$$

and similarly the structure of a Q -group obtained by inverse image from the group P on e ,

$$P \times Q = P_Q \longrightarrow Q. \tag{2.0.3}$$

Thus on P we shall have to consider the two groups G_P and Q_P , and on Q the two groups G_Q and P_Q .

Let E be a torsor on $P \times Q$ under the group $G_{P \times Q}$. It may be regarded indifferently as a torsor on Q_P under the group $G_{P \times Q} = (G_Q)_P$, or as a torsor on P_Q under the group $G_{P \times Q} = (G_P)_Q$. It therefore makes sense to consider on E a structure of commutative extension of Q_P by G_P compatible with its torsor structure, or, symmetrically, a structure of extension of P_Q by G_Q compatible with its torsor structure. Moreover, applying the preceding discussion on extensions to the induced topoi $\mathcal{T}_P$ and $\mathcal{T}_Q$ gives a convenient way to make such structures explicit. Using the canonical isomorphisms

$$\mathcal{T}_{P \times Q} \simeq (\mathcal{T}_P)_{/P \times Q}, \quad \mathcal{T}_{P \times Q} \simeq (\mathcal{T}_Q)_{/P \times Q},$$

one obtains the following descriptions.

A structure of extension of Q_P by G_P on the torsor E is equivalent to giving, for every object S of $\mathcal{T}$ and elements $p, p' \in P(S)$, $q \in Q(S)$, an isomorphism of torsors under G ,

$$\rho_{p,p';q} : E_{p,q} \cdot E_{p',q} \xrightarrow{\sim} E_{pp',q}, \quad (2.0.4)$$

satisfying the conditions recalled below. As usual, for a point $(p, q) \in (P \times Q)(S) = P(S) \times Q(S)$, the notation $E_{p,q}$ denotes the torsor under G obtained by inverse image of the torsor E under $G_{P \times Q}$ by the morphism $(p, q) : S \rightarrow P \times Q$. The data (2.0.4) are required to be functorial in S , that is, compatible with inverse images by morphisms $S' \rightarrow S$ in $\mathcal{T}$, and to satisfy associativity, expressed by the commutativity of diagrams of the form

$$\begin{array}{ccc} E_{p,q} \cdot E_{p',q} \cdot E_{p'',q} & \longrightarrow & E_{pp',q} \cdot E_{p'',q} \\ \downarrow & & \downarrow \\ E_{p,q} \cdot E_{p'p'',q} & \longrightarrow & E_{pp'p'',q}, \end{array} \quad (2.0.5)$$

where $p, p', p'' \in P(S)$ and $q \in Q(S)$. They must also satisfy commutativity, expressed by the commutativity of the diagrams

$$\begin{array}{ccc} E_{p,q} \cdot E_{p',q} & \xrightarrow{\rho} & E_{pp',q} \\ \downarrow \text{sym} & & \uparrow \\ E_{p',q} \cdot E_{p,q} & \longrightarrow & E_{p'p,q}, \end{array} \quad (2.0.6)$$

where $p, p' \in P(S)$ and $q \in Q(S)$.

One similarly makes explicit a structure of extension of P_Q by G_Q on E by means of isomorphisms of torsors under G

$$\gamma_{p;q,q'} : E_{p,q} \cdot E_{p,q'} \xrightarrow{\sim} E_{p,qq'} \quad (2.0.7)$$

for $p \in P(S)$ and $q, q' \in Q(S)$. The γ are functorial in S and satisfy the associativity and commutativity conditions symmetric to (2.0.5) and (2.0.6), expressed by the commutativity of the

diagrams

$$\begin{array}{ccc}
E_{p,q} \cdot E_{p,q'} \cdot E_{p,q''} & \longrightarrow & E_{p,qq'} \cdot E_{p,q''} \\
\downarrow & & \downarrow \\
E_{p,q} \cdot E_{p,q'q''} & \longrightarrow & E_{p,qq'q''},
\end{array} \tag{2.0.8}$$

where $p \in P(S)$ and $q, q', q'' \in Q(S)$, and

$$\begin{array}{ccc}
E_{p,q} \cdot E_{p,q'} & \xrightarrow{\gamma} & E_{p,qq'} \\
\downarrow \text{sym} & & \uparrow \\
E_{p,q'} \cdot E_{p,q} & \longrightarrow & E_{p,q'q}.
\end{array} \tag{2.0.9}$$

2.1. Definition of the notion of biextension

Definition 0.54. Let P , Q , and G be abelian groups of $\mathcal{T}$, and let E be a torsor under $G_{P \times Q}$. A *biextension of (P, Q) by G* is such a torsor E , endowed with a structure of extension of Q_P by G_P defined by isomorphisms (2.0.4), functorial in S and subject to the conditions expressing the commutativity of (2.0.5) and (2.0.6), and with a structure of extension of P_Q by G_Q defined by isomorphisms (2.0.7), functorial in S and subject to the conditions expressed by the commutativity of (2.0.8) and (2.0.9). These two structures are said to be *compatible* if, for every object S of $\mathcal{T}$ and for $p, p' \in P(S)$, $q, q' \in Q(S)$, the following diagram is commutative:

$$\begin{array}{ccc}
E_{p,q} \cdot E_{p,q'} \cdot E_{p',q} \cdot E_{p',q'} & \longrightarrow & E_{p,qq'} \cdot E_{p',qq'} \\
\downarrow & & \downarrow \rho \\
E_{pp',q} \cdot E_{pp',q'} & \xrightarrow{\gamma} & E_{pp',qq'}.
\end{array} \tag{2.1.1}$$

Here one of the vertical arrows is deduced from ρ and γ by symmetry, and the other uses the symmetry isomorphism

$$E_{p,q} \cdot E_{p',q'} \xrightarrow{\sim} E_{p',q'} \cdot E_{p,q}.$$

2.1.1. Equivalently, a biextension structure on the $G_{P \times Q}$ -torsor E is defined by two partially defined laws of composition $\oplus(x, y)$ and $\otimes(x, y)$ on E . The first is defined when $\text{pr}_1 j(x) = \text{pr}_1 j(y)$, and the second when $\text{pr}_2 j(x) = \text{pr}_2 j(y)$, where $j : E \rightarrow P \times Q$ is the structural morphism of the torsor. In addition to the usual compatibilities with the action of G , such as

$$g \oplus (x, y) = \oplus(x, gy) = \oplus(gx, y) \quad (x, y \in E(S), g \in G(S)), \tag{2.1.1.1}$$

and the analogous relations for $\otimes$, the axioms are the associativity and commutativity of these partial laws of composition, together with the relation

$$\otimes(\oplus(x, y), \oplus(z, t)) = \oplus(\otimes(x, z), \otimes(y, t)) \tag{2.1.1.2}$$

for x, y, z, t respectively in $E_{p,q}(S)$, $E_{p,q'}(S)$, $E_{p',q}(S)$, and $E_{p',q'}(S)$.

There is also reason to consider the unit section of E when E is regarded as an extension of Q_P

by G_P ,

$$e_{P/E} \in \Gamma(E/P) = \text{Hom}_P(P, E), \quad (2.2.1)$$

and, symmetrically, the unit section of E when it is regarded as an extension of P_Q by G_Q ,

$$e_{Q/E} \in \Gamma(E/Q) = \text{Hom}_Q(Q, E). \quad (2.2.2)$$

In particular, the torsor E over $P \times Q$, with group $G_{P \times Q}$, is canonically trivialized above $1_P \times Q$ by $e_{P/E}$, and above $P \times 1_Q$ by $e_{Q/E}$. These two trivializations moreover coincide above the unit section $1_P \times 1_Q$ of $P \times Q$, and therefore define a single section

$$e \in \Gamma(E/e) = \text{Hom}_e(e, E), \quad (2.2.3)$$

which, by abuse of language, will be called the unit section, or neutral section, of E .

To prove the asserted equality of the two sections e, e' of E obtained respectively from $e_{P/E}$ and $e_{Q/E}$, note that they are characterized respectively as the unit sections in the inverse image $E|_e$ of the unit section of $P \times Q$ in E , for the two laws of composition $\oplus$ and $\otimes$. Applying (2.1.1.2) to the sections e, e', e', e of $E|_e$, one finds that the two sides are respectively equal to e and e' , whence $e = e'$. It also follows that the two composition laws $\oplus$ and $\otimes$ on $E|_e$ are identical. Indeed, it is immediate that they correspond to the group law of G transported by means of the morphisms $g \mapsto g \cdot e$ and $g \mapsto g \cdot e'$ respectively; hence they agree since $e = e'$.

Finally, the extension E of Q_P by G_P over the base P , when restricted above the section 1_P of P , gives a structure of extension of Q by G on the inverse image of $1_P \times Q$ in E . The homomorphism $e_{P/E}$ is immediately seen to be multiplicative for this extension structure; hence this extension is canonically split. Symmetrically, the extension of P by G obtained by restricting the extension over Q above the section 1_Q is canonically split by $e_{Q/E}$.

SGA 7-I, Exposé VII.2.3 through Exposé IX.6.6

Exposé VII. Biextensions of sheaves of groups, continued

2.3. Morphisms of biextensions

Let E be a biextension of (P, Q) by G , and likewise let E' be a biextension of (P', Q') by G' , where P', Q' , and G' are again commutative groups of the topos $\mathcal{T}$. A *morphism of biextensions* from E to E' is a system

$$(u, v, w, f)$$

where

$$u : P \longrightarrow P', \quad v : Q \longrightarrow Q', \quad w : G \longrightarrow G'$$

are homomorphisms of groups, and where

$$f : E \longrightarrow E'$$

is a morphism of the underlying sheaves of sets, such that f is at the same time a homomorphism of extensions over P , respectively over P' , associated with the corresponding homomorphisms of relative groups, and a homomorphism of extensions over Q , respectively over Q' , associated with the corresponding homomorphisms of relative groups. In other words the two group-extension structures carried by a biextension must be respected simultaneously.

The maps u, v, w are already determined by f . Passing to the quotient recovers $u \times v$, hence u and v by restriction to $P \times 1_Q$ and $1_P \times Q$. Restricting f to the copy of G through the neutral section recovers w . Thus no ambiguity results from identifying the morphism (u, v, w, f) with the single map f .

With the usual precaution of recording source and target, biextensions in $\mathcal{T}$ are therefore the objects of a category. We denote it by **BIEXT**($\mathcal{T}$). The assignment

$$(E; P, Q, G) \longmapsto (P, Q, G)$$

defines a canonical functor

$$\mathbf{BIEXT}(\mathcal{T}) \longrightarrow \mathbf{GRAB}(\mathcal{T}) \times \mathbf{GRAB}(\mathcal{T}) \times \mathbf{GRAB}(\mathcal{T}),$$

where **GRAB**($\mathcal{T}$) denotes the category of abelian groups in $\mathcal{T}$. It is often useful to decompose this functor into the two functors

$$E \longmapsto (P, Q), \quad E \longmapsto G.$$

2.4. The category $\mathbf{BIEXT}(P, Q; G)$ with P, Q, G variable

The first of the two preceding functors is fibred, and the second is cofibrant. Concretely, if E is a biextension of (P, Q) by G , and if

$$u : P_1 \rightarrow P, \quad v : Q_1 \rightarrow Q$$

are homomorphisms of abelian groups, then there is a biextension E' of (P_1, Q_1) by G and a morphism of biextensions

$$E' \longrightarrow E$$

compatible with u, v, id_G , having the usual universal property. This biextension is the inverse image of E by (u, v) .

Dually, if $w : G \rightarrow G'$ is a homomorphism of abelian groups, there is a biextension E' of (P, Q) by G' and a morphism of biextensions

$$E \longrightarrow E'$$

compatible with $\text{id}_P, \text{id}_Q, w$, again characterized by its universal property. It is the push-out of E by w . These constructions are obtained by paraphrasing the corresponding constructions for ordinary extensions of groups.

For fixed P, Q, G we write

$$\mathbf{BIEXT}(P, Q; G)$$

for the fibre category over the triple (P, Q, G) . The preceding description gives the usual functorialities:

$$(u, v)^* : \mathbf{BIEXT}(P, Q; G) \rightarrow \mathbf{BIEXT}(P_1, Q_1; G),$$

and

$$w_* : \mathbf{BIEXT}(P, Q; G) \rightarrow \mathbf{BIEXT}(P, Q; G').$$

These are compatible in the evident sense.

2.5. The structure of $\mathbf{BIEXT}(P, Q; G)$

For fixed P, Q, G , the category $\mathbf{BIEXT}(P, Q; G)$ is a strictly commutative Picard category. The sum of two biextensions is defined by taking the contracted product over $P \times Q$:

$$E + E' = (E \times_{P \times Q} E')/G,$$

where G is embedded diagonally with the usual opposite action on one factor. The two extension laws on E and E' induce the two required extension laws on $E + E'$, and the compatibility axiom for a biextension is inherited from the corresponding axioms for E and E' .

The neutral object is the trivial biextension

$$P \times Q \times G$$

with its evident partial addition laws. The inverse of a biextension is obtained by replacing the G -torsor E by its opposite torsor. Associativity and commutativity are inherited from the corresponding facts for contracted products of torsors. Thus the isomorphism classes form an abelian group, denoted

$$\mathrm{Biext}^1(P, Q; G).$$

The automorphism group of the neutral biextension is also useful. An automorphism of the trivial biextension over $P \times Q$ is given by a morphism

$$P \times Q \longrightarrow G.$$

The condition that it respect both extension structures says precisely that this morphism is additive separately in P and in Q . Thus

$$\mathrm{Aut}^{\mathbf{BIEXT}(P, Q; G)}(0) \simeq \mathrm{Bilin}(P, Q; G),$$

the group of biadditive morphisms $P \times Q \rightarrow G$. We denote this group by

$$\mathrm{Biext}^0(P, Q; G).$$

The two groups Biext^0 and Biext^1 behave functorially in all three arguments. Pulling back in P and Q , and pushing out in G , gives the corresponding homomorphisms of Picard categories and hence of their homotopy groups.

2.6. Opposite and transposed biextensions

Let E be a biextension of (P, Q) by G . One may form the opposite biextension, obtained by replacing the G -torsor by its inverse. It represents the negative of the class of E in $\mathrm{Biext}^1(P, Q; G)$.

One may also transpose the two variables. The transposed biextension

tE

is a biextension of (Q, P) by G , obtained by pulling E back along the symmetry $Q \times P \simeq P \times Q$ and interchanging the two extension structures. This operation is involutive and is compatible with inverse images and direct images.

2.7. Symmetric biextension associated with a given biextension

The sum of E with its transpose, when the two variables are identified, gives the natural symmetric construction. More generally, whenever a symmetry condition is imposed on P and Q , a biextension may be equipped with an isomorphism to its transpose satisfying the usual cocycle condition. This gives the category of symmetric biextensions. In the applications to pairings and to duality, this is the construction which records the symmetry or alternating behaviour of the pairing arising from

a biextension.

2.8. Change of topos. Localization

Let $u : \mathcal{T}' \rightarrow \mathcal{T}$ be a morphism of topoi. If E is a biextension of (P, Q) by G in $\mathcal{T}$, then u^*E is a biextension of (u^*P, u^*Q) by u^*G in $\mathcal{T}'$. This construction is functorial and respects sums, inverses, and transposition.

If X is an object of $\mathcal{T}$, the localization $\mathcal{T}/X$ gives a special case. The restriction of E to X is a biextension of the restrictions of P and Q by the restriction of G . Conversely, assertions about biextensions which are local for the topology of $\mathcal{T}$ may be checked after such localizations. This is used repeatedly below when questions of existence or uniqueness are reduced to the case of a torsor admitting a section.

The formation of $\text{Biext}^i(P, Q; G)$ is therefore compatible with change of topos in the expected way:

$$u^* \text{Biext}^i(P, Q; G) \longrightarrow \text{Biext}^i(u^*P, u^*Q; u^*G).$$

For $i = 0$ this is the usual pullback of biadditive morphisms; for $i = 1$ it is obtained by pulling back representatives.

2.9. Relation with bitrivialized torsors

A biextension may be described equivalently as a torsor over $P \times Q$ under G , together with two compatible partial group laws. If one chooses trivializations along the two axes $P \times 1_Q$ and $1_P \times Q$, agreeing at the neutral section, the remaining datum is often encoded by a torsor equipped with two rigidifications. This is the bitrivialized-torsor description.

This point of view is especially useful for explicit computations. Given a G -torsor E over $P \times Q$, a rigidification along $P \times 1_Q$ and one along $1_P \times Q$ allow one to speak of the defect of additivity in each variable. The compatibility condition for a biextension says exactly that these defects are themselves compatible; in a Čech description it is the usual two-variable cocycle condition.

The equivalence between the two descriptions is natural in P, Q, G , compatible with change of topos, and preserves the Picard structure.

2.10. Variants of the notion of biextension

The definition admits evident variants. One may replace abelian groups by group objects subject to commutativity restrictions sufficient for the two partial extension structures to commute. One may also introduce rigidified, symmetrized, or alternating biextensions, according to whether additional trivializations or symmetry constraints are imposed.

There are also relative variants, in which P, Q, G are objects over a base object of the topos. They are obtained by applying the preceding definitions in the localized topos. All these variants are

stable under pullback and push-out and will be used without further comment when the context requires them.

3. Complements of homological algebra

3.1. The additive cofibration $\mathcal{F}_{L_\bullet}$

Let $\mathcal{C}$ be an additive category in which the extensions under consideration are defined, and let

$$L_\bullet : \cdots \longrightarrow L_2 \longrightarrow L_1 \longrightarrow L_0$$

be a chain complex of objects of $\mathcal{C}$. To $L_\bullet$ one associates a cofibred additive category

$$\mathcal{F}_{L_\bullet}$$

over $\mathcal{C}$. The fibre over an object A is the category of pairs consisting of an extension

$$0 \longrightarrow A \longrightarrow X \longrightarrow L_0 \longrightarrow 0$$

together with a lifting of the differential $L_1 \rightarrow L_0$ to X , subject to the condition imposed by the next differential $L_2 \rightarrow L_1$.

More explicitly, an object of the fibre over A is a pair (X, α) , where X is an extension of L_0 by A and

$$\alpha : L_1 \longrightarrow X$$

satisfies

$$j\alpha = d_1, \quad \alpha d_2 = 0$$

with $j : X \rightarrow L_0$ the quotient morphism. Morphisms are morphisms of extensions compatible with α . The resulting category is additive and cofibred in A , by push-out of extensions.

There is a canonical interpretation of its connected components and automorphism groups in terms of Ext groups:

$$\pi_0 \mathcal{F}_{L_\bullet}(A) \simeq \text{Ext}^1(L_\bullet, A), \quad \pi_1 \mathcal{F}_{L_\bullet}(A) \simeq \text{Ext}^0(L_\bullet, A),$$

where the Ext groups are computed from the complex $L_\bullet$.

3.2. Determination of $\mathcal{F}_{L_\bullet}^1(A)$

The group $\mathcal{F}_{L_\bullet}^1(A)$ is the group of isomorphism classes of objects of the fibre over A . The preceding description shows that it is exactly the group of extensions of the complex $L_\bullet$ by A , modulo the usual equivalence relation. In formulas,

$$\mathcal{F}_{L_\bullet}^1(A) \simeq \text{Ext}^1(L_\bullet, A).$$

Under this correspondence, the Baer sum of extensions agrees with the addition in Ext^1 . Likewise, automorphisms of the zero object correspond to

$$\text{Hom}(H_0(L_\bullet), A)$$

or, more generally, to $\text{Ext}^0(L_\bullet, A)$ in the derived sense.

The construction is compatible with morphisms of complexes. A morphism $L'_\bullet \rightarrow L_\bullet$ gives an inverse-image functor

$$\mathcal{F}_{L_\bullet} \longrightarrow \mathcal{F}_{L'_\bullet},$$

and the induced maps on $\mathcal{F}^i$ are the usual maps on Ext groups.

3.3. Exactness properties of $\mathcal{F}_{L_\bullet}$

Suppose a morphism of complexes

$$L'_\bullet \longrightarrow L_\bullet$$

induces isomorphisms on H_0 and H_1 . Then for every A the induced functor

$$\mathcal{F}_{L_\bullet}(A) \longrightarrow \mathcal{F}_{L'_\bullet}(A)$$

is an equivalence of categories. This is the categorical form of the fact that extensions of degree 0 and 1 depend only on the truncation of the complex through H_1 .

For a short exact sequence of complexes

$$0 \longrightarrow L'_\bullet \longrightarrow L_\bullet \longrightarrow L''_\bullet \longrightarrow 0,$$

one obtains the exact sequence

$$0 \rightarrow \text{Ext}^0(L''_\bullet, A) \rightarrow \text{Ext}^0(L_\bullet, A) \rightarrow \text{Ext}^0(L'_\bullet, A) \rightarrow \text{Ext}^1(L''_\bullet, A) \rightarrow \text{Ext}^1(L_\bullet, A) \rightarrow \cdots$$

The connecting homomorphism admits a direct categorical description: an element of $\text{Ext}^0(L'_\bullet, A)$, viewed as an automorphism of the neutral object in $\mathcal{F}_{L'_\bullet}(A)$, measures the obstruction to descending the corresponding trivialized object along the exact sequence of complexes. This obstruction is the image in $\text{Ext}^1(L''_\bullet, A)$.

The sign conventions are those of the derived category. Since only the resulting Picard categories and exactness assertions will be used later, the possible ambiguities in signs do not affect the applications that follow.

3.4. Variance of $\mathcal{F}_{L_\bullet}$ with respect to $L_\bullet$

Let

$$0 \rightarrow L'_\bullet \rightarrow L_\bullet \rightarrow L''_\bullet \rightarrow 0$$

be an exact sequence of complexes. The functor

$$\mathcal{F}_{L_\bullet} \longrightarrow \mathcal{F}_{[L'_\bullet \rightarrow L''_\bullet]}$$

which sends an extension of $L_\bullet$ to its restriction to $L'_\bullet$, equipped with the induced trivialization after pullback to $L''_\bullet$, is an equivalence of categories. Here $\mathcal{F}_{[L'_\bullet \rightarrow L''_\bullet]}$ denotes the category of objects of $\mathcal{F}_{L'_\bullet}$ together with a trivialization of their inverse image.

Indeed, an object in the target consists of an extension

$$0 \rightarrow A \rightarrow X' \rightarrow L'_0 \rightarrow 0$$

together with the auxiliary maps encoding the liftings from L'_1 and L''_1 , satisfying the evident relations. These data combine uniquely into an object of $\mathcal{F}_{L_\bullet}$, and the construction is inverse to restriction. This proves the asserted equivalence and gives the left-exactness in the complex-variable which complements the left-exactness in A .

The preceding equivalence identifies the connecting homomorphism in the long exact Ext sequence with the geometric operation of taking the object whose trivialization fails to descend. This is the compatibility needed below.

3.5. A canonical partial flat resolution of an arbitrary $\mathbb{Z}$ -module

We apply the preceding results to the category of commutative groups of the topos $\mathcal{T}$, that is, to the category of $\mathbb{Z}$ -modules of $\mathcal{T}$. Let P be such a group. Define a chain complex $L_\bullet(P)$, augmented toward P , by

$$0 \longrightarrow \mathbb{Z}[P \times P \times P] \xrightarrow{d_2} \mathbb{Z}[P \times P] \xrightarrow{d_1} \mathbb{Z}[P] \xrightarrow{\epsilon} P \longrightarrow 0.$$

For a section p of P , write $[p]$ for the corresponding section of the free module $\mathbb{Z}[P]$, and use analogous notation $[p, q]$ and $[p, q, r]$. Then

$$\epsilon[p] = p,$$

$$d_1[p, q] = [p + q] - [p] - [q],$$

and

$$d_2[p, q, r] = [p + q, r] - [p, q + r] + [p, r] - [q, r].$$

These formulas define a complex:

$$\epsilon d_1 = 0, \quad d_1 d_2 = 0.$$

The first identity is immediate. The second decomposes into the identities expressing commutativity and associativity of the group law on P .

The construction is functorial in P . Each component $L_i(P)$ is a free $\mathbb{Z}$ -module, hence flat.

Proposition 0.55. *One has $H_1(L_\bullet(P)) = 0$, and the augmentation induces an isomorphism*

$$H_0(L_\bullet(P)) \xrightarrow{\sim} P.$$

It is enough to prove that for every commutative group G of $\mathcal{T}$, the functor

$$\mathcal{F}_{L_\bullet(P)}(G) \longrightarrow \mathcal{F}_P(G)$$

is an equivalence. The left hand side identifies with the category of extensions of the $\mathbb{Z}$ -module P by G . Interpreting extensions of the free modules $\mathbb{Z}[P]$, $\mathbb{Z}[P \times P]$, and $\mathbb{Z}[P \times P \times P]$ as torsors over P , $P \times P$, and $P \times P \times P$, respectively, an object of the right hand side is a G_P -torsor on P together with a trivialization of its pullback to $P \times P$, satisfying the compatibility imposed by $d_1 d_2 = 0$. This is exactly the classical description of an extension of the group object P by G . The two constructions are inverse to one another.

3.6. Application to the cofibred category of biextensions of a fixed pair

Let P, Q, G be commutative groups in $\mathcal{T}$. Apply the preceding construction to both P and Q , and set

$$L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q).$$

The complex $L_\bullet(P, Q)$ is a flat resolution, in the relevant low degrees, of $P \otimes^{\mathbf{L}} Q$. The general formalism yields a Picard category

$$\mathcal{F}_{L_\bullet(P, Q)}(G).$$

The category $\mathbf{BIEXT}(P, Q; G)$ is canonically equivalent to this Picard category. Under this equivalence, the class of a biextension corresponds to the class of the associated extension datum for the complex $L_\bullet(P, Q)$. Therefore

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

and

$$\mathrm{Biext}^0(P, Q; G) \simeq \mathrm{Ext}^0(P \otimes^{\mathbf{L}} Q, G).$$

The latter is precisely the group of biadditive maps $P \times Q \rightarrow G$.

This description is functorial in G , by push-out, and in P, Q , by pullback. It is compatible with the Picard structure and with change of topos.

3.7. Various compatibilities

The equivalence just described is compatible with the usual exactness properties in each variable. For an exact sequence

$$0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0$$

and fixed Q, G , the sequence of Picard categories of biextensions is left-exact. On isomorphism classes and automorphism groups it gives the beginning of the long exact Ext sequence associated with

$$0 \rightarrow P' \otimes^{\mathbf{L}} Q \rightarrow P \otimes^{\mathbf{L}} Q \rightarrow P'' \otimes^{\mathbf{L}} Q \rightarrow 0.$$

Equivalently, a biextension of (P, Q) whose inverse image to (P', Q) is trivial has an obstruction class in the group of biextensions of (P'', Q) .

Similarly, exactness in Q is obtained by symmetry, and exactness in G is the exactness of push-outs of extensions. The connecting morphisms are the boundary maps in the corresponding Ext sequences.

Thus the geometrically defined group $\mathrm{Biext}^1(P, Q; G)$ is expressed as the simple homological invariant $\mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G)$, and all the natural operations on biextensions are the operations induced on Ext.

3.8. Biextensions of (P, Q) trivialized on extensions of P and Q

Consider two exact sequences of abelian groups

$$0 \rightarrow L_1 \xrightarrow{\partial} L_0 \xrightarrow{u} P \rightarrow 0, \quad 0 \rightarrow M_1 \xrightarrow{\partial} M_0 \xrightarrow{v} Q \rightarrow 0.$$

We describe the category of biextensions E of (P, Q) by an abelian group G whose pullback $(u, v)^*E$ is trivial. A trivialization is a section

$$s : L_0 \times M_0 \longrightarrow (u, v)^*E$$

additive in each variable.

Once s is chosen, the two partial restrictions of the trivial biextension give homomorphisms

$$f : L_1 \otimes M_0 \rightarrow G, \quad g : L_0 \otimes M_1 \rightarrow G.$$

They must agree on $L_1 \otimes M_1$, in the sense that

$$f(x_1 \otimes \partial y_1) = g(\partial x_1 \otimes y_1).$$

Conversely, such a pair (f, g) reconstructs the required descent datum for the trivialized biextension.

Changing the trivialization by a bilinear form

$$h : L_0 \otimes M_0 \rightarrow G$$

changes f and g by the restrictions of h :

$$f'(x_1 \otimes y_0) = f(x_1 \otimes y_0) + h(\partial x_1 \otimes y_0),$$

$$g'(x_0 \otimes y_1) = g(x_0 \otimes y_1) + h(x_0 \otimes \partial y_1).$$

Proposition 0.56. *The subgroup of $\text{Biext}^1(P, Q; G)$ consisting of classes whose pullback to (L_0, M_0) is trivial is canonically isomorphic to the quotient of the group of pairs (f, g) , with*

$$f : L_1 \otimes M_0 \rightarrow G, \quad g : L_0 \otimes M_1 \rightarrow G,$$

satisfying the compatibility condition on $L_1 \otimes M_1$, by the subgroup of pairs obtained by restricting a form $h : L_0 \otimes M_0 \rightarrow G$.

Interpreting the two exact sequences as complexes

$$L_\bullet : 0 \rightarrow L_1 \rightarrow L_0 \rightarrow 0, \quad M_\bullet : 0 \rightarrow M_1 \rightarrow M_0 \rightarrow 0,$$

their tensor product is

$$0 \rightarrow L_1 \otimes M_1 \rightarrow L_1 \otimes M_0 \oplus L_0 \otimes M_1 \rightarrow L_0 \otimes M_0 \rightarrow 0.$$

The preceding quotient is precisely

$$H^1(\text{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)).$$

The canonical morphism

$$L_\bullet \otimes M_\bullet \longrightarrow P \otimes^{\mathbf{L}} Q$$

therefore induces

$$H^1(\text{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)) \longrightarrow \text{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

and the isomorphism with $\text{Biext}^1(P, Q; G)$ sends the class described by (f, g) to this image.

In the important special case

$$L_0 = \mathbb{Z}[P], \quad M_0 = \mathbb{Z}[Q],$$

with L_1 and M_1 the kernels of the augmentations to P and Q , the data (f, g) can be written explicitly as functions

$$P \times P \times Q \rightarrow G, \quad P \times Q \times Q \rightarrow G.$$

The five compatibility conditions are exactly the five axioms which define the two partial group laws of a biextension. Thus the explicit cocycle description and the intrinsic definition agree.

Bibliography to Exposé VII

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Exposé VIII. Complements on biextensions; general properties of biextensions of group schemes

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3. Biextensions of group schemes (P, Q) by $\mathbb{G}_m$: generalities.
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7. Canonical prolongations of extensions and biextensions by $\mathbb{G}_m$.

0. Introduction

We continue the general study of biextensions begun in the preceding exposé. We first add several complements in the case of an arbitrary base topos, developing in particular the formalism of pairings associated with biextensions. This formalism is well known to be important in the theory of abelian schemes. We then begin the study of biextensions of general group schemes. The case of greatest interest for the sequel is that of biextensions by the multiplicative group $\mathbb{G}_m$; nevertheless, as technical intermediates, we shall also study biextensions by other types of group schemes.

The most useful result for what follows is the criterion of Section 7: under suitable hypotheses, if P and Q are smooth over a trait with generic point η , a biextension of (P_η, Q_η) by $\mathbb{G}_m$ extends to a biextension of (P, Q) by $\mathbb{G}_m$, uniquely up to a unique isomorphism. This will be convenient in the following exposé, which is devoted to the Néron model of abelian varieties. It is likely that one could proceed there by a shorter route; nevertheless the developments of Exposés VII and VIII are expected to be useful elsewhere as well.

When working with group schemes over a scheme S , the ambient topos is the fppf topos of S , associated with the site of schemes locally of finite presentation over S , endowed with the faithfully flat topology. The word “local” without further qualification means local for this topology.

Most statements about extensions or biextensions by $\mathbb{G}_m$ remain valid with $\mathbb{G}_m$ replaced by a torus G . Since such a torus is locally isomorphic, for the fppf topology and even for the étale topology, to a power $\mathbb{G}_m^r$, the proofs are obtained by descent.

1. Various special cases over an arbitrary topos

In this section, as in Exposé VII, T is a topos and P, Q, G are abelian groups in T . We use the canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G).$$

It gives immediate vanishing and comparison results in a number of special situations.

If

$$\mathrm{Ext}^1(P \otimes Q, G) = 0$$

and

$$\mathrm{Hom}(\mathrm{Tor}_1(P, Q), G) = 0,$$

then every biextension of (P, Q) by G is trivial. More precisely, the spectral sequence for the derived tensor product gives the exact sequence

$$0 \rightarrow \mathrm{Ext}^1(P \otimes Q, G) \rightarrow \mathrm{Biext}^1(P, Q; G) \rightarrow \mathrm{Hom}(\mathrm{Tor}_1(P, Q), G) \rightarrow \mathrm{Ext}^2(P \otimes Q, G),$$

and the assertion follows under the corresponding vanishing hypotheses.

If P is a flat $\mathbb{Z}$ -module, then

$$P \otimes^{\mathbf{L}} Q \simeq P \otimes Q,$$

hence

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes Q, G).$$

The same conclusion holds, by symmetry, if Q is flat. In particular, if both P and Q are locally free abelian sheaves, biextensions are ordinary extensions of $P \otimes Q$ by G .

Suppose P is killed by an integer $n > 0$. The exact triangle associated with multiplication by n on Q relates $P \otimes^{\mathbf{L}} Q$ to $\mathrm{Tor}_1(P, Q)$ and to $P \otimes Q$. This provides the exact sequences used below to extract torsion pairings from a biextension.

When both P and Q are n -torsion, the derived tensor product is controlled by $\mathrm{Tor}_1(P, Q)$ and $P \otimes Q$, and the biextension group is described by the same Ext sequence. These elementary cases will be applied to finite flat group schemes and to the finite layers of Tate modules.

2. Pairings defined by a biextension

Let P and Q be abelian groups of T , and let $n > 0$ be an integer. Write ${}_nP$ for the kernel of multiplication by n on P , and similarly for Q . There is a canonical homomorphism

$${}_nP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1(P, Q).$$

It is obtained from the exact sequence

$$0 \rightarrow {}_nQ \rightarrow Q \xrightarrow{n} Q$$

by applying $P \otimes^{\mathbf{L}} -$. Equivalently, a pair (p, q) , with $np = 0$ and $nq = 0$, defines the standard Tor class represented by the relation $nq = 0$ pulled against p .

One may describe this homomorphism explicitly. For a section q of ${}_nQ$, lift it in the two-term resolution

$$Q \xrightarrow{n} Q.$$

Tensoring with P , a section $p \in {}_nP$ yields a cycle. Its class in $\mathrm{Tor}_1(P, Q)$ is the image of $p \otimes q$. The construction is functorial in both variables and is bilinear.

Interchanging P and Q gives a second construction

$${}_nQ \otimes {}_nP \longrightarrow \mathrm{Tor}_1(Q, P).$$

Under the canonical symmetry

$$\mathrm{Tor}_1(P, Q) \simeq \mathrm{Tor}_1(Q, P),$$

the two constructions differ by the usual sign. This is the standard Koszul sign arising from the symmetry of the derived tensor product. Thus the induced pairing is alternating in the expected graded sense.

Now let G be a further abelian group and let

$$\xi \in \mathrm{Biext}^1(P, Q; G).$$

Through the isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

and the canonical map

$${}_nP \otimes {}_nQ \rightarrow \mathrm{Tor}_1(P, Q),$$

the biextension ξ defines a pairing

$${}_nP \times {}_nQ \longrightarrow G/nG$$

whenever the target is interpreted through the connecting morphism determined by multiplication by n on G . More invariantly, it defines a morphism

$${}_nP \otimes {}_nQ \longrightarrow \mathrm{Coker}(G \xrightarrow{n} G).$$

If $G = \mathbb{G}_m$, this is the familiar Weil-type pairing attached to a biextension.

When n varies through powers of a prime ℓ , the preceding pairings form a compatible projective system. For any commutative group E , write

$$T_\ell(E) = \varprojlim_{\nu} {}_\ell{}^\nu E.$$

A biextension ξ then defines an ℓ -adic pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(G)$$

or, after the standard shift appropriate to the multiplicative group, a pairing with values in the Tate module of $\mathbb{G}_m$. These pairings are functorial in all variables and compatible with pullback, push-out, and change of topos.

In concrete terms, take local sections $p \in {}_n P$ and $q \in {}_n Q$. Restrict the biextension E to the subgroup generated by p and q . The relations $np = 0$ and $nq = 0$ provide two ways of comparing the neutral section with itself after n successive additions. Their quotient is an element of G/nG . This element is the value of the pairing at (p, q) . The biextension axiom is precisely what ensures that the value is independent of the choices and is bilinear.

2.3. Relation with Cartier duality

The description just given, through the isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

is not the original definition of the pairing attached to a biextension. In the classical setting of finite locally free commutative group schemes and $G = \mathbb{G}_m$, the pairing is usually described by Cartier duality.

Let P and Q be finite locally free commutative group schemes over a base S , killed by n , and let E be a biextension of (P, Q) by $\mathbb{G}_m$. Fixing a section p of P , the restriction of E to $\{p\} \times Q$ is an extension of Q by $\mathbb{G}_m$, hence determines a section of the Cartier dual $Q^\vee$. In this way E gives a morphism

$$P \longrightarrow Q^\vee.$$

Equivalently it gives a bilinear pairing

$$P \times Q \longrightarrow \mathbb{G}_m.$$

The homological construction above recovers the same morphism. The verification is local on the base and reduces to the explicit description of the Tor class associated with p and q , together with the concrete computation of the commutator or monodromy element in the restricted biextension.

Thus the two viewpoints agree: the pairing obtained by Cartier duality is the pairing obtained from the Ext-class of the biextension by applying the canonical map from torsion to Tor_1 . This compatibility is the bridge used below between the formal theory of biextensions and the duality theory of finite flat group schemes.

2.4. Pairings attached to torsion and passage to limits

Let n and m be positive integers. The constructions of the previous section are compatible with the transition morphisms

$$_{mn}P \longrightarrow _nP, \quad _{mn}Q \longrightarrow _nQ$$

and with the corresponding transition maps on roots of unity or on the cokernels of multiplication. Thus an element $\xi \in \text{Biext}^1(P, Q; \mathbb{G}_m)$ gives, for every n , a pairing

$$_nP \times _nQ \longrightarrow \mu_n$$

whenever the relevant finite flatness hypotheses hold. These pairings are compatible in n , and hence for every prime ℓ yield a pairing of Tate modules

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1).$$

Equivalently, after tensoring with $\mathbb{Q}_\ell$, one obtains

$$V_\ell(P) \times V_\ell(Q) \longrightarrow \mathbb{Q}_\ell(1).$$

When P and Q are dual abelian schemes, and ξ is the Poincaré biextension, this is the usual Weil pairing. The preceding construction shows that it is not an additional structure but the image, under the formalism of derived tensor products, of the universal biextension class.

3. Biextensions of group schemes (P, Q) by $\mathbb{G}_m$: generalities

3.1. The basic example

Recall first the most important example. If A is an abelian scheme over S , and A' is its dual, then the Poincaré invertible sheaf on $A \times_S A'$, rigidified along the two zero sections, defines a biextension of (A, A') by $\mathbb{G}_m$. Its universal property identifies A' with the functor which classifies extensions of A by $\mathbb{G}_m$. Thus the Poincaré biextension is the geometric representative of the identity morphism

$$A' \longrightarrow A'.$$

The partial extension structures are the theorem of the square in each variable. Their compatibility is the theorem of the cube. This example is the model for the general discussion.

3.2. The case in which one variable is an abelian scheme

Suppose Q is an abelian scheme. For a commutative group scheme P , a biextension of (P, Q) by $\mathbb{G}_m$ may be viewed, by fixing the first variable, as a morphism from P to the sheaf of extensions of

Q by $\mathbb{G}_m$. Since this sheaf is represented by the dual abelian scheme Q' , one obtains a description

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q')$$

under the hypotheses which ensure the usual representability and rigidity. In the reverse direction, a morphism $P \rightarrow Q'$ pulls the Poincaré biextension on $Q \times Q'$ back to a biextension of (P, Q) .

This construction is compatible with base change, since the dual abelian scheme and the Poincaré biextension commute with base change. It is also compatible with the pairings on torsion defined above.

3.3. The case in which one variable is finite locally free

Assume now that Q is finite locally free over S . A biextension of (P, Q) by $\mathbb{G}_m$ is equivalently an extension of P by the Cartier dual $Q^\vee$, under the standard hypotheses. Indeed, restricting the biextension to $P \times \{q\}$ for sections q of Q gives a system of extensions of P by $\mathbb{G}_m$, additive in q . By Cartier duality this is the same as an extension by $Q^\vee$.

Similarly, if P is finite locally free, biextensions of (P, Q) by $\mathbb{G}_m$ correspond to extensions of Q by $P^\vee$. The two descriptions are identified by the symmetry of biextensions and the symmetry of Cartier duality.

This gives a useful bridge between the theory of biextensions and the duality theory of finite flat group schemes. In particular, for finite flat n -torsion group schemes the pairing of Section 2 is exactly the pairing obtained from the corresponding Cartier-dual extension.

3.4. The unipotent case

If one of P or Q is unipotent and the other is of the type considered above, many biextension groups vanish. The reason is that homomorphisms from a unipotent group to a torus, or from an abelian variety to an affine unipotent group, vanish under the usual hypotheses. Consequently, the Ext descriptions obtained above collapse to zero in the relevant degrees.

These vanishings will be used in reduction arguments: after decomposing a smooth connected group scheme into its abelian, toric, and unipotent parts, only the abelian and toric components contribute to biextensions by $\mathbb{G}_m$ in the situations needed later.

3.5. Summary of the preceding cases

The preceding discussion may be summarized as follows. If P and Q are group schemes over S belonging locally to the standard classes of abelian schemes, tori, finite locally free groups, and unipotent groups, then $\mathrm{Biext}^1(P, Q; \mathbb{G}_m)$ is computed by the corresponding duality objects: dual abelian schemes, character groups of tori, and Cartier duals of finite flat groups. The comparison is functorial and compatible with the pairings on torsion.

3.6. Two abelian schemes

Let P and Q be abelian schemes over S , and let Q' be the dual of Q . The preceding construction gives a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q').$$

By symmetry, this is also canonically

$$\mathrm{Hom}(Q, P').$$

The Poincaré biextension corresponds to the identity in the case $P = Q'$, and pullback of the Poincaré biextension gives the inverse map.

In particular, a biextension of (P, Q) by $\mathbb{G}_m$ determines, and is determined by, the bilinear correspondence between P and Q expressed by the associated homomorphism to the dual abelian scheme.

3.7. Semi-abelian variants

Suppose P is an extension of an abelian scheme by a torus, and Q is of a similar type. The computation of $\mathrm{Biext}^1(P, Q; \mathbb{G}_m)$ is obtained by devissage from the abelian and toric parts. The abelian–abelian part is governed by the dual abelian scheme; the toric part is governed by character groups; and the mixed parts are the evident homomorphism groups between the corresponding lattices and dual abelian schemes.

This form of the result is the one needed for Néron models. It is stable under smooth base change and local for the fppf topology.

3.8. Principal consequences

For smooth group schemes over S whose geometric fibres are built from abelian varieties, tori, and additive groups, the category of biextensions by $\mathbb{G}_m$ is controlled by the duality data of those pieces. In particular, if one of the relevant Hom or Ext groups vanishes for structural reasons, the biextension is uniquely trivial. If an abelian part remains, the biextension is the pullback of the Poincaré biextension by the corresponding morphism to the dual.

4. Biextensions and the theorem of the cube

4.1. The cube condition

Let P, Q, G be sheaves of commutative groups on a topos. A biextension of (P, Q) by G can be characterized as a G -torsor on $P \times Q$, rigidified along the two axes, for which the two partial difference torsors satisfy the cocycle conditions in both variables. This is an abstract form of the theorem of the cube.

For group schemes, especially for abelian schemes, the theorem of the cube says that an invertible sheaf on a product which is rigidified on the coordinate axes and satisfies the usual compatibility on triples carries a unique biextension structure. Thus biextensions by $\mathbb{G}_m$ may often be recognized purely in terms of rigidified line bundles.

4.2. Smooth connected groups

If P and Q are smooth connected group schemes over a base S , the cube condition is checked fibrewise and then descended. The connectedness ensures that a morphism to a discrete group is determined by its value on the identity component, hence often vanishes. This is the mechanism behind the full faithfulness results used later.

4.3. Abelian varieties over a field

Let P and Q be abelian varieties over a field k . A rigidified line bundle on $P \times Q$ satisfying the cube condition defines a biextension of (P, Q) by $\mathbb{G}_m$. Conversely, every such biextension has this form. The theorem of the cube gives the necessary associativity and compatibility of the two partial group laws.

For abelian varieties this recovers the classical description:

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q').$$

4.4. On the automatic nature of the cube condition

It is natural to ask whether the cube condition imposed above follows automatically from the two rigidifications and from weaker additivity conditions. The answer depends on the class of group schemes. For abelian varieties the theorem of the cube gives the desired uniqueness. In more general situations, especially when disconnected fibres or affine parts intervene, one must retain the cube condition explicitly or verify it by devissage.

4.5–4.8. Devissage

The computations for smooth connected groups are obtained by filtering the group schemes by affine, toric, unipotent, and abelian parts. The biextension functor is exact enough in each variable to allow passage from the pieces to the whole. Vanishing for unipotent pieces and the standard duality for tori and abelian schemes give the required formulas.

4.9–4.13. Invertible sheaves and ampleness

Let P be a smooth connected group scheme over k , extension of an abelian variety by a torus or by a unipotent group. Rigidified invertible sheaves on P are controlled by the corresponding

Picard functor. If an invertible sheaf $\mathcal{L}$ on P is ample, the associated morphism to the dual is an isogeny on the abelian quotient. The biextension attached to $\mathcal{L}$ is obtained by the usual formula

$$m^*\mathcal{L} \otimes p_1^*\mathcal{L}^{-1} \otimes p_2^*\mathcal{L}^{-1},$$

where m is the group law. The cube theorem identifies the compatibility needed for this line bundle to define a biextension.

5. Extensions and biextensions by constant truncated group schemes

Let S be a scheme, $T \subset S$ a closed subscheme, and Z a set or a group. Denote by Z_S the corresponding constant sheaf or constant group scheme on S , and by Z_T its restriction to T .

When Z is a group, the sheaf Z_S is an étale group scheme. The basic computations of Hom , Ext , and Biext with coefficients in such constant groups reduce to computations on connected components. If P is proper over S , then morphisms $P \rightarrow Z_S$ are locally constant and hence factor through the component group. For an abelian scheme P , this gives

$$\text{Hom}(P, Z_S) = 0$$

when Z is torsion-free and the fibres of P are connected.

More generally, if P is an abelian scheme over S , the Ext groups $\text{Ext}^i(P, Z_S)$ are controlled by the component data on the boundary. Analogous formulas hold for Tor_i , and all these constructions commute with base change. These facts will be applied with $Z = \mathbb{Z}$, supported on a closed subscheme, in the study of the Néron model of $\mathbb{G}_m$.

Proposition 0.57. *Let S be a scheme, $T \subset S$ a closed subscheme, Z a torsion-free constant group, and P, Q abelian schemes over S . Under the standard connectedness hypotheses, the groups of extensions and biextensions by the constant group supported on T are described by the corresponding Hom groups from the component groups of the special fibres to Z , and by their derived functors.*

The proposition is a direct consequence of the exact sequences for Ext and Biext and the fact that proper connected fibres admit no non-constant maps to a discrete group.

6. Extensions and biextensions by the Néron model of $\mathbb{G}_m$

Let S be a scheme, $T \subset S$ a closed subscheme, and $U = S - T$. Let $\mathcal{N}$ be the Néron–Raynaud model of $\mathbb{G}_m$ relative to U . It fits, under the hypotheses used here, into an exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathcal{N} \longrightarrow i_*\mathbb{Z}_T \longrightarrow 0,$$

where $i : T \hookrightarrow S$.

For a commutative group scheme P over S , the extensions of P by $\mathcal{N}$ are easier to control than extensions by $\mathbb{G}_m$, because $\mathcal{N}$ has the Néron mapping property. If P is an abelian scheme, one gets

$$\mathrm{Ext}^1(P, \mathcal{N}) \simeq \mathrm{Ext}^1(P_U, \mathbb{G}_m)$$

by restriction to U . Similarly, for two abelian schemes P, Q ,

$$\mathrm{Biext}^1(P, Q; \mathcal{N}) \simeq \mathrm{Biext}^1(P_U, Q_U; \mathbb{G}_m).$$

The obstruction to replacing $\mathcal{N}$ by $\mathbb{G}_m$ lies in the corresponding groups with coefficients in $i_*\mathbb{Z}_T$. If L is an invertible sheaf on $P \times Q$, rigidified along the axes, its associated biextension by $\mathbb{G}_m$ extends to a biextension by $\mathcal{N}$ when restricted to the open U . The failure of this $\mathcal{N}$ -biextension to come from a $\mathbb{G}_m$ -biextension is measured by the component group term $i_*\mathbb{Z}_T$.

Proposition 0.58. *Let S be regular and let T be a divisor with suitable normal-crossing or unibranch hypotheses. Then the quotient sheaf*

$$j_*(\mathbb{G}_m)_U / (\mathbb{G}_m)_S$$

is represented, étale locally, by the constant group supported on T , and its formation is compatible with smooth base change.

In the case where T is geometrically unibranch, this quotient is simply

$$i_*\mathbb{Z}_T.$$

Consequently the exact sequence above gives explicit obstruction groups for the extension of $\mathbb{G}_m$ -extensions and $\mathbb{G}_m$ -biextensions.

Theorem 0.59. *Let S be regular, let T be a divisor of the preceding type, and let P, Q be abelian schemes over S . The obstruction to extending a biextension of (P_U, Q_U) by $\mathbb{G}_m$ to a biextension of (P, Q) by $\mathbb{G}_m$ is the image of its canonical extension by $\mathcal{N}$ in*

$$\mathrm{Biext}^1(P, Q; i_*\mathbb{Z}_T).$$

The corollary used later is the corresponding criterion for vanishing of this obstruction in terms of component groups.

7. Canonical prolongations of extensions and biextensions by $\mathbb{G}_m$

Let $S = \mathrm{Spec} V$ be a trait, with generic point η , closed point s , and residue field k . Let P be a smooth commutative group scheme over S , and write

$$\Phi = P_s / P_s^0$$

for the group of connected components of the special fibre.

Theorem 0.60. *Assume Φ is a torsion group, for instance P is of finite type over S . Then the restriction functor from extensions of P by $\mathbb{G}_m$ over S to extensions of P_η by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$, then to every extension E_η of P_η by $\mathbb{G}_m$ there is associated a canonical homomorphism*

$$d(E_\eta) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k$$

whose vanishing is necessary and sufficient for E_η to be the generic fibre of an extension of P by $\mathbb{G}_m$. In particular, if P_s is connected, restriction is an equivalence.

For biextensions the statement is analogous. Let P, Q be smooth commutative group schemes over S , and put

$$\Phi = P_s/P_s^0, \quad \Psi = Q_s/Q_s^0.$$

Theorem 0.61. *Assume that Φ or Ψ is torsion. Then restriction from biextensions of (P, Q) by $\mathbb{G}_m$ over S to biextensions of (P_η, Q_η) by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$ or $n\Psi = 0$, then every generic biextension E_η has a canonical obstruction pairing*

$$d(E_\eta) : \Phi \otimes \Psi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k,$$

and this pairing vanishes if and only if E_η extends to a biextension of (P, Q) by $\mathbb{G}_m$ over S . If P_s or Q_s is connected, restriction is an equivalence.

The proof uses the Néron–Raynaud model $\mathcal{N}$ of $(\mathbb{G}_m)_\eta$. Restriction for biextensions by $\mathcal{N}$ is an equivalence by the Néron mapping property. Thus the question is reduced to passing from $\mathcal{N}$ to $\mathbb{G}_m$, using the exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathcal{N} \longrightarrow \mathbb{Z}_s \longrightarrow 0.$$

For extensions, full faithfulness is equivalent to

$$\mathrm{Hom}(P, \mathbb{Z}_s) = 0,$$

which identifies with $\mathrm{Hom}(\Phi, \mathbb{Z})$ and therefore vanishes when Φ is torsion. The obstruction group is

$$\mathrm{Ext}^1(P, \mathbb{Z}_s) \simeq \mathrm{Hom}(\Phi, \mathbb{Q}/\mathbb{Z}).$$

For biextensions the same argument, with the exact sequence of Biext's, gives the obstruction group

$$\mathrm{Biext}^1(P, Q; \mathbb{Z}_s) \simeq \mathrm{Hom}(\Phi \otimes \Psi, \mathbb{Q}/\mathbb{Z}).$$

7.3. Complements

In the situation of the extension theorem, with $n\Phi = 0$, the obstruction $d(E_\eta)$ admits a torsion interpretation. The restriction of E_η to ${}_nP_\eta$ gives an extension of ${}_nP_\eta$ by μ_n . Passing to component

groups gives

$${}_n\Phi \longrightarrow (\mathbb{Z}/n\mathbb{Z})_k,$$

which is the restriction of $d(E_\eta)$.

In the biextension case, a generic biextension E_η defines the torsion pairings discussed in Section 2. These recover the obstruction pairing

$$d(E_\eta) : \Phi \otimes \Psi \rightarrow (\mathbb{Q}/\mathbb{Z})_k.$$

If $q \in Q(S)$ is a section and q_s its image in $\Psi(k)$, then the extension of P_η obtained from E_η by restriction along q_η has obstruction

$$d_q(x) = d(E_\eta)(x, q_s).$$

Thus the biextension obstruction is characterized by the extension obstructions obtained by fixing one variable.

The obstruction is symmetric under transposition. If ${}^tE_\eta$ denotes the transposed biextension of (Q_η, P_η) , then the diagram

$$\begin{array}{ccc} \Phi \otimes \Psi & \xrightarrow{d(E_\eta)} & (\mathbb{Q}/\mathbb{Z})_k \\ \downarrow & & \parallel \\ \Psi \otimes \Phi & \xrightarrow{d({}^tE_\eta)} & (\mathbb{Q}/\mathbb{Z})_k \end{array}$$

commutes. In particular, if $P = Q$ and E_η is symmetric, then the obstruction form on Φ is symmetric.

Finally, let $S' \rightarrow S$ be a dominant local morphism of traits and let

$$e(S'/S)$$

be the ramification index. Under base change one has

$$d(E_{\eta'}) = e(S'/S) d(E_\eta) \otimes_k k'.$$

When $e(S'/S) = 1$, for example for étale extensions, henselizations, strict henselizations, or completions, formation of the obstruction commutes with base change. In general, after a suitable finite extension of traits, the obstruction is killed and the extension or biextension prolongs.

Proposition 0.62. *Let S be reduced. If P is a smooth group scheme over S whose fibres at the maximal points are connected, then the functor from extensions of P by $\mathbb{G}_m$ to rigidified $\mathbb{G}_m$ -torsors on P is fully faithful. If S is normal and P has connected fibres, a rigidified torsor lies in the essential image if and only if this is true over every maximal point of S .*

The analogous statement holds for biextensions of (P, Q) : the functor to bitrivialized $\mathbb{G}_m$ -torsors on $P \times_S Q$, rigidified along the two unit sections, is fully faithful under the same connectedness assumptions, and over a normal base its essential image is detected at the maximal points.

Corollary 0.63. *Let S be normal and let P, Q be smooth group schemes over S , with connected maximal fibres and with one of P, Q having connected fibres. Assume that over every maximal point one of P_η, Q_η is an extension of an abelian variety by a torus, or more generally has a composition series whose factors are abelian varieties, tori, or additive groups. Then every bitrivialized torsor of the preceding kind comes from a unique biextension.*

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Exposé IX. Néron models of abelian schemes and monodromy

0. Introduction

Let S be a trait with generic point $\eta = \operatorname{Spec} K$, closed point s , and residue field k . Let A_K be an abelian variety over K , and let A be its Néron model over S . The object of this exposé is to express the structure of the special fibre A_s , and in particular its group of connected components, in terms of the action of inertia on the Tate module of A_K .

The dual abelian variety is denoted A'_K , and its Néron model by A' . The Poincaré biextension of (A_K, A'_K) by $\mathbb{G}_m$ has a canonical prolongation to a biextension

$$W$$

of (A^0, A'^0) by $\mathbb{G}_m$. From it one obtains a canonical pairing of component groups

$$\Phi \times \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_k,$$

where

$$\Phi = A_s/A_s^0, \quad \Phi' = A'_s/A'^0_s.$$

This pairing will be related to the monodromy pairing on Tate modules.

Fix a prime ℓ invertible on S . The ℓ -adic Tate module $T_\ell(A_K(\overline{K}))$ is equipped with an action of

$$\pi = \operatorname{Gal}(\overline{K}/K),$$

and in particular of the inertia group I . The main constructions below single out two canonical submodules: the fixed part and the toric part. The orthogonality theorem identifies the toric part as the orthogonal complement of the fixed part for the Weil pairing attached to the canonical biextension.

1. Néron model of an abelian scheme

1.0. Notation and the canonical pairing

Let A_K be an abelian variety over K , A'_K its dual, and A, A' their Néron models over S . Write

$$\Phi = A_s/A_s^0, \quad \Phi' = A'_s/A'^0_s.$$

The canonical biextension W_K of (A_K, A'_K) by $\mathbb{G}_m$ extends canonically to a biextension

$$W$$

of (A^0, A'^0) by $\mathbb{G}_m$. Applying the obstruction formalism of Exposé VIII to this biextension gives a pairing

$$\Phi \times \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_k.$$

This is the component pairing associated with the Poincaré biextension. It is functorial in A_K and compatible with duality.

The restriction of W to the generic fibre is W_K . The uniqueness part of the canonical prolongation theorem ensures that W is uniquely determined by W_K , and hence that the pairing on component groups is canonical.

1.1. Non-degeneracy properties

The pairing

$$\Phi \times \Phi' \rightarrow (\mathbb{Q}/\mathbb{Z})_k$$

is expected, and in the situations used below is known, to be non-degenerate in the appropriate sense. More precisely, after restricting to the finite component groups which arise for Néron models of abelian varieties over a trait, it induces homomorphisms

$$\Phi \longrightarrow \mathrm{Hom}(\Phi', \mathbb{Q}/\mathbb{Z}), \quad \Phi' \longrightarrow \mathrm{Hom}(\Phi, \mathbb{Q}/\mathbb{Z}).$$

Raynaud's results identify these maps as isomorphisms under the standard hypotheses. This is one of the principal inputs in the orthogonality theorem.

1.2. Behaviour under isogenies

Let C_K be another abelian variety over K , with Néron model C , and let

$$u_K : C_K \rightarrow A_K$$

be a homomorphism. By the Néron mapping property it extends uniquely to

$$u : C \rightarrow A.$$

If u_K is an isogeny, then u induces isogenies on the connected components of the special fibres and on the corresponding toric and abelian parts. In particular the numerical invariants attached to the reduction of A_K are invariant under isogeny. The same applies to A_K and its dual.

1.3. The invariants λ, μ, α

Let A_s^0 be the identity component of the special fibre. Its Chevalley decomposition has an affine part whose maximal torus has rank μ , an abelian quotient of dimension α , and a unipotent part of rank λ . These integers satisfy

$$\dim A_K = \lambda + \mu + \alpha.$$

The dual abelian variety has the same invariants. This follows from the isogeny-invariance just mentioned, together with the duality between the toric and component data.

1.4. The canonical biextension on the identity components

The canonical biextension W on (A^0, A'^0) restricts on the special fibre to a biextension

$$W_s$$

of $(A_s^0, A'_s{}^0)$ by $\mathbb{G}_m$. This biextension is the object through which the ℓ -adic pairings on the fixed parts of the Tate modules are compared with the geometry of the special fibre. Its non-degeneracy on the abelian quotients and its vanishing on toric directions are the essential facts used below.

2. Fixed and toric parts of $T_\ell(A)$. Criteria for good reduction

2.1. The Tate module and inertia

Let ℓ be a prime distinct from the residue characteristic p . The Tate module

$$T_\ell(A_K) = T_\ell(A_K(\overline{K}))$$

is a free $\mathbb{Z}_\ell$ -module of rank $2 \dim A_K$, equipped with a continuous action of $\pi = \text{Gal}(\overline{K}/K)$. Let $I \subset \pi$ be the inertia group.

If S is henselian, the finite étale group schemes ${}_{\ell^\nu}A$ over S identify their sections over the maximal unramified extension with the inertia invariants in ${}_{\ell^\nu}A_K(\overline{K})$. Passing to the inverse limit gives the fixed part:

$$T_\ell(A_K)^f = T_\ell(A_K)^I.$$

2.2. Definition of the fixed part

The fixed part is the ℓ -adic sheaf on S whose generic fibre is the inertia-invariant submodule of $T_\ell(A_K)$. Equivalently, when S is henselian, it is

$$T_\ell(A_K)^f = \varprojlim_{\nu} {}_{\ell^\nu}A(K^{\text{nr}}),$$

where K^{nr} is the maximal unramified extension of K in $\overline{K}$.

The specialization map identifies this module with the Tate module of the identity component of the special fibre:

$$T_\ell(A_K)^f \simeq T_\ell(A_s^0)$$

after the standard passage through the Néron model. More precisely, the fixed part is characterized as the lisse $\mathbb{Z}_\ell$ -sheaf whose special fibre is $T_\ell(A_s^0)$.

Proposition 0.64. *Assume S henselian and $\ell \neq p$. The submodule of $T_\ell(A_K(\overline{K}))$ corresponding to the fixed part is the submodule of invariants:*

$$T_\ell(A_K)^f = T_\ell(A_K(\overline{K}))^I.$$

For every integer $m > 0$ prime to p one has

$${}_mA(K^{\text{nr}}) = ({}_mA(\overline{K}))^I.$$

This is because ${}_mA$ is finite étale over S . Passing to the inverse limit over $m = \ell^\nu$ proves the proposition.

2.2.6. Isogenies and the fixed part

Let C_K be another abelian variety with Néron model C , and let $u_K : C_K \rightarrow A_K$ be an isogeny. Then

$$T_\ell(u_K) : T_\ell(C_K) \rightarrow T_\ell(A_K)$$

induces an isogeny on fixed parts

$$T_\ell(C_K)^f \longrightarrow T_\ell(A_K)^f.$$

The induced morphism on the identity components of special fibres gives an isogeny

$$T_\ell(C_s^0) \longrightarrow T_\ell(A_s^0).$$

It also induces an isogeny on maximal tori and, after a harmless extension of the residue field, on the abelian quotients of the identity components.

Consequently the invariants

$$\lambda, \quad \mu, \quad \alpha$$

of the reduction are invariant under isogeny. In particular A_K and its dual A'_K have the same invariants.

2.2.9. Good reduction

Let A_K be an abelian variety over K , A its Néron model, and $\ell \neq p$. The following conditions are equivalent.

- (i) There exists an abelian scheme over S with generic fibre A_K .
- (ii) The Néron model A is an abelian scheme.
- (iii) The identity component A^0 is an abelian scheme.
- (iv) The special fibre A_s is an abelian variety.

- (v) The identity component A_s^0 is an abelian variety, equivalently $\lambda = \mu = 0$.
- (vi) The Néron model A is proper over S .
- (vii) The inertia group I acts trivially on $T_\ell(A_K(\overline{K}))$.
- (viii) Equivalently, for every ν , I acts trivially on ${}_{\ell^\nu}A_K(\overline{K})$.

This is the Néron–Ogg–Shafarevich criterion for good reduction. An abelian scheme satisfying (i) is the Néron model of its generic fibre and is therefore unique up to unique isomorphism.

The proof uses the Néron mapping property for abelian schemes over a regular base, the structure of A_s^0 , and the equality between the fixed part of the Tate module and the inertia invariants. Trivial inertia action forces the fixed part to be the whole Tate module, which is equivalent to the absence of toric and unipotent contribution in the special fibre.

2.3. The toric part

Assume again that S is henselian and $\ell \neq p$. Let T_0 be the maximal torus of A_s^0 . The inclusion

$$T_\ell(T_0) \subset T_\ell(A_s^0)$$

defines a canonical lisse $\mathbb{Z}_\ell$ -subsheaf of $T_\ell(A^0)$, called the toric part:

$$T_\ell(A^0)^t \subset T_\ell(A^0) \subset T_\ell(A).$$

Its special fibre is

$$T_\ell(A^0)^t \times_S s = T_\ell(T_0).$$

On the generic fibre this gives a π -stable submodule

$$T_\ell(A_K)^t \subset T_\ell(A_K(\overline{K})),$$

the toric submodule of the Tate module. Thus

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K)$$

is a canonical three-step filtration. The same construction applies to the dual abelian variety.

2.4. The orthogonality theorem

Let A'_K be the dual abelian variety. The Poincaré biextension gives the Weil pairing

$$e_\ell : T_\ell(A_K) \times T_\ell(A'_K) \longrightarrow \mathbb{Z}_\ell(1).$$

Theorem 0.65. *Assume S henselian and $\ell \neq p$. With respect to the pairing e_ℓ , the toric part of $T_\ell(A_K)$ is the orthogonal complement of the fixed part of $T_\ell(A'_K)$ inside the fixed part of $T_\ell(A_K)$:*

$$T_\ell(A_K)^t = T_\ell(A_K)^f \cap (T_\ell(A'_K)^f)^\perp.$$

The symmetric statement with A and A' interchanged also holds.

The first inclusion follows by extending the Poincaré biextension to (A^0, A'^0) . On special fibres, it is enough to prove that the pairing

$$T_\ell(T_0) \times T_\ell(A_s'^0) \rightarrow \mathbb{Z}_\ell(1)$$

is zero; this is a consequence of the vanishing of biextensions between a torus and the relevant connected part. For the reverse inclusion, one reduces to the special fibre and uses the non-degeneracy, due to Raynaud, of the biextension W_s on the abelian quotient.

2.5. Polarized form

If ξ is a polarization of A_K , it identifies $T_\ell(A_K)$ with a submodule of $T_\ell(A'_K)$ up to isogeny and gives an alternating form

$$e_\ell^\xi : T_\ell(A_K) \times T_\ell(A_K) \rightarrow \mathbb{Z}_\ell(1).$$

Let

$$V = T_\ell(A_K)^f, \quad W = T_\ell(A_K)^t.$$

The orthogonality theorem becomes the simple formula

$$V^\perp = W.$$

The successive ranks in the induced filtration recover the numerical invariants λ, μ, α of the special fibre.

3. The case of semi-stable reduction

The semi-stable case is characterized by the absence of unipotent part in the identity component of the special fibre. Equivalently, the identity component is an extension of an abelian variety by a torus. The Tate module filtration then has only the fixed and toric contributions described above.

We first record a general result in the style of SGA 3. Let G be a commutative S -group scheme, flat, separated, and of finite presentation, let H be a quasi-separated group scheme locally of finite type over S , and let

$$u : G \rightarrow H$$

be a homomorphism with kernel K . For $s \in S$, and for n prime to the residue characteristic, the

group scheme ${}_nK$ is étale of finite type near s , and its geometric ranks specialize divisibly:

$$\text{rank}({}_nK_{\bar{s}}) \mid \text{rank}({}_nK_{\bar{t}}).$$

This gives inequalities among the toric, abelian, and unipotent ranks of the fibres. In particular, if the generic connected kernel is unipotent, then so is the special connected kernel. If G_s has no unipotent part, H is smooth, and u_s is an isogeny, then u is flat and quasi-finite near s .

These facts are used to prove the semi-stable reduction theorem: after a finite separable extension of K , an abelian variety acquires semi-stable reduction. In the language of Tate modules this is reflected in the quasi-unipotence of the inertia action and in the existence, after finite extension, of the canonical filtration whose graded pieces are the toric and abelian parts.

Further continuation of Exposé IX

The following paragraphs continue the discussion of semistable reduction, conductors, and the filtrations on Tate modules. **4.4.** Place ourselves under the conditions of 4.3 b). Then for every element g in π' whose image in π_0 is of the form f_q^n with $n \geq 0$, the eigenvalues of the induced automorphisms of $U^{\text{ét}}$, $U^{\text{ét}}/U^{\text{ét}}$, $U/U^{\text{ét}}$ are algebraic integers whose absolute values are respectively q^n , $q^{n/2}$ and 1. Concerning the two extreme pieces, one can say, more precisely, that the corresponding eigenvalues are respectively $(q^n \varepsilon_i)_{1 \leq i \leq \mu}$ and $(\varepsilon_i)_{1 \leq i \leq \mu}$, where the ε_i ($1 \leq i \leq \mu$) are roots of unity.

Indeed, formula (4.1.2) gives for the last piece the eigenvalues $q^n/(q^n \varepsilon_i) = \varepsilon_i^{-1}$, but as ε_i is a root of unity, we have $\varepsilon_i^{-1} = \overline{\varepsilon_i}$ (complex conjugate), and as the family of ε_i is stable under complex conjugation (as the family of solutions of an equation with integer coefficients), the family of ε_i^{-1} is also that of the ε_i , which proves the last assertion of 4.4.

4.5. We shall give an application of 4.3 a) to the theory of local conductor of abelian schemes. Suppose k algebraically closed (*). Recall (SGA 5 X 6.2 or [20]) that for every commutative étale ℓ -group scheme F_K over K , ($\ell \neq p$), one defines an integer

$$a_s(F_K) = a_s(i_*(F_K)) \quad (1)$$

($i: \eta \rightarrow S$ denoting the inclusion), in the following way : one chooses a finite Galois extension K' of K , with group Γ , which splits F_K , so that F_K is defined by a finite ℓ -group M on which Γ acts. Moreover, one associates to the extension K' of K a certain *projective* $\mathbb{Z}_\ell[\Gamma]$ -module (defined up to isomorphism)

$$\text{Sw} = \text{Sw}(K'/K, \mathbb{Z}_\ell) , \quad (2)$$

and one sets

$$a_s(F_K) = \text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(\text{Sw}, M) . \quad (3)$$

One verifies (loc. cit.) that this expression does not depend on the choice of the extension K'/K . Besides, the fact that Sw is a projective $\mathbb{Z}_\ell[\Gamma]$ -module implies that $a_s(F_K)$ behaves additively for exact sequences.

We shall apply this definition to the case of the ℓ -adic sheaf $\mathcal{A}_K$, where $\mathcal{A}_K$ is our abelian scheme over K . One has thus, in this case, an integer

$$a_s(\mathcal{A}_K; \ell) = \text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(\text{Sw}, U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell) , \quad U_\ell = T_\ell(A(\overline{K})) , \quad (4)$$

Γ being chosen as was said (choice depending a priori on ℓ). The integer (4.5.4) is called the *wild conductor exponent* (or “sauvage”) (*) of the abelian scheme $\mathcal{A}_K$ with respect to the discrete valuation ring $V \subset K$; by passing to the henselisation, it is of course defined without supposing V henselian, and it is immediately reduced to the case where S is *strictly local*. We propose to show the

Corollary 0.66 (4.6). *The wild conductor exponent (4.5.4) does not depend on the choice of the*

prime number $\ell \neq p$.

We may assume S strictly local. We shall give the calculation of (4.5.4), replacing Sw by any other projective $\mathbb{Z}_\ell[\Gamma]$ -module L . Note that the filtration (4.1.1) of U_ℓ is formed by $\mathbb{Z}_\ell$ -submodules which are manifestly direct factors, so this filtration induces an analogous filtration of $U(\ell) = U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell$, whose successive factors are deduced from the factors V_i ($i = 1, 2, 3$)

$$U^{\text{ét}}, \quad U^{\text{ét}}/U^{\text{ét}}, \quad U/U^{\text{ét}}$$

appearing in the filtration (4.1.1) by tensorisation with $\mathbb{F}_\ell$. Of course, this filtration is stable under Γ , and gives by additivity a decomposition of $\text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell))$ into a sum of three terms, lengths of the groups

$$\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell) \simeq (\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i)) \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell, \quad (5)$$

the isomorphism written in (*) resulting again from the projectivity hypothesis made on L ; one only assumes K' chosen large enough for Γ to operate on the V_i . Now $\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i)$ is evidently a torsion-free $\mathbb{Z}_\ell$ -module, hence its *rank* is equal to the right hand side of (*). If φ_i is the character of the representation of Γ in V_i , and ψ the character of the representation of Γ in $\check{L}$, one therefore finds

$$\begin{aligned} \text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell)) &= \sum_{1 \leq i \leq 3} N^{-1} \sum_{g \in \Gamma} \psi(g) \varphi_i(g) \\ &= N^{-1} \sum_{g \in \Gamma} \psi(g) \varphi(g), \quad \text{with } N = \text{card } \Gamma, \end{aligned} \quad (6)$$

where $\varphi(g) = \sum \varphi_i(g)$ is also the function on Γ deduced by passage to the quotient from the function on the inertia group I , character of the monodromy representation of I on U . Now by virtue of 4.3 a) this character is a function with *rational integer values independent of ℓ* . Taking $L = \text{Sw}$, it is the same for the character $\psi(g)$ of L , which is equal to $a - N\delta + 1$, where δ is the “Dirac function” of Γ , and a the “Artin conductor” of K'/K [26]. This therefore proves 4.6, while exhibiting at the same time the following expression, independent of ℓ , for the wild conductor exponent :

$$a_s(\mathcal{A}_K) = N^{-1} \sum_{g \in \Gamma} (a(g) - N\delta(g) + 1) \cdot \varphi(g), \quad (7)$$

where Γ is the Galois group of a finite Galois extension K' of K splitting $U^{\text{ét}}$ (for example the smallest such extension, cf. 4.1), $a(g)$ the Artin conductor on Γ , and $\varphi(g)$ the trace function on Γ made explicit in (4.3.0). In the case of semi-stable reduction, one can take $K' = K$, hence $\Gamma = e$, and one finds

$$a_s(\mathcal{A}_K) = 0 \quad (\text{semi-stable reduction case}). \quad (8)$$

(*) It would in fact suffice that k be *perfect*, but by passing to the strict henselisation of S , one immediately reduces to the case under consideration.

4.7. To conclude, let us make explicit a criterion of semi-stable reduction which follows from the

general reflections of 4.2 :

Proposition 0.67 (4.7). (M. RAYNAUD's criterion for semi-stable reduction). *Let S be a trait, A_K an abelian scheme defined over its function field K , n an integer ≥ 1 prime to the residual characteristic p (*), and such that ${}_nA_K$ is unramified with respect to S , i.e. such that the inertia group acts trivially on ${}_nA(\overline{K})$. Then, either A_K has semi-stable reduction over S , or $n = 2$, and if S is strictly local, A_K acquires semi-stable reduction over a finite Galois extension K' of K with Galois group isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$.*

Proof. We may obviously assume S strictly local. With the notation of 4.1 and 4.2, let K' be the smallest possible, i.e. assume that Γ acts *faithfully* on A_0^{*0} . By hypothesis, Γ acts trivially on ${}_nA(\overline{K})$, and a fortiori on ${}_nA(\overline{K})^{\text{ét}} \simeq {}_nA_0^*(k) \supset {}_nA_0^{*0}(k)$. We conclude by means of the two following lemmas :

Lemma 0.68 (4.7.1). (J.P. SERRE). *Let G be a commutative group scheme over a field k , extension of an abelian scheme by a torus, n an integer ≥ 2 . Then every automorphism of finite order of G inducing the identity on ${}_nG$ is reduced to the identity if $n \neq 2$, and is the identity or of order two if $n = 2$.*

Lemma 0.69 (4.7.2). *A finite group in which every element $\neq 1$ is of order 2 is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$, for a suitable $r \geq 0$.*

4.7.3. Let us recall the proof of 4.7.1, which is well known. One notes that thanks to the fact that $n \cdot \text{id}_G$ is an epimorphism, the hypothesis that u induce the identity on ${}_nG$, i.e. that $1 - u$ induce 0 on ${}_nG$, is expressed by the condition that one has $1 - u = nv$ in the ring of endomorphisms of G . To prove the announced conclusion, one is reduced (decomposing the finite cyclic group generated by u into its p -primary components, for the various prime numbers p dividing the order of u) to the case where u is of order p^N , a power of a prime number. Let E be the subring of $\text{End}(G)$ generated by v , which is a quotient of $\mathbb{Z}[T]/(T^{p^N} - 1)$, hence of finite type over $\mathbb{Z}$, and which moreover is torsion-free (this is a formal consequence of the fact that G is divisible). It thus embeds into $E_{\mathbb{Q}} = E \otimes_{\mathbb{Z}} \mathbb{Q}$, which appears as a quotient of $\mathbb{Q}[T]/(T^{p^N} - 1)$, which is a direct sum of finite field extensions of $\mathbb{Q}$. It is thus the same for $E_{\mathbb{Q}}$. Considering the images of E, u, v in the component fields K_i of $E_{\mathbb{Q}}$, one is thus reduced to seeing that if $K = \mathbb{Q}(\mu)$ is a field generated by a primitive p^n -th root of unity μ , and if n is an integer ≥ 2 such that $\mu - 1 \equiv 0 \pmod{n}$ (in the ring of integers of K), then one has $\mu = 1$, or $\mu = -1$ and $n = 2$. Now this follows immediately from the well-known fact that if $\mu \neq 1$ i.e. $m \geq 1$, the ideal $(\mu - 1)$ is prime and the ideal (p) is a power thereof, whose exponent is > 1 except if $p = 2, m = 1$ i.e. $\mu = -1$. From the fact that $(\mu - 1)$ is prime and divides p it follows indeed that $n \geq 2$ can divide $\mu - 1$ only if $n = p$, and p divides $\mu - 1$ only if $(\mu - 1) = (p)$ i.e. $\mu = -1$, hence $n = 2$.

4.7.4. Let us prove 4.7.2. It suffices to prove that Γ is commutative. Now if $x, y \in \Gamma$, one will have $(xy)(xy) = 1$, which can also be written, since $x^2 = 1, y^2 = 1$,

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx, \quad \text{qed.}$$

□

Remark 0.70 (4.7.5). The proof given of 4.7.3, together with 4.7.2, shows us that if G is a divisible commutative group of a topos, $n \geq 2$ an integer, and Γ a finite group, acting faithfully on G while inducing the identity on ${}_nG$, then Γ is reduced to the identity if $n \geq 3$, and is isomorphic to a group $(\mathbb{Z}/2\mathbb{Z})^r$ if $n = 2$.

§ 5. The orthogonality theorem in the case $\ell = p$ (semi-stable case). Duality of abelian schemes B_0 and B'_0 . Characterisation of the fixed part. Criteria for good reduction, essentially good reduction, and semi-stable reduction.

5.1. Place ourselves in the case 2.2.3 where S is henselian and A^0 is ℓ -divisible, so that the *fixed part*

$$T_\ell(A^0)^f \hookrightarrow T_\ell(A)$$

is defined as the Barsotti–Tate pro- ℓ -group formed by the $(\ell^\nu A^0)^f$. We shall now give a definition of the toric part (2.3.2) of $T_\ell(A^0)$, independent of the hypothesis $\ell \neq p$ made in 2.3, but assuming on the other hand that S is *complete* (*). For this, introduce for every integer $n \geq 0$ the local artinian scheme

$$S_n = \operatorname{Spec}(V/\mathfrak{m}^{n+1}) , \quad (9)$$

n -th infinitesimal neighbourhood of s in S , and for every S -scheme X let $X_n = X \times_S S_n$. As $(A^0)_n = (A_n)^0$ is smooth over S_n , it is known (SGA 3 IX 3.6 bis) that T_0 is the special fibre of a unique subtorus

$$T_n \subset A_n^0 . \quad (10)$$

According to the uniqueness of the T_n , these are deduced from one another by the base changes $S_{n'} \rightarrow S_n$. In other terms, they define a *formal torus* $\hat{T}$ of the formal group $\hat{A}^0$ (in the category of formal schemes over S) formal completion of A^0 along the special fibre :

$$\hat{T} \subset \hat{A}^0 . \quad (11)$$

As the functor $X \mapsto \hat{X}$ (formal completion along the special fibre) induces an equivalence between the category of finite schemes over S , and that of formal finite schemes over S (EGA III 4.8), we shall simply identify a finite scheme over S with the corresponding formal scheme. With this identification, the definition of the “fixed parts” immediately gives us for every $\nu \geq 0$:

$$(\ell^\nu A^0)^f = \ell^\nu(\hat{A}^0) , \quad (12)$$

which can also be written as an identity of projective systems

$$T_\ell(A^0)^f = T_\ell(\hat{A}^0) . \quad (13)$$

One thus obtains a sub-pro-Barsotti–Tate group of the fixed part of $T_\ell(A^0)$ by taking $T_\ell(\hat{T}) \subset T_\ell(\hat{A}^0)$. We then define the *toric part* of $T_\ell(A^0)$ as

$$T_\ell(A^0)^t \stackrel{\text{def}}{=} T_\ell(\hat{T}) \subset T_\ell(\hat{A}^0) = T_\ell(A^0)^f . \quad (14)$$

So by definition, one will have

$$T_\ell(A^0)^t = (\ell^\nu \hat{T})_{\nu \geq 0} , \quad (15)$$

the finite and flat S -group $\ell^\nu \hat{T}$ itself being described as the inductive limit of the finite and flat

S_n -groups

$$\ell^\nu(T_n) , \quad n \text{ variable} , \quad (16)$$

i.e. its affine algebra is described as the projective limit of the affine algebras of the groups (5.1.8). Of course, one will have

$$(T_\ell(A^0)^t)_n = T_\ell(T_n) , \quad (17)$$

which for $n = 0$ is none other than formula (2.3.3) by which we had characterised the toric part (as pro-finite-étale group) in the case $\ell \neq p$. This shows that the definition (5.1.6) is compatible with that given in 2.3 when $\ell \neq p$.

Passing to the generic fibres, the fixed and toric parts of $T_\ell(A^0)$ define fixed and toric parts in $T_\ell(A_K)$:

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K) . \quad (18)$$

These are inclusions of pro- ℓ -Barsotti–Tate groups. Consider the analogous filtration, corresponding to the dual abelian scheme A'_K :

$$T_\ell(A'_K)^t \subset T_\ell(A'_K)^f \subset T_\ell(A'_K) . \quad (19)$$

Theorem 0.71 (5.2). *Suppose that S is a complete trait, and let A_K be an abelian scheme over K , A'_K the dual abelian scheme, ℓ a prime number such that the connected Néron model A^0 is ℓ -divisible (cf. 2.2.1), so that we then dispose of the filtrations (5.1.10) and (5.1.11) of the pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$. This being posited, the toric parts are reconstructed in terms of the fixed parts by the formula*

$$T_\ell(A_K)^t = T_\ell(A_K)^f \cap (T_\ell(A'_K)^f)^\perp \quad (20)$$

and the analogous formula obtained by exchanging the roles of A_K and A'_K , where the sign $\perp$ denotes the orthogonal with respect to the canonical pairing (1.0.2).

Finally, when A_K has semi-stable reduction over S (which results from the other hypotheses in the case $\ell = p$), then one even has the formula

$$T_\ell(A_K)^t = (T_\ell(A'_K)^f)^\perp . \quad (21)$$

5.2.3. The meaning of formula (5.2.1) being quite clear if ℓ is prime to the characteristic of K (where it can be interpreted by passing to ordinary $\mathbb{Z}_\ell$ -modules, with action of $\pi = \text{Gal}(\overline{K}/K)$, formed of the $\overline{K}$ -points of the pro-groups in play, as in 2.4), let us make it explicit in the general case, using the notion of dual

$$D(G) = \text{Hom}(G, T_\ell(\mathbb{G}_m)) \quad (22)$$

of a pro- ℓ -Barsotti–Tate group $G = (G(\nu))$, which by definition is the projective system of Cartier duals $D(G(\nu))$ (SGA 3 VII 3.3.1 and VIII 1), where the transition morphisms are defined in an evident way (by transposition from the morphisms of *inclusion* between the $G(\nu)$, for example). The duality theory of abelian varieties teaches us that the pairing mentioned in 5.2 in fact defines

an *isomorphism*

$$T_\ell(A'_K) \xrightarrow{\sim} D(T_\ell(A_K)) , \quad (23)$$

whence by composition a morphism

$$T_\ell(A_K)^f \longrightarrow D(T_\ell(A'_K)^f) , \quad (24)$$

and formula (5.2.1) asserts that the kernel of the preceding homomorphism (which a priori is a sub-pro-group of the first term) is none other than the toric part $T_\ell(A_K)^t$. We shall note that *it is an isomorphism of pro-objects, and not of projective systems* : we do not assert (and it would be false in general) that for every integer ν , $T_\ell(A_K)^t(\nu)$ is equal to the kernel of $T_\ell(A_K)^f(\nu) \rightarrow D(T_\ell(A'_K)^f)(\nu)$; the latter could a priori be larger. (We shall see however below that it is not so if A_K has semi-stable reduction, which is in particular the case, according to the hypotheses made in 5.2, when $\ell = p$.)

5.2.4. The proof of (5.2.1) merely repeats, practically without change, that of 2.4. One notes that the induced pairing

$$T_\ell(A_K)^f \times T_\ell(A'_K)^f \longrightarrow T_\ell(\mathbb{G}_{mK}) \quad (25)$$

giving rise to (5.2.3.3) is induced by a pairing

$$T_\ell(A^0)^f \times T_\ell(A'^0)^f \longrightarrow T_\ell(\mathbb{G}_{mS}) \quad (26)$$

of Barsotti–Tate pro-groups over S , itself induced by the pairing of pro-groups

$$T_\ell(A^0) \times T_\ell(A'^0) \longrightarrow T_\ell(\mathbb{G}_{mS}) \quad (27)$$

deduced from the biextension W of (A^0, A'^0) by $\mathbb{G}_{mS}$. Hence, the pairing

$$T_\ell(A_0^0) \times T_\ell(A_0'^0) \longrightarrow T_\ell(\mathbb{G}_{mk}) , \quad (28)$$

deduced from (5.2.4.2) by passage to the special fibres, is none other than that deduced from the biextension W_0 of $(A_0^0, A_0'^0)$ by $\mathbb{G}_{mk}$. By virtue of VIII 4.11 and of 1.5, the left annihilator of the pairing (5.2.4.4) is therefore equal to $T_\ell(T_0)$, which is none other than the special fibre of the toric part $T_\ell(A^0)^t$. One thus concludes, returning to the definition of the left annihilators made explicit in 5.2.3, by means of the

Lemma 0.72 (5.2.5). *Let $u: G \rightarrow H$ be a homomorphism of pro- ℓ -Barsotti–Tate groups over S . If $u_0: G_0 \rightarrow H_0$ is zero, then u is zero. If $u_0: G_0 \rightarrow H_0$ is an isogeny (resp. an isomorphism), u is an isogeny (resp. an isomorphism).*

We indeed apply the first assertion of 5.2.5 to the composed morphism

$$T_\ell(A^0)^t \longrightarrow T_\ell(A^0)^f \longrightarrow D(T_\ell(A'^0)^f) ,$$

where the second morphism is that deduced from the pairing (5.2.4.2); one finds that this composition is zero, hence that the pairing under consideration vanishes on the toric part on the

left. By symmetry, it therefore vanishes on the toric part on the right, hence it comes from a pairing on the quotients, or “abelian pieces”

$$T_\ell(A^0)^{\text{ab}} \times T_\ell(A'^0)^{\text{ab}} \longrightarrow T_\ell(\mathbb{G}_m) ,$$

whose quotients are again Barsotti–Tate pro-groups, as one easily verifies. It remains to prove that the morphism corresponding to this pairing

$$T_\ell(A^0)^{\text{ab}} \longrightarrow D(T_\ell(A'^0)^{\text{ab}})$$

is an isogeny, which again results from 5.2.5.

We shall leave to the reader the (easy) proof of 5.2.5, which is a result valid over every noetherian connected base. We shall find moreover below (7.4.7), by a somewhat more precise analysis of the biextensions W_n of the $(A_n^0, A_n'^0)$, a fairly obvious direct proof of the two particular cases where we had to use 5.2.5.

5.2.6. Suppose now that A_K has semi-stable reduction over S , so that we may take for ℓ any prime number (distinct or not from p). Moreover, by virtue of the Galois criterion for semi-stable reduction 3.5, one has for $\ell \neq p$

$$(T_\ell(A'_K)^f)^\perp (= T_\ell(A_K)^t) \subset T_\ell(A_K)^f . \quad (29)$$

It remains to prove, to finish the proof of 5.2, that this inclusion is valid without restriction on ℓ . Indeed, the left hand side is canonically isomorphic to $D(T_\ell(A'_K)/T_\ell(A'_K)^f)$, and is thus of “rank” (or “height”) equal to $2n - 2\alpha - \mu = \mu$ (notation of (2.1.9), where by hypothesis $\lambda = 0$), which is also that of $T_\ell(A_K)^t$. As the inclusion

$$T_\ell(A_K)^t \subset T_\ell(A_K)$$

is *strict*, in the sense that it already induces a monomorphism for all terms of the projective systems under consideration, it follows easily that the first member of (*) is maximal among the sub-pro-Barsotti–Tate groups of $T_\ell(A_K)$ having the same rank as it, so that one indeed has the equality of (5.2.7), which is equivalent to the inclusion between the extreme terms of 5.2.7 by the orthogonality theorem.

5.3. Using the correspondence by orthogonality between strict sub-pro-Barsotti–Tate groups (in the precise sense explained above) of the dual pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$, one deduces formally from (5.2.2) that the pairing (1.0.2) between these pro-groups induces a pairing

$$T_\ell(A_K)^{\text{ab}} \times T_\ell(A'_K)^{\text{ab}} \longrightarrow T_\ell(\mathbb{G}_{mk}) \quad (30)$$

which is a *perfect duality*, i.e. identifying each of the two factors of the first member with the dual of the other (in the sense of 5.2.2). (NB the exponents ab denote the “abelian parts”, quotient of

the fixed part by the toric part). It is clear moreover that (5.3.2) is induced by the pairing

$$T_\ell(A^0)^{\text{ab}} \times T_\ell(A'^0)^{\text{ab}} \longrightarrow T_\ell(\mathbb{G}_{mS}) \quad (31)$$

deduced by passage to the quotients from the pairing (5.2.4.2). This being a perfect duality on the generic fibres, i.e. the corresponding homomorphism

$$T_\ell(A^0)^{\text{ab}} \longrightarrow D(T_\ell(A'^0)^{\text{ab}}) \quad (32)$$

being an isomorphism on the generic fibres, we obviously want to deduce from this that it is the same over all of S , or equivalently, over the special fibres. This deduction is valid, trivially, if $\ell \neq p$; it is also valid, at least in the case of unequal characteristics, if $\ell = p$, thanks to a deep theorem of TATE [33, th. 4]. We shall however be able to establish this *directly*, without the help of Tate's theorem, below (7.4.6).

To make precise the geometric meaning of the result we desire to establish, note that the pairing deduced from (5.3.3) by passage to the special fibres is none other than the pairing

$$T_\ell(B_0) \times T_\ell(B'_0) \longrightarrow T_\ell(\mathbb{G}_{mk}) \quad (33)$$

deduced from the biextension W_0 of $(A_0^0, A_0'^0)$ by passage to the quotient, where B_0 and B'_0 denote respectively the abelian parts of A_0^0 and $A_0'^0$. As, by virtue of VIII 4.8, the biextension W_0 comes from a biextension W_0^{ab} of (B_0, B'_0) by $\mathbb{G}_{mk}$, determined up to unique isomorphism, the pairing (5.3.5) can also be interpreted as deduced from the divisor correspondence W_0^{ab} , and the corresponding homomorphism

$$T_\ell(B_0) \longrightarrow D(T_\ell(B'_0)) \xrightarrow{\sim} T_\ell(D(B'_0)) \quad (34)$$

(where in the last term, the letter D denotes the dual abelian scheme) is none other than that deduced by functoriality of T_ℓ from the homomorphism

$$B_0 \longrightarrow D(B'_0) \quad (35)$$

defined by W_0^{ab} . Hence the fact that (5.3.5) is a perfect duality also means, by the well-known dictionary (cf. [5]) that the kernel of (5.3.7), which is a finite group over k by virtue of 1.5, has a null ℓ -primary component. Saying that this is so for all ℓ therefore means that the isogeny (5.3.7) is an *isomorphism*, i.e. that the divisor correspondence W_0^{ab} is a perfect duality between B_0 and B'_0 . The preceding considerations only show that the kernel of (5.3.7) is a p -group. Let us therefore state the more complete result :

Theorem 0.73 (5.4). *Let S be a trait, and A_K an abelian scheme over S having semi-stable reduction over S . Denote by B_0 , B'_0 the abelian parts of the Néron models of A_K and of the dual abelian scheme A'_K respectively. Then the canonical divisor correspondence W_0^{ab} on (B_0, B'_0) , deduced by VIII 4.8 from the canonical biextension W_0 of $(A_0^0, A_0'^0)$ (special fibre of the biextension W of 1.4), is a perfect duality between B_0 and B'_0 , i.e. defines an isomorphism of each of these*

abelian schemes with the dual of the other.

The proof of 5.4 in the general case will be given in No. 7, as a consequence of a very natural and important construction due to RAYNAUD. Let us only note here that to prove 5.4, one may assume S complete, in view of 3.3.

5.5. To express conveniently the extreme factors in the filtrations (5.1.10) and (5.1.11), in the case of semi-stable reduction, let us introduce the twisted constant group schemes over S

$$\underline{M}' = D(T_0) , \quad \underline{M} = D(T_0') , \quad (36)$$

duals of the maximal tori T_0, T_0' of the special fibres of A, A' (SGA 3 X 5.7). As S is local complete, these group schemes may be considered as the special fibres of twisted constant group schemes

$$\underline{M}' , \quad \underline{M} \quad \text{on } S ; \quad (37)$$

thus, if the torus T_0 is split (for example if k is separably closed) i.e. if $\underline{M}'_0$ is the constant group defined by the ordinary group

$$M' = \text{Hom}_{k\text{-gr}}(T_0, \mathbb{G}_{mk}) , \quad (38)$$

then $\underline{M}'$ is the constant group defined by M' :

$$\underline{M}' = \underline{M}'_S .$$

One may also note that, as T_0 and T_0' are isogenous (2.2.6), T_0' is split if T_0 is, so that there is reason to also introduce in this case

$$M = \text{Hom}_{k\text{-gr}}(T_0', \mathbb{G}_{mk}) , \quad (39)$$

and one will then have

$$\underline{M} = \underline{M}_S .$$

This being posited, one will evidently have, for every integer n , isomorphisms

$$\underline{M}'_n \simeq D(T_n) , \quad \underline{M}_n \simeq D(T'_n) , \quad (40)$$

where T'_n is defined in terms of A' as T_n in terms of A (5.1.2), whence one deduces canonical isomorphisms (where one sets $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{mS})$) :

$$T_\ell(T_n) \simeq \check{\underline{M}}'_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) , \quad T_\ell(T'_n) \simeq \check{\underline{M}}_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) , \quad (41)$$

and by passage to the limit on n canonical isomorphisms :

$$\begin{cases} T_\ell(A^0)^t = T_\ell(\hat{T}) \simeq \check{\underline{M}}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) \\ T_\ell(A'^0)^t = T_\ell(\hat{T}') \simeq \check{\underline{M}} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) . \end{cases} \quad (42)$$

In the semi-stable reduction case, one concludes by duality, using the orthogonality theorem 5.2 and relation (5.2.2), canonical isomorphisms :

$$\begin{cases} T_\ell(A_K)/T_\ell(A_K)^f \simeq \underline{M} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \\ T_\ell(A'_K)/T_\ell(A'_K)^f \simeq \underline{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell . \end{cases} \quad (43)$$

One finds likewise, thanks to the same formula (5.2.2), canonical isomorphisms :

$$\begin{cases} T_\ell(A_K)/T_\ell(A_K)^t \simeq D(T_\ell(A'_K)^f) \\ T_\ell(A'_K)/T_\ell(A'_K)^t \simeq D(T_\ell(A_K)^f) , \end{cases} \quad (44)$$

where one will recall that the groups $T_\ell(A_K)^f$ and $T_\ell(A'_K)^f$ appearing in the second member are the restrictions to η of Barsotti–Tate pro- ℓ -groups defined over S , $T_\ell(A^0)^f$ and $T_\ell(A'^0)^f$ respectively; the letter D denotes here the duals in the sense of Barsotti–Tate pro-groups, namely $\text{Hom}(-, \mathbb{Z}_\ell(1))$.

Proposition 0.74 (5.6). *Suppose S complete and that A_K has semi-stable reduction over S . Then for every prime number ℓ (which may be equal to the residual characteristic of S), the pro- ℓ -Barsotti–Tate group $T_\ell(A_K)^t$ is the generic fibre of a Barsotti–Tate group over S , of “toric type”, given by (5.5.7), while $T_\ell(A_K)/T_\ell(A_K)^f$ is the generic fibre of a Barsotti–Tate group over S of pro-étale type, given by (5.5.8). Finally the groups $T_\ell(A_K)^f$ and $T_\ell(A_K)/T_\ell(A_K)^t$ are the generic fibres of Barsotti–Tate groups defined over S , equal respectively to $T_\ell(A^0)^f$ and to $D(T_\ell(A^0)^f) = \text{Hom}(T_\ell(A^0)^f, \mathbb{Z}_\ell(1))$.*

5.6.1. One will note that it amounts to the same for the étale covering S' to split $T_\ell(A_K)^t$ or to split $T_\ell(A_K)/T_\ell(A_K)^f$, for this means respectively that S' splits the twisted constant groups $\underline{M}$ and $\underline{M}'$, which are isogenous, since T_0 and T'_0 are so. One sees at the same time that there exists a smallest étale extension S' of S which does the job.

Let us also mention the following easy consequence of the definitions of 2.2.3 :

Proposition 0.75 (5.7). *Suppose A^0 is ℓ -divisible, S henselian. Let F_K be a constructible ℓ -adic sheaf on K which is “unramified” i.e. which extends to a constant twisted ℓ -adic sheaf F on S , and let $u: F_K \rightarrow T_\ell(A_K)$ be a homomorphism of pro-groups, then u factorises into a homomorphism from F into $T_\ell(A_K)^f$.*

Proof. Indeed, one will have

$$u = (u(\nu))_{\nu \geq 0} , \quad \text{with } u(\nu): F_K(\nu) \longrightarrow T_\ell(A_K)(\nu) = {}_\ell \nu A_K . \quad (45)$$

By the universal property of A , as $F(\nu)$ is étale hence smooth over S , the morphism $u(\nu)$ comes from a homomorphism

$$v(\nu): F(\nu) \longrightarrow {}_\ell \nu A ,$$

and by definition of the fixed part, this homomorphism factorises as

$$v(\nu): F(\nu) \longrightarrow (\ell^\nu A)^f .$$

As the image of the induced morphism on the special fibres falls into the infinitely ℓ -divisible part of A_0 , one concludes that $v(\nu)$ even factorises through $(\ell^\nu A^0)^f$. By passage to the limit on ν , one finds a homomorphism

$$v: F \longrightarrow T_\ell(A^0)^f ,$$

and the morphism induced on the generic fibres evidently gives the sought factorisation. $\square$

5.8. Using the orthogonality theorem, we shall now obtain the main variants of the Néron–Ogg–Shafarevich criterion :

Theorem 0.76 (5.8). *Let A_K be an abelian scheme over K . Suppose that S is strictly local, and let ℓ be a prime number $\neq p$. The following conditions are equivalent :*

- (i) A_K has good reduction over S , i.e. A_K extends to an abelian scheme over S .
- (ii) The specialisation homomorphism $I \rightarrow \text{Aut}(T_\ell(A_K))$ is trivial.

Theorem 0.77 (5.9). *With the notation of 5.8, suppose A_K has semi-stable reduction over S . Then the following equivalent conditions :*

- (i) A_K has good reduction over S .
- (ii) $T_\ell(A_K)^t = 0$.
- (iii) $T_\ell(A_K)^t = 0$ for one prime number $\ell \neq p$.

Theorem 0.78 (5.10). *With the notation of 5.8 (but without necessarily supposing A_K to have semi-stable reduction), the following equivalent conditions :*

- (i) A_K has good reduction over S .
- (ii) $T_\ell(A_K)^f = 0$ for one prime number $\ell \neq p$.
- (iii) $T_\ell(A_K)^f = 0$ for every prime number $\ell \neq p$.

Proof. (i) $\Rightarrow$ (ii) is clear, for if A_K has good reduction, then $A^0 = A$ and $T_\ell(A^0)^f = T_\ell(A)$, hence $T_\ell(A_K)^f = 0$.

(ii) $\Rightarrow$ (iii) is trivial.

(iii) $\Rightarrow$ (i) : If $T_\ell(A_K)^f = 0$, then A_0^0 is unipotent (2.2.5). But for it to be zero, it is necessary and sufficient that the euclidean dual of $T_\ell(A_K)/T_\ell(A_K)^f$ be zero, which still means that the maximal torus of A_0^0 is zero, i.e. A'_K has good reduction. As $A''_K = A_K$, this means that A_K has good reduction. $\square$

5.11. One also finds criteria for good reduction as well as for semi-stable reduction in terms of finite groups :

Theorem 0.79 (5.12). *Suppose S strictly local, and let n be an integer ≥ 1 prime to p . The following conditions are equivalent :*

- (i) A_K has good reduction over S .
- (ii) The group ${}_nA_K$ is unramified over S , i.e. I acts trivially on ${}_nA(\overline{K})$.

Proof. If A_K has good reduction, ${}_nA$ is finite and smooth over S , hence ${}_nA_K$ is unramified. Conversely, suppose that ${}_nA_K$ is unramified. Then ${}_nA$ is a finite and flat scheme over S (SGA 3 X 5.9), and as ${}_nA_K$ is unramified, the special fibre ${}_nA_0$ is étale over k . Hence ${}_nA$ is étale over S . For $p > 1$, this implies that A_0 is an abelian scheme, hence A_K has good reduction. $\square$

Theorem 0.80 (5.13). *With the notation of 5.12 (but without supposing A_K to have semi-stable reduction), the following equivalent conditions :*

- (i) A_K has semi-stable reduction over S .
- (ii) For every element $g \in I$, the multiplicity of 1 as a root of the characteristic polynomial of the automorphism of $V_\ell(A_K)$ defined by g is equal to 2α , where α denotes the rank of the abelian scheme $B_0^{A_K}$ appearing in the decomposition (2.2.2) of the special fibre of the Néron model of A_K .

Proof. (i) $\Rightarrow$ (ii) : If A_K has semi-stable reduction, then the characteristic polynomial of $g \in I$ is of the form $(t - 1)^{2\alpha}P(t)$, where $P(1) \neq 0$, whence (ii).

(ii) $\Rightarrow$ (i) : This is a criterion of M. RAYNAUD, whose proof will be given elsewhere. $\square$

Theorem 0.81 (5.14). *Suppose S strictly local. Let ℓ be a prime number $\neq p$, A_K an abelian scheme over K , and suppose that A_K has semi-stable reduction. Consider the canonical injection*

$$T_\ell(A_K)^f \hookrightarrow T_\ell(A_K), \quad (46)$$

and let μ be the rank of the toric factor of the special fibre of the Néron model of A_K . Then the following equivalent conditions :

- (i) $\mu = 0$, i.e. A_K has good reduction over S .
- (ii) I acts trivially on $T_\ell(A_K)/T_\ell(A_K)^f$, and the characteristic polynomial of the automorphism defined by an element $g \in I$ on $T_\ell(A_K)/T_\ell(A_K)^f$ has rational integer coefficients independent of ℓ .

Theorem 0.82 (5.15). *With the notation of 5.14, suppose k finite. Then the following conditions are equivalent :*

- (i) A_K has semi-stable reduction over S .
- (ii) $T_\ell(A_K)/T_\ell(A_K)^f$ and $T_\ell(A_K)^t$ are defined over K (i.e. I acts trivially on these groups).

§ 6. Remarks on the construction of the toric and fixed parts in the non-semi-stable case.

6.1. Let us now return to the general situation of § 2, that is, we do not suppose A_K to have semi-stable reduction. One then has, for every prime number $\ell \neq p$, the filtrations (1.3.2) and (2.3.0) :

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K). \quad (47)$$

The fundamental result of 3.6 teaches us that the toric part $T_\ell(A_K)^t$ is zero for ℓ sufficiently large, i.e. if K' is a finite extension of K such that $A_{K'}$ has semi-stable reduction over S' , then $I' = \text{Gal}(\overline{K}/K')$ acts trivially on $T_\ell(A_K)^t$, so that $T_\ell(A_K)^t$ is the generic fibre of a Barsotti–Tate pro-group of toric type over S . Likewise, the fixed part $T_\ell(A_K)^f$ is defined for all $\ell \neq p$, and we have seen in 2.2 that $T_\ell(A_K)^f$ is the generic fibre of a Barsotti–Tate pro-group over S , restriction to η of $T_\ell(A^0)^f$, when S is strictly local and A^0 is ℓ -divisible.

We now propose to show how the results of §§ 4, 5 allow in their turn to simplify and make precise the structure of the filtration (6.1.1) in the general case, and to construct the toric and fixed parts in an intrinsic way, without passing through the semi-stable reduction.

6.2. Let S be strictly local. Recall (1.4) that we have constructed a *biextension* W of (A^0, A'^0) by $\mathbb{G}_{mS}$, where A' is the Néron model of the dual A'_K of A_K . This biextension reduces on the special fibre to a biextension W_0 of $(A_0^0, A_0'^0)$ by $\mathbb{G}_{mk}$, and this biextension defines a divisor correspondence, identifying B_0' with the dual of B_0 (and reciprocally), where B_0 and B_0' denote the “abelian parts” of A_0^0 and $A_0'^0$ respectively (2.2.2).

Let T_0 be the maximal torus of A_0^0 , and T_0' the maximal torus of $A_0'^0$. According to 2.2.6, the twisted constant group schemes $M = D(T_0')$ and $M' = D(T_0)$ are isogenous. The fact that W_0 vanishes on $T_0 \times T_0'$ (VIII 4.10) means that the pairing

$$T_\ell(T_0) \times T_\ell(T_0') \longrightarrow T_\ell(\mathbb{G}_{mk}) \quad (48)$$

is trivial, which is evident since $T_\ell(T_0)$ and $T_\ell(T_0')$ are Barsotti–Tate pro-groups of toric type, and every homomorphism from such a group to $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{mS})$ is trivial. The vanishing of (6.2.1) allows one to pass to the quotient and obtain a pairing :

$$T_\ell(B_0) \times T_\ell(B_0') \longrightarrow T_\ell(\mathbb{G}_{mk}), \quad (49)$$

which is a *perfect duality* according to 5.4.

6.3. Let ℓ be a prime number $\neq p$. We shall now show that the factors of the filtration (6.1.1) can be reconstructed in terms of the filtration (1.3.2) and of the biextension pairing. Indeed, consider the exact sequence of Galois cohomology

$$0 \longrightarrow T_\ell(A_K)^t \longrightarrow T_\ell(A_K)^f \longrightarrow T_\ell(A_K)^{\text{ab}} \longrightarrow 0, \quad (50)$$

where $T_\ell(A_K)^{\text{ab}}$ denotes the quotient $T_\ell(A_K)^f/T_\ell(A_K)^t$, and the dual exact sequence

$$0 \longrightarrow T_\ell(A'_K)^t \longrightarrow T_\ell(A'_K)^f \longrightarrow T_\ell(A'_K)^{\text{ab}} \longrightarrow 0 . \quad (51)$$

The biextension pairing defines a perfect duality

$$T_\ell(A_K)^{\text{ab}} \times T_\ell(A'_K)^{\text{ab}} \longrightarrow \mathbb{Z}_\ell(1) , \quad (52)$$

so that $T_\ell(A_K)^{\text{ab}}$ and $T_\ell(A'_K)^{\text{ab}}$ determine one another by duality.

6.4. We have seen in 2.2.6 that the ranks of the tori T_0 and T'_0 are equal. More precisely, the constant twisted groups $M = D(T'_0)$ and $M' = D(T_0)$ are isogenous. In fact, the divisor correspondence W_0^{ab} allows us to define a *canonical isomorphism* between the $\mathbb{Q}_\ell$ -vector spaces

$$V_\ell(A_K)^t \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \simeq V_\ell(A'_K)^t \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell , \quad (53)$$

and the structure of the factors of the filtration (6.1.1) is completely determined by the biextension pairing.

6.5. To conclude, let us point out that the construction of the toric part in the case where $\ell = p$ (hence in residual characteristic p) is done by the same method as in 5.1 : one defines the toric part by

$$T_p(A^0)^t = T_p(\hat{T}) \subset T_p(A^0)^f , \quad (54)$$

where $\hat{T}$ is the formal torus defined in (5.1.3). This construction is valid under the hypothesis that A^0 is p -divisible, which is the case when A_K has semi-stable reduction over S (3.5). In the general case, one can always define $T_p(A^0)^t$ as the largest sub-pro-Barsotti–Tate group of toric type contained in $T_p(A^0)^f$.

6.6. Let us summarise the main properties of the monodromy filtration in the general case : to every abelian scheme A_K over K , one has associated a canonical filtration :

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K) , \quad (\ell \neq p) , \quad (55)$$

whose successive factors are :

- $T_\ell(A_K)^t$: toric part, of rank $\mu = \dim T_0$, defined over a finite extension of K (semi-stable reduction) ;
- $T_\ell(A_K)^{\text{ab}} = T_\ell(A_K)^f/T_\ell(A_K)^t$: abelian part, of rank $2\alpha = 2 \dim B_0$, in perfect duality with the abelian part of the dual ;
- $T_\ell(A_K)/T_\ell(A_K)^f$: étale part, of rank μ , isomorphic to $M' \otimes \mathbb{Z}_\ell$ for the semi-stable reduction.

The ranks of these groups are related to the invariants of the decomposition (2.2.2) by :

$$\dim A_K = \alpha + \lambda + \mu , \quad (56)$$

where $\alpha = \dim B_0$, $\lambda = \dim U_0$, $\mu = \dim T_0$.

In the semi-stable reduction case ($\lambda = 0$), one has $\dim A_K = \alpha + \mu$, and the monodromy filtration reduces to :

$$0 \subset T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K) , \quad (57)$$

with :

- $\text{rank } T_\ell(A_K)^t = \mu$,
- $\text{rank } T_\ell(A_K)^f = 2\alpha + \mu$,
- $\text{rank } T_\ell(A_K)/T_\ell(A_K)^f = \mu$.